

- 6 To recover concentrated antibody, invert Microcon into a new vial. Spin at 1000g for 3 min.

4. Notes

1. The 90% rejection standard and the significance of molecular shape rather than weight make it very important that the selected membrane cutoff be well below the molecular weight of the solute to be retained. As a rule, it is best to choose a membrane or device with cutoff at about half of the molecular weight of the protein to be concentrated. This provides a balance between high protein recovery and minimal filtration time. For example, if bovine serum albumin ($M_r = 67,000$) were the protein of interest, a 30,000 MWCO (rather than 50,000 MWCO) membrane or device would result in the most efficient concentration and recovery of the protein in the retentate. For highest possible retention, a membrane cutoff one-tenth of the M_r of the solute should be used (10,000 in the example above). Of course, the tighter the membrane, the slower the filtration rate. When species of low molecular weight are to be exchanged, the membrane cutoff should be substantially above that of the passing solute.
2. Recovery of protein can sometimes be improved by passivation of the membrane. For very dilute protein solutions (in the range of 1 $\mu\text{g/mL}$), concentrate recovery is often not quantitative (Fig. 5). This loss of protein can be caused by nonspecific binding of the protein to exposed binding sites on the plastic of the concentration device. The extent of nonspecific binding varies with the relative hydrophobicity of individual protein conformations. Pretreating (passivating) the plastic by blocking the available binding sites before concentration can often improve the recovery yield from dilute protein solutions. For passivation procedures, manufacturers' recommendations should be followed. The procedure is simple, requiring the filling of devices with a prescribed solution, leaving them overnight, and then rinsing thoroughly. Among recommended passivation solutions are 1% IgG in PBS, Tween-20, or Triton-X in distilled water, bovine serum albumin, and even powdered milk in distilled water.
3. Highly viscous solutions filter slowly, as do solutions containing particulate matter, such as colloids. Where a viscous agent (sucrose, glycerin, etc.) is to be removed, flow can often be increased by predilution.
4. For rapid desalting (diafiltration), the UF cell can be connected to an auxiliary reservoir containing a diafiltrate solution with the desired microsolite concentration, if any. The auxiliary reservoir is then gas pressurized to 2.7–3.4 atm (40–50 psi). The cell's fluid volume, as well as the macrosolute concentration, remain constant as filtrate is automatically replaced by diafiltrate solution. This technique provides a simple means for rapid microsolite exchange. It is typically used as substitute for dialysis. A Concentration/Dialysis Selector Switch (Amicon) permits simple switching between concentration and diafiltration (Fig. 3).

- 5 Fully membrane-permeating molecules pass at the same rate as salt because their transport through the membrane is independent of their individual molecular sizes. Proteins of $M_r > 3000$ may be separated from each other and from smaller molecules if they differ by a factor of about 10 in molecular size and are not associated in solution. For example, one cannot diafilter bovine serum albumin ($M_r = 67,000$) from IgG ($M_r = 160,000$), but can easily diafilter biotin ($M_r = 244$) from cytochrome c ($M_r = 12,400$). Small volumes should be prediluted to assist in diafiltration.
- 6 UF is not primarily a fractionation technique. It can only separate molecules that differ by at least an order of magnitude in size and is best done in a diafiltration arrangement, with multiple washes to separate the mixture of macromolecules. Molecules of similar size cannot be separated by UF. During concentration polarization, the gel layer on the membrane surface superimposes its own rejection characteristics on those of the membrane. Usually, concentration polarization increases rejection of lower M_r species. A membrane with a nominal 100,000 MWCO may reject 10–20% of albumin ($M_r = 67,000$) in a 0.1% solution of pure albumin. However, in the presence of larger solutes, such as IgG, it may reject 90% of the albumin.
- 7 For example, the monomer of Triton X-100 ($M_r = 500$ –650) should pass readily through a 3000 MWCO membrane. However, at concentrations above its Critical Micelle Concentration of 0.2 mM, Triton X-100 forms micelles composed of approx 140 monomeric units. During UF, the micelles behave like globular proteins of $M_r = 70,000$ –90,000. Therefore, above the Critical Micelle Concentration of Triton X-100, a 100,000 MWCO membrane would be required to remove the detergent effectively.
8. Use of compressed air can cause large pH shifts owing to dissolution of carbon dioxide. With sensitive solutions, oxidation can occur, leading to other potential problems.
- 9 Because of their unique design, which uses gravitational force to counteract deposition of suspended particles on the membrane, certain concentrators (Centriprep®, Amicon) are especially useful for UF of solutions with high solid content. The feature counteracts membrane fouling and eliminates the need for prefiltration of samples. It can also be obtained with a microporous filter for separation of antibodies from cell-culture supernatant or for desalting of bacteria. Another centrifugal UF device uses a low-adsorptive, polypropylene sample reservoir, which minimizes nonspecific adsorption of solutes to the walls of the devices. It also contains a low-adsorptive, hydrophilic membrane. No passivation is required before processing very dilute protein solutions. By employing a final inverted-spin feature, high solute recovery is possible, even from dilute solutions in the 1 µg/mL range (Centriplus™).

CHAPTER 13

Bulk Purification by Fractional Precipitation

Shawn Doonan

1. Introduction

The solubility of a particular protein in aqueous solution depends on the solvent composition and on the pH; hence, variation in these parameters provides a way of purifying proteins by fractional precipitation. The factors that dictate solubility are complex, since the surface of a protein is itself complex containing ionized residues, polar regions, and hydrophobic patches, all of which will interact with the solvent in ways that are not completely understood. Hence, it is not possible to elaborate a theoretical approach to fractional precipitation (1); rather, the methods are used in an essentially empirical fashion. The most widely used procedures are precipitation by addition of salt (salting out) and by addition of organic solvents; these are the focus of this chapter (*see* Note 1).

Addition of high concentrations of salt to a protein solution causes precipitation largely by removing water of solvation from hydrophobic patches on the protein's surface, thus allowing these patches to interact with resulting aggregation. For a pure protein, the relationship between solubility S (in g/kg of water) and the ionic strength I (in mol/kg water) is given by:

$$\log S = \beta - K_s[(I/2)] \quad (1)$$

where β and K_s are constants for a particular protein at a particular pH and temperature. The point here is that a protein will precipitate over a range of ionic strength values (determined by the value of K_s) and that

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different proteins will precipitate over different, but frequently overlapping, ranges. This highlights the fact that it is rarely possible to purify a particular protein from a complex mixture by using fractional precipitation alone (although heroic attempts were made to do this in the early days of protein purification). Rather, the value of the method is that it provides a simple procedure for enrichment of the protein of interest from large volumes of extracts and, at the same time, can be used to concentrate the fraction (*see* Chapter 10); with large-scale purifications in particular, this is important in reducing the problem to a manageable scale. Hence, fractional precipitation by salt is almost invariably used at an early stage of a purification procedure, often on the clarified extract, to obtain an initial purification and concentration (*see* Note 2). Further purification is then achieved by chromatographic methods, which have higher resolving power, but generally lower capacity.

A variety of salts has been used in the past for this purpose, including NaCl, Na₂SO₄, KCl, CaCl₂, and MgSO₄, and these are still sometimes used for particular applications. By far the most frequently used salt is, however, ammonium sulfate ((NH₄)₂SO₄). The reasons for this include its high solubility in water (about 4M at saturation), its low heat of solution, the fact that the density of saturated solutions (1.235 g/mL) is less than that of proteins, hence allowing for their collection by centrifugation, and the essentially innocuous nature to proteins of its constituent ions. Concentrations of the salt are traditionally expressed in terms of percentage saturation at a particular temperature; Table 1 gives the amounts of solid ammonium sulfate required to obtain the required concentration at 0°C (2), the temperature at or near which fractional precipitation is usually carried out (*see* Note 3).

The precipitation of proteins by addition of organic solvents is a more complex process. Factors involved probably include the decrease in dielectric constant, which promotes aggregation by charge interaction, as well as sequestration of water of solvation of the protein. The same caveats about the usefulness of the method apply as discussed for salt fractionation, although solvent fractionation has achieved some noteworthy successes, such as the classical Cohn fractionation of plasma proteins (summarized in ref. 3). There is with this method, however, the added problem that organic solvents can cause protein denaturation by interaction with hydrophobic residues in the protein's interior. Hence, it is usually essential to carry out solvent precipitation at a low temperature to minimize denaturation. The solvents used need

Table 1
Amounts of Solid Ammonium Sulfate Required to Change the Concentration
of a Solution from a Given Starting Value to a Desired Target Value at 0°C

Initial percentage saturation at 0°C	Target percentage saturation at 0°C ^a																
	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100
0	106	134	164	194	226	258	291	326	361	398	436	476	516	559	603	650	697
5	79	108	137	166	197	229	262	296	331	368	405	444	484	526	570	615	662
10	53	81	109	139	169	200	233	266	301	337	374	412	452	493	536	581	627
15	26	54	82	111	141	172	204	237	271	306	343	381	420	460	503	547	592
20		27	55	83	113	143	175	207	241	276	312	349	387	427	469	512	557
25			27	56	84	115	146	179	211	245	280	317	355	395	436	478	522
30				28	56	86	117	148	181	214	249	285	323	362	402	445	488
35					28	57	87	118	151	184	218	254	291	329	369	410	453
40						29	58	89	120	153	187	222	258	296	335	376	418
45							29	59	90	123	156	190	226	263	302	342	383
50								30	60	92	125	159	194	230	268	308	348
55									30	61	93	127	161	197	235	273	313
60										31	62	95	129	164	201	239	279
65											31	63	97	132	168	205	244
70												32	65	99	134	171	209
75													32	66	101	137	174
80														33	67	103	139
85															34	68	105
90																34	70
95																	35

^ag of solid ammonium sulfate/L of solution

to be miscible with water in all proportions and nontoxic; acetone and ethyl alcohol best meet these requirements (*see* Note 4).

Whichever of the two methods is used, there will necessarily be a trade-off between degree of purification achieved and yield. An ammonium sulfate or solvent "cut" over a 10% concentration range might give a yield of 70% of the desired protein with a purification factor of five-fold. Increasing the concentration range may increase the yield, but with a corresponding decrease in purification factor. For an example of the use of both methods in purification of the same protein, *see* ref. 4.

2. Materials

2.1. Fractional Precipitation with Ammonium Sulfate

1. Solid ammonium sulfate (*see* Note 5).
2. Magnetic stirrer or electrical paddle stirrer.

3. Ice bath
4. Refrigerated centrifuge and rotor, e.g., a 6×250 -mL angle rotor
5. Screw-top plastic bottles for the rotor to be used.
6. Visking dialysis tubing (14- or 19-mm inflated diameter; 26- or 31-mm flat width).

2.2. Fractional Precipitation with Acetone

1. Analar acetone precooled to -20°C (*see* Note 6).
2. Electrical paddle stirrer.
3. Ice-salt cooling bath.
4. Centrifuge, rotor, bottles, and dialysis tubing as in Section 2 1

3. Methods

3.1. Fractional Precipitation with Ammonium Sulfate

It is assumed that a trial experiment has been carried out in order to determine the optimal concentration range of ammonium sulfate for the particular protein sample to be fractionated (*see* Note 7). Suppose that the trial showed the best results to be obtained with the fraction precipitated between 35 and 50% saturation. Proceed as follows.

1. Measure the volume of the protein solution to be fractionated, and pour it into a glass beaker of capacity about twice the measured volume of solution
2. Weigh out 0.194 g of solid ammonium sulfate for every 1 mL of protein solution (Table 1) If the ammonium sulfate contains lumps, then these should be broken up using a mortar and pestle.
3. Place the beaker containing the protein on ice, and stir either with a magnetic stirrer or with a paddle stirrer A slow rate of stirring should be used to avoid foaming.
4. Add the ammonium sulfate to the protein solution in small batches over a period of several minutes ensuring that one lot of ammonium sulfate has dissolved before adding the next (*see* Note 8).
5. After addition is complete, leave to stand for 10 min to ensure equilibrium, and then remove precipitated protein by centrifugation at about 5000g and 4°C for 30 min using screw-top plastic tubes (*see* Note 9).
6. Decant off the supernatant solution from each tube into a measuring cylinder, and determine the total volume (*see* Note 10).
7. Pour the combined supernatants into a beaker in the ice bath and add 0.087 g of ammonium sulfate/mL of protein solution (the amount required to take the concentration from 35 to 50% saturation; *see* Table 1) using the same precautions as in steps 2–4.
8. Recover the precipitated protein by centrifugation as in step 5, decant the supernatant solution into a beaker (*see* Note 10), and suspend the protein

pellets in the minimum volume of water or of an appropriate buffer (*see* Note 11).

9. Transfer the protein suspension to a sack of Visking dialysis tubing, and dialyze against at least two changes of buffer using 100 times the volume of the sample and allowing 3–4 h for equilibration (*see* Note 12).
10. After dialysis, remove any precipitated material by centrifugation.

3.2. Fractional Precipitation with Acetone

Again, a trial experiment must be carried out to determine the optimum precipitation range for the particular protein fraction (*see* Note 7). Assuming this to be between 37.5 and 50% (v/v), proceed as follows.

1. Measure the volume of the protein solution, pour it into a beaker (*see* Note 13) immersed in an ice-salt bath, and stir until the temperature reaches 0°C.
2. For each 1 mL of protein solution, add 0.60 mL of acetone (precooled to –20°C) dropwise with constant stirring and at such a rate that the temperature does not rise above 0°C (*see* Notes 8 and 14). After addition of acetone is complete, continue stirring for 10 min with constant control of temperature.
3. Remove precipitated protein by centrifugation at 0°C for 10 min at 3000g using precooled centrifuge tubes (*see* Note 15).
4. Measure the volume, and return the combined supernatant solutions to the beaker in the ice-salt bath. Add a further 0.25 mL of acetone/mL of protein solution using the precautions described in steps 1 and 2.
5. Recover the precipitated protein by centrifugation as in step 3. Pour off the supernatant solutions, and invert the centrifuge tubes over filter paper to drain; blot off any drops of solution adhering to the walls of the tubes.
6. Resuspend the pellets in water or an appropriate buffer, and remove residual acetone by dialysis, membrane filtration, or gel filtration (*see* Note 16).

4. Notes

1. An alternative that is sometimes used is precipitation by alteration of pH. Proteins will generally have minimum solubility at their isoelectric points, where the net charge is zero and there is no electrostatic repulsion. The approach is to incubate the protein fraction at various pH values to see if the species of interest precipitates; obviously, if the isoelectric point of the protein is known, then that pH should be used. A problem with the method is that the protein may not be stable at its isoelectric point if this is far removed from neutrality. In addition, the protein may not precipitate, particularly if its concentration is low, in the absence of added salt; this makes establishing the conditions for pH fractionation quite difficult.
2. There is a further advantage in using ammonium sulfate fractionation immediately after homogenization and clarification (*see* Chapter 2). Some

tissues give homogenates that are very difficult to clarify because of the presence of membrane fragments and nucleoprotein complexes, which resist sedimentation. This particulate matter usually aggregates at low concentrations of ammonium sulfate and sediments readily in the first fraction. Hence, unless the protein of interest is contained in this fraction, the protein solution obtained from this procedure will be devoid of suspended matter as required for use of such techniques as column chromatography. Fractionation with organic solvents confers the same advantage. In this case, it is because of the low density and viscosity of water/solvent mixtures, which facilitate sedimentation of particulate matter

3. Reference 3 gives a corresponding table for 25°C, which may be used if it is preferable to carry out fractionation at room temperature. Alternatively, the following formula can be used:

$$g = [533 (P_2 - P_1)/(100 - 0.3 P_2)] \quad (2)$$

where g is the number of grams of ammonium sulfate required to change the concentration of 1 L of solution from P_1 to P_2 % saturation at 20°C.

4. An alternative to using organic solvents is provided by fractionation with water-soluble polymers of which polyethylene glycol (PEG) is the most commonly employed. The mechanism of precipitation seems to be by steric exclusion, i.e., the protein is concentrated in the extra-polymer space of the solution until its solubility limit is exceeded and precipitation occurs (5). Consistently, larger proteins tend to precipitate earlier than smaller ones, and precipitation is relatively insensitive to pH and ionic strength. The commonly used precipitants are PEG 4000 or PEG 6000 (i.e., PEGs with mol-wt averages of 4000 or 6000). The great advantage of these precipitants is that they have little tendency to denature proteins. For further details, *see* ref. 5.
5. Analar ammonium sulfate is more than twice as expensive as the general-purpose grade (around \$100 for 5 kg as compared to about \$45 for GPR). It is doubtful whether the extra cost is worth it when dealing with large volumes of crude protein mixtures; Aristar grade is certainly not worth using at about \$225/kg. The major problem with cheaper grades is the presence of low amounts of heavy metal contaminants; if the protein of interest is metal-sensitive, then EDTA (10 mM) can be added to the solution to remove these contaminants.
6. Use Analar grade, which is only marginally more expensive than the general-purpose grade. If carrying out fractionation with ethanol, then either absolute ethanol (>99%) or the more commonly available 96% variety may be used; in the latter case, the volume added to achieve the desired concentration would need to be adjusted to allow for the water content.

7. Optimum in this context means the range of ammonium sulfate concentrations that gives the desired balance between yield and purification. If the protein of interest has been purified previously, then the required range may be available in the literature, but it should be borne in mind that precipitation will depend on pH, on buffer composition, and on the protein concentration and composition of the fraction. Hence, unless these are identical with those in published procedures, it is unwise to assume that the protein will behave identically in your purification.

Ideally, the trial ammonium sulfate fractionation should be carried out exactly as described in Section 3.1., but using a small volume (~20 mL) of fraction. That is, ammonium sulfate should be added to concentrations of 0–20, 20–30%, and so on, in 10% steps at each stage removing precipitated protein by centrifugation before increasing the salt concentration. The recovered precipitates should be dissolved in buffer and assayed for the protein of interest and for total protein; the latter can be done most simply by measuring the absorbance of a suitably diluted sample at 280 nm and using the approximate relationship that an absorbance of one corresponds to a protein concentration of 1 mg/mL (accurate enough for present purposes). If the test for activity of the protein is sensitive to NH_4^+ or SO_4^{2-} ions, then the dissolved pellets will have to be dialyzed before assay. This trial should give the necessary information to proceed to large-scale work, but if, for example, the protein of interest precipitates equally in the ranges 20–30% and 30–40%, it may be worthwhile checking to see if the range 25–35% gives better results than simply taking a 20–40% cut. If the protein precipitates over a very broad range so that only a low degree of purification can be achieved, then it is probably better to abandon the idea of fractionation and use precipitation only as a means of concentration if large volume is a problem (see Chapter 10).

A quicker way of carrying out a trial is to take several samples of the protein fraction and add 20% ammonium sulfate to the first, 30% to the second, and so on. Then, after equilibration, recover the precipitated protein by centrifugation, dissolve, and assay as above. The problem with this is that the precipitation behavior of a particular protein will depend on the precise protein composition of the solution, so that the conditions in a trial of this sort will not properly reflect the conditions in the large-scale procedure. The use of this method is not, therefore, recommended. The above considerations apply equally to trial solvent fractionation experiments.

8. Important: If the salt or organic solvent is added too quickly, then high local concentrations will develop and proteins will be precipitated that would remain soluble at the target ammonium sulfate or solvent concentration. Such proteins may not readily redissolve, and the result will be

decreased purity of the active fraction and possible loss of the protein of interest in lower fractions. With solvents, there is also the increased risk of denaturation. Some authors recommend the use of saturated solutions of ammonium sulfate for fractionation, but this has the disadvantage of leading to large volume increases and should not be necessary if due care is taken when adding the solid salt.

- 9 Care must be taken with this step to protect both the centrifuge and the operator! The tubes must be balanced to within 0.1–0.2 g across the rotor axis (*see* Note 3, Chapter 2), and it is particularly important not to counter-balance a tube full of ammonium sulfate solution with a tube full of water because of the large difference in density between the two. Either the ammonium sulfate/protein suspension should be divided between two tubes, or the suspension should be balanced using a solution of ammonium sulfate of the same concentration. It is also most important to avoid spillage of ammonium sulfate solutions into the centrifuge head. Such solutions are extremely corrosive to the materials of which rotors are constructed and can lead to irreversible damage, rendering the rotors unsafe to use. After centrifugation of ammonium sulfate (or other salt) solutions, rotors should always be removed and washed in warm water.

Note that proteins precipitated in ammonium sulfate are usually stable, and hence, it is often convenient to store the suspension overnight at 4°C in this form before proceeding with the purification schedule.

- 10 The pellets of precipitated protein from this step may be discarded, but to be on the safe side, it is worth keeping them until it has been established that the protein of interest is indeed obtained in the next fraction. If something has gone wrong, then it is easier to reprocess the 0–35% precipitate than to go back to the beginning of the preparation and start again! Similarly, do not throw the supernatant from the 35–50% cut away until you are sure that your protein has been precipitated.
- 11 This step is crucial to obtaining a concentrated fraction. Add a very small volume of water or buffer (about 10 mL if using a 250-mL tube), and resuspend the protein pellet using a glass rod. If several centrifuge tubes have been used, then transfer the protein suspension from tube to tube, resuspending the pellet each time; only add more water or buffer if the suspension becomes too thick to transfer readily. After transfer of the final suspension to the dialysis bag, a further small aliquot of water or buffer can be used to rinse out the tubes and the rinsings combined with the suspension. A common error is to attempt to redissolve the protein pellets after centrifugation, since the pellets contain considerable quantities of ammonium sulfate, this takes a large volume of buffer and it is easy to end up with a volume comparable to that of the original fraction. When doing a

trial fractionation (*see* Note 7), it is acceptable to redissolve the precipitates, since the volume will not generally be important.

12. Visking tubing comes in several sizes. For a given volume of solution, the larger the diameter of the dialysis tubing used, the shorter the piece that will be required; attainment of equilibrium, however, will be slower with short, fat bags than with long, thin ones. The sizes recommended are a compromise between these two factors. For most applications, it is only necessary to soak the dry tubing in water or in the buffer to be used for dialysis, and it is ready to be used. The tubing does, however, contain significant quantities of sulfur compounds and of heavy metal ions; the latter may be a problem if the protein of interest is metal sensitive. They may be removed by boiling the dialysis tubing in 2% (w/v) sodium bicarbonate, 0.05% (w/v) EDTA for about 15 min, washing with distilled water, and then boiling in distilled water twice for 15-min periods. Prepared tubing can be stored indefinitely at 4°C in water or buffer containing 0.1% (w/v) sodium azide. For most applications, Visking tubing is perfectly adequate, but problems will arise if the protein of interest is small. The pores in this type of tubing are of such a size that the nominal mol-wt cutoff (NMWC) is about 15,000, although larger proteins may still pass through the pores if they have an elongated shape. As a rule of thumb, Visking tubing can be used with confidence if $M_r > 20,000$. For dialysis of smaller proteins, Spectropor tubings with a range of NMWC values starting at 1000 are available (e.g., from Pierce and Warriner, Chester, UK), and the appropriate tubing should be selected. These tubings are much more expensive than Visking tubing, and the rate of dialysis decreases with pore size; hence, they should only be used if strictly necessary.

To remove ammonium sulfate from the protein suspension, a piece of dialysis tubing with a volume about twice that of the suspension is taken and securely closed with a double knot at one end. The suspension is then poured into the bag using a funnel, air is removed from the top part of the bag by running it between the fingers, and the top secured with a double knot. It is very important to have this space in the bag to allow for expansion, since water will flow in while the internal salt concentration is high. If insufficient space is left, the bag can become very tight owing to this inflow. Bags rarely burst, since the membranes are quite strong, but they are tricky to open in this state; the best way is to insert one end into a measuring cylinder, and then prick the bag with a scalpel. Equilibrium will be reached in about 3–4 h, but only if the system is stirred; otherwise, about 6 h should be allowed. Care should be taken when stirring to ensure that the magnetic pellet or stirring paddle does not tear the dialysis bag. The volume of dialysis solution to be used depends, of course, on the

sample volume and the final concentration of ammonium sulfate desired. If a ratio of 1:100 (sample:dialysis fluid) is used, then at equilibrium, the ammonium sulfate concentration will have been reduced 100-fold (this is not strictly true, because it ignores the Donnan effect, but will do as an approximation). A second dialysis will then result in a total decrease of 10,000-fold, and so on. The buffer to be used for dialysis is usually dictated by the requirements of the next step in the purification schedule. For example, if this is to be ion-exchange chromatography, then column equilibration buffer is the logical choice.

13. Glass is acceptable, but stainless steel is to be preferred because of the more rapid heat transfer. Plastic will not do.
14. Addition of ethanol or acetone to water leads to a volume reduction of about 5%, but this is usually ignored in calculating percentage concentrations. To calculate the volume (v) in mL of solvent required to change the concentration of 1 L of solution at P_1 to $P_2\%$, use the formula:

$$v = [1000 (P_2 - P_1)/(100 - P_2)] \quad (3)$$

15. Only short centrifugation times are required because of the low density and viscosity of water/solvent mixtures.
16. Do not attempt to resuspend the precipitates in too small a volume (in distinction to the practice with salt fractionation), since the result would be a high solvent concentration with attendant risk of denaturation. Similarly, if the protein is sensitive to solvents, it is necessary to remove residual solvent quickly, so membrane filtration or gel filtration may be preferable to dialysis.

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CHAPTER 14

Ion-Exchange Chromatography

David Sheehan and Richard FitzGerald

1. Introduction

Proteins contain charged groups on their surfaces, which enhance their interactions with solvent water and hence their solubility. At physiological pH, some of these charged groups are cationic (positively charged, e.g., Lysine), whereas others are anionic (negatively charged, e.g., Aspartate). Since proteins differ from each other in their amino acid sequence, the net charge possessed by a protein at physiological pH is determined ultimately by the balance between these charges (i.e., negatively charged proteins possess more negatively charged groups than positively charged groups). This also underlies the different pIs of proteins (*see* Chapter 23). Ion-exchange chromatography (1) separates proteins first on the basis of their charge type (cationic or anionic) and, second, on the basis of relative charge strength (e.g., strongly anionic from weakly anionic).

The basis of ion-exchange chromatography (Fig. 1) is that charged ions can freely exchange with ions of the same type. In this context, the mass of the ion is irrelevant. Therefore, it is possible for a bulky anion like a negatively charged protein to exchange with chloride ions. This process can later be reversed by washing with chloride ions in the form of a NaCl or KCl solution. Such washing removes weakly bound proteins first, followed by more strongly bound proteins with a greater net negative charge.

Like most column chromatography techniques, ion-exchange chromatography requires a stationary phase which is usually composed of insoluble, hydrated polymers, such as cellulose, dextran, and Sephadex (2).

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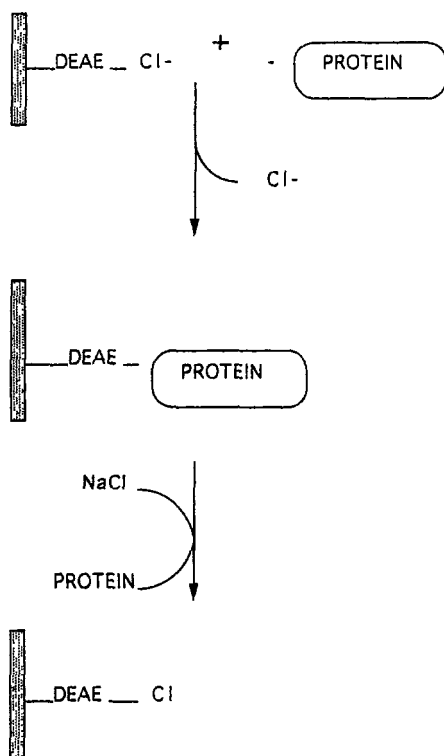


Fig. 1. Ion-exchange chromatography of anionic protein. The protein exchanges with the Cl^- ion and binds to the resin. This process is reversed when an NaCl gradient is applied to the column. Each protein has slightly different charge characteristics and, therefore, elutes at a slightly different Cl^- ion concentration. This, therefore, results in separation of proteins.

The ion-exchange group is immobilized on this stationary phase, and some of the chemical structures of commonly used groups are shown in Table 1.

In this chapter, the use of microgranular diethylaminoethyl (DEAE)-cellulose manufactured by Whatman (Maidstone, UK) is described. Variations on this method suitable for other resins are summarized in Section 4. A novel format for ion exchange is the immobilization of DEAE or carboxymethyl (CM) groups on filters packed in a cartridge (MemSep,TM PerSeptive Biosystems, Farmingham, MA). These cartridges may be used in a low-pressure format or with an FPLC system (*see* Chapter 26), and offer the advantages of high resolution and high flow rates with very low back pressure. This is owing to the use of thin

Table 1
Some Commonly Used Ion Exchange Groups^a

Group	pH range	Structure
Anion exchangers		
Q (quaternary ammonium)	2–12	$\text{—CH}_2\text{—}\overset{\text{CH}_3}{\underset{\text{CH}_3}{\overset{+}{\text{N}}}}\text{—CH}_3$
DEAE (diethylaminoethyl)	2–9	$\text{—O—CH}_2\text{—CH}_2\text{—}\overset{\text{C}_2\text{H}_5}{\underset{\text{C}_2\text{H}_5}{\overset{+}{\text{N}}}}\text{H}$
QAE (quaternary aminoethyl)	2–12	$\text{—O—(CH}_2\text{)}_2\text{—}\overset{\text{C}_2\text{H}_5}{\underset{\text{C}_2\text{H}_5}{\overset{+}{\text{N}}}}\text{—CH}_2\text{—CHOH—CH}_3$
Cation exchangers		
SP (sulphopropyl)	2–12	$\text{—(CH}_2\text{)}_2\text{—CH}_2\text{—SO}_3^-$
S (methyl sulphonate)	2–12	$\text{—CH}_2\text{—SO}_3^-$
CM (carboxymethyl)	6–11	$\text{—O—CH}_2\text{—COO}^-$

^aThese groups may be immobilized on various stationary phases, such as cellulose, dextran, agarose, and glass beads

filters rather than beads as the stationary phase and can result in separations in as little as 3–5 min.

2. Materials

- The buffers used will depend on the characteristics of the protein of interest (*see* Note 1). Buffers are prepared fresh before use, and all reagents are of Analar grade. Deionized/distilled water is used. Typically, the following buffers may be employed for a column that requires operation at pH 8.0
 Buffer A: 200 mM Tris-HCl, pH 8.0.
 Buffer B: 10 mM Tris-HCl, pH 8.0.
 Buffer C: 10 mM Tris-HCl, pH 8.0, 100 mM NaCl.
- Microgranular DEAE-cellulose 52 is obtained from Whatman. Desalting resin, such as Sephadex G-25, is from Pharmacia LKB (Uppsala, Sweden). Both resins are stored in water containing 0.1% Na azide. Sephadex resins may be reused after extensive washing with large volumes of water, whereas DEAE-cellulose requires regeneration before reuse (*see* Note 2).
- A 500-mL sintered glass funnel (no. 1 sinter) is useful for washing resin during equilibration and regeneration. This is usually connected to water suction.

- 4 For desalting and ion-exchange chromatography (column volumes of 300 mL), sintered glass columns (45×3 cm in diameter) are required. Smaller columns could be used for smaller column volumes.
- 5 A gradient maker (2×500 mL) is used to generate salt gradients
- 6 A calibrated conductivity meter is required.

3. Methods

3.1. Sample Preparation

Protein samples require desalting before being applied to the ion-exchange column. This may be achieved either by dialysis or by gel filtration on a Sephadex G-25 column (3). Pharmacia produces a PD10 column for rapid desalting of small volumes.

3.2. Resin Equilibration

1. Before use, wash 100–200 g resin in 1–2 L water followed by removal of fines. This is achieved by stirring the resin with a glass rod, allowing the bulk of the resin to settle, and then aspirating supernatant, which contains the fines. Repeat this process three times. It is sometimes convenient to do this procedure in a graduated cylinder.
- 2 Wash the defined resin in 500 mL buffer A (*see* Note 3) to achieve equilibration. Again, following settlement of the resin, aspirate off the supernatant. Repeat this process twice (i.e., a total of three washes). Gently stir the resin into a slurry, and transfer it into a sintered glass column that contains a small volume of water (this prevents air bubbles from being trapped in the sinter) Pass buffer B through the column until it is completely equilibrated (*see* Note 4)

3.3. Chromatography

1. Apply the desalted sample to the equilibrated resin, and collect the effluent. This should be assayed for the protein of interest in case it has not bound (*see* Note 5).
2. Then elute bound protein by applying a gradient (2×500 mL) of 0–100 mM NaCl in 10 mM Tris buffer, pH 8.0 (*see* Note 1).
3. Collect fractions in a fraction collector, and assay for the protein of interest, protein concentration (4), and conductivity. The conductivity meter should be carefully calibrated (Note: this property is temperature-dependent), so that conductivity measurements may be expressed as NaCl concentrations. In this way, the reproducibility of each chromatography experiment may be assessed. These measurements can also be made on-line, using on-line UV and conductivity detectors connected to a PC or chart recorder.
- 4 Pool appropriate fractions, and concentrate for further study or purification (*see* Note 6).

3.4. Resin Regeneration

The resin may usually be regenerated by washing with 10 mM Tris-HCl, pH 8.0, buffer containing 2M NaCl after use. Three to four uses of a batch of resin are possible in most cases before extensive regeneration is necessary (see Note 2). However, owing to column compaction as a consequence of high-salt concentration gradients, in some cases, only single usage of a packed resin is possible before extensive regeneration is required.

1. Place the resin in a beaker containing 15 vol of 0.5M HCl for 30 min, and then decant the supernatant.
2. Wash the resin extensively on a glass sinter until the pH of the wash-through reaches an intermediate value of 4.0, then place the resin in a beaker containing 15 vol of 0.5M NaOH for 30 min, and decant the supernatant. Again, the resin is extensively washed with water until a pH of 8.0–7.0 is achieved. The resin may now be stored in 0.1% Na azide until required.

4. Notes

1. For proteins not previously purified by DEAE-cellulose chromatography, buffer selection may be helped if the pI of the protein of interest is known. In general, proteins will bind to anion exchangers at pH values above their pIs, while binding to cation exchangers, such as CM-cellulose, at pH values below their pIs.

Phosphate, Tris, and other common buffers are used in ion-exchange chromatography in concentrations of 10–50 mM. Urea (desalted on a mixed-bed column of Dowex or else molecular biology grade) may be included in buffers at concentrations up to 8M if separation of denatured samples is required (e.g., chromatography of CNBr-generated peptides). Care should be taken that the urea does not precipitate in the column during chromatography (especially at low temperatures).

2. Fibrous DEAE-cellulose is exposed to 0.5M HCl for 60 min rather than 30 min. Regeneration of CM-cellulose is achieved by a similar process, except that the resin is washed first with 0.5M NaOH for 30 min, followed later by 0.5M HCl. The intermediate pH is 8, whereas the final pH is 5.5–6.5.
3. Concentrated buffer is used to equilibrate and charge the exchange groups, because much lower volumes are required than if more dilute buffer is used. This results in more rapid equilibration.
4. Equilibration is assessed by measuring the pH and conductivity of buffer B and the column eluate. When these values are the same, the column is equilibrated.
5. For proteins not previously purified by this method, it is often necessary to develop a method. The main variables one can use to do this include

choice of ion-exchange group, pH (see Note 1), salt concentration, and gradient steepness. In general, for example, around 70% of rat liver cytosolic proteins will bind to DEAE-cellulose at pH 7.5. If the protein of interest has an alkaline pI, then passage through such a column at this pH might be a useful first step in the purification. Conversely, if the protein of interest binds to DEAE-cellulose at pH 7.5, then it is worth repeating the experiment at progressively higher pH values, e.g., 8.5 and 9.5. Only highly acidic proteins will bind at pH 10.0. It will also be advantageous to assess a range of salt gradients. Generally, shallower gradients (e.g., 0–100, 300–400 mM) give better resolution. However, steep gradients (e.g., 0–1M) can be used in initial experiments to identify elution positions of proteins of interest. It is also possible to use stepwise washes (e.g., 100 mM NaCl followed by 200 mM NaCl) rather than continuous gradients. The best approach to developing a new method, therefore, is to assess the binding (i.e., binding vs nonbinding) of the protein of interest on a small scale on both CM-cellulose and DEAE-cellulose at a range of pH values in the alkaline and acidic ranges, respectively. Then develop chromatograms where the protein of interest has bound with a steep gradient. Finally, shallower gradients are assessed at the most promising pH values. It is also possible to use nonlinear gradients for more difficult separations.

- 6 A major factor in the success of chromatography is the pooling of fractions. In general, fractions are never pooled in a new purification until SDS-PAGE analysis (5,6) has been carried out on aliquots taken from active fractions. This gives a good indication of fraction purity. It is best to accept some losses of the protein of interest if this means removing significant contaminants. Carrying out the chromatography at different pH values (see Note 5) usually gives quite different chromatograms and this is often useful for displacing contaminant peaks.

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CHAPTER 15

Hydrophobic Interaction Chromatography

Paul O'Farrell

1. Introduction

During the development of affinity chromatography, spacer arms consisting of alkyl chains of 2–10 carbons were used to hold the affinity ligand at some distance from the solid matrix so as to minimize ligand/support interactions. Control experiments were carried out to determine the effect of the spacer itself where the spacer alone was attached to the matrix. It was found that columns prepared from such material in fact retained certain proteins. These observations lead to the first systematic studies of hydrophobic interaction chromatography (HIC).

Although the nature of the hydrophobic interaction is still not fully understood, it is clear that it has its origins largely in the properties of the solvent. When a nonpolar solute molecule is dissolved in a polar solvent, such as water, the solvent is forced to form a structural cavity around it. This causes a decrease in entropy and is therefore thermodynamically unfavorable. When two such solute molecules come into contact they may aggregate as a result of the consequent favorable increase in entropy, since less solvent molecules are required for structure formation around the dimer. Thus hydrophobic interaction is not an attractive force as such, rather it is forced on a nonpolar solute by the structure of the polar solvent. It follows that the addition of modifiers to the solvent which change its ability to form structure will have an effect on hydrophobic interactions. Chaotropic salts decrease hydrophobic interactions (because there is less overall structure in the solvent in the presence of such agents, the

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energetic advantage gained by the aggregation of nonpolar solute molecules is reduced), while so-called "structure-forming" salts increase the strength of the interaction (*see* Note 1).

It has been estimated that as much as 50% of the solvent accessible surface of a soluble protein can consist of nonpolar atoms (1), but this percentage varies over a wide range from one protein to another. This characteristic represents a general difference between proteins which can be exploited for their separation and which provides a third dimension in addition to size exclusion and ion exchange as chromatographic methods.

Although the exact nature of hydrophobic interaction is not fully understood, nevertheless its exploitation in column chromatography is a straightforward matter. In general, protein is loaded onto the matrix in high-salt buffer, and eluted by decreasing the salt concentration, either by a gradient or in a stepwise fashion. The high salt concentration of the loading buffer means that it is not necessary to remove salt present in the protein sample by dialysis or gel filtration. Consequently, HIC is an excellent choice of procedure following initial purification by ion-exchange chromatography or if the protein of interest is in the supernatant fraction after partial purification by precipitation with ammonium sulfate; it is necessary only to add enough salt to ensure binding to the column. In addition, the presence of salts in the sample solution can exert a stabilizing effect and prevent loss of activity.

If a decreasing salt gradient is used for elution, the gradient in the column is stabilized by density effects. However a number of problems also exist. Proteins may tend to aggregate during elution; conditions are, after all, similar to those used for fractionation by salting out. Because of this and other factors, such as slow association/dissociation, peaks can be broad and separation is not always ideal. Such effects are generally reduced if a step elution method is used.

The method presented here describes the final stage in the purification of aspartate aminotransferase (AAT) from *Thermus aquaticus*. The extract applied to the column had already been partially purified by a combination of ammonium sulfate precipitation, protamine sulfate precipitation, and anion- and cation-exchange chromatography (2). The protein concentration was approx 150 $\mu\text{g}/\text{mL}$ and the AAT specific activity was 25 U/mg.

2. Materials

1. Buffer 1: 10 mM Tris-HCl, pH 8.0, 0.6M NaCl.
2. Buffer 2: 10 mM Tris-HCl, pH 6.5 (*see* Note 2)

3. Phenyl Sepharose CL-4B (Pharmacia, Uppsala, Sweden) (*see* Note 3). The ligand density is 40 $\mu\text{mol/mL}$ of gel.
4. Chromatography column: 10 cm long $\times$ 1.5 cm in diameter, total bed vol 18 mL.

3. Method

Previous steps in the purification schedule had been carried out at 4°C, the HIC step was carried out at room temperature (*see* Note 4).

1. Wash the gel, which is supplied preswollen in 20% ethanol, with at least 10 vol of water to ensure that all the ethanol is removed; this is important since ethanol strongly interferes with hydrophobic interactions.
2. Allow the gel to settle and decant the supernatant.
3. Prepare a slurry of gel in buffer 1 with a ratio of 75% settled gel to 25% buffer.
4. Pour the slurry into the column and, after fitting the column top piece, equilibrate by pumping buffer 1 through the column at 80 mL/h for 1.5–2 h.
5. Pool active fractions from the previous purification step (chromatography on DEAE-cellulose, DE-52 in the present case) and adjust to approx 0.6M NaCl.
6. Dialyze this material overnight against buffer 1 (*see* Note 5).
7. Apply the resultant solution to the column at 60 mL/h and collect 3 mL fractions.
8. Remove unbound proteins by washing the column with buffer 1 at 60 mL/h.
9. When the column effluent is free of protein, determined by measuring the absorbance at 280 nm, elute the AAT by changing the running buffer to buffer 2 (*see* Note 6). Assay fractions for AAT activity and pool active fractions.
10. Clean the column by passing 3–4 bed volumes of distilled water through it and then re-equilibrate with loading buffer (*see* Note 7).

4. Notes

1. The Hofmeister series lists ions in the order from those which favor hydrophobic interactions to those which decrease the strength of the interaction:
 - a. Anions: PO_4^{3-} , SO_4^{2-} , CH_3COO^- , Cl^- , Br^- , NO_3^- , ClO_4^- , I^- , SCN^- .
 - b. Cations: NH_4^+ , Rb^+ , K^+ , Na^+ , Cs^+ , Li^+ , Mg^{2+} , Ca^{2+} , Ba^{2+} .
2. Both buffers contained 4 μM pyridoxal-5'-phosphate and 2 mM 2-oxoglutarate but these were added specifically to stabilize the AAT and, of course, would not be used in other applications. Other additives may be needed in particular cases and it is necessary to confirm that they do not interfere with chromatographic behavior.
3. A wide variety of hydrophobic resins are currently commercially available. In addition there are a number of products which combine hydrophobic characteristics with others such as size exclusion or affinity. The most commonly used purely hydrophobic matrices are substituted with alkyl

(generally octyl) or phenyl ligands. The choice of which of these matrices to use is empirical and the preferred method for a particular separation should be determined by small-scale screening experiments. Reference 3 gives excellent advice on developing and interpreting such screening experiments. In general, a weakly hydrophobic resin should be used with an uncharacterized protein, since a strongly hydrophobic protein may not be eluted easily from a strongly binding resin (e.g., octyl-substituted resins)

4. As HIC is entropy based, it can be expected that temperature will have an effect on strength of binding. In general, hydrophobic interactions increase in strength with increasing temperature; there can be a 20–30% reduction in binding when the temperature is changed from 20 to 4°C (4) However, temperature also affects protein conformation and solubility and consequently can affect a protein's interaction with a hydrophobic matrix. It is difficult to predict the effect of temperature on a particular separation, but it should be remembered that changes in temperature can cause changes in elution patterns and that in some cases a protein which binds to a particular hydrophobic matrix at room temperature may not bind in the cold room.
5. Generally, the strength of the interaction between proteins and hydrophobic ligands decreases with increasing pH. This is presumably because of an increase in the hydrophilicity of proteins due to the titration of charged groups. The effect of pH is different for different proteins and thus it may be expected that elution profiles may be improved by carrying out elution at different pH values. However, the effect of changing the pH is not great at moderate pH values, and for most applications it is probably better simply to work at a pH where the protein of interest is known to be stable.
6. Elution can be achieved in a number of ways and can be either via a gradient or a stepwise method. Although gradient elution should in principle give better separation of components, as noted in the Introduction, peaks can be broad and the elution volume may therefore be larger than desired. This may be of little consequence if HIC is to be followed by another adsorption procedure since the next step will concentrate the sample. However, if HIC is to be followed by, for example, size exclusion chromatography, or if the protein of interest is unstable in dilute solution, a stepwise method which will result in a smaller elution volume may be indicated.

A variety of salts may be used for loading and elution (5) In general the effects of various salts on HIC mimic their effects on salting out. Those salts which are most effective in salting out ("structure-forming" salts) are most effective at binding proteins to hydrophobic matrices, whereas salts used for salting in decrease the interaction. Thus, elution may be achieved by decreasing the concentration of structure forming salt, or increasing the

concentration of a chaotropic agent. In fact, a combination of the two methods has been shown to result in better separations and sharper peaks (6)

If a protein is very strongly bound, then alcohols (e.g., ethylene glycol, gradient of 0–80%) or detergents (e.g., Triton X-100, 1% [w/v]) may be used to effect elution. However, such agents can have deleterious effects on protein structure and in addition, detergents can prove very difficult to remove from hydrophobic matrices. If it is found to be necessary to resort to such extremes, it is probably best to consider using a more weakly hydrophobic ligand

7. For more stringent cleaning, 4 bed volumes of 0.5–1.0M NaOH, followed by 2–3 bed vol of distilled water may be used. The column can be stored filled with 0.01M NaOH, after storage, 2–3 bed vol of water are passed through the column which is then ready to be equilibrated with the loading buffer.

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CHAPTER 16

Affinity Chromatography

Paul Cutler

1. Introduction

1.1. General Principles

Affinity chromatography is a method of selectively and reversibly binding proteins to a solid support matrix based on the exploitation of known biological affinities between molecules. The interaction between the target protein and the matrix is not based on general properties, such as the isoelectric point (pI) or hydrophobicity, which are more commonly used in adsorption chromatography, but on individual structural properties, such as the interaction of antibodies with antigens, enzymes with substrate analogs, nucleic acids with binding proteins, and hormones with receptors. In order to exploit the interaction for the purposes of purification, one of the components, the ligand, must be immobilized onto a solid matrix in a manner that renders it stable and active. Once produced, the matrix can be used to purify its specific target from a suitable biological extract (1).

Affinity chromatography is a particularly powerful technique, since it offers the potential of purifying target proteins that exist in very low titer from complex mixtures with a very high degree of purification in a single step. Since the aim is to purify material on the basis of biological function, it is even possible to separate active and inactive forms of the same material selectively.

Affinity matrices can be formed from ligands that are either mono-specific or group-specific. Mono-specific ligands recognize a single form of a protein, such as a receptor with affinity for a specific hormone or an

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Table 1
Some Commonly Employed
Group-Specific Affinity Ligands

Ligand	Target protein
5'AMP, ATP	Dehydrogenases
NAD, NADP	Dehydrogenases
Protein A	Antibodies
Protein G	Antibodies
Lectins	Polysaccharides, glycoproteins
Histones	DNA
Heparin	Lipoproteins, DNA, RNA
Gelatin	Fibronectin
Lysine	rRNA, dsDNA, plasminogen
Arginine	Fibronectin
Benzamidine	Serine proteases
Polymyxin	Endotoxins
Calmodulin	Kinases
Cibacron Blue	Kinases, phosphatases, dehydrogenases, albumin

enzyme recognizing an inhibitor. These matrices are often "tailor-made" for particular separations. Group-specific ligands include enzyme cofactors, plant lectins, and protein A from *Staphylococcus aureus* (Table 1). Group-specific matrices owing to their generic nature, are frequently available commercially from a range of suppliers, often in prepacked columns.

Originally, in order to purify a protein by affinity chromatography, it was necessary to find a naturally occurring ligand. In recent years, the definition of affinity chromatography has expanded to include the separation of proteins by specific interactions other than purely biological interactions. Included in this category are dye affinity ligands, such as Cibacron Blue (2), and immobilized metal ion-affinity matrices (3). With the advent of monoclonal antibodies (MABs) and recombinant protein technology, it is possible to manipulate a biological affinity. By using hybridoma technology, it is possible to produce economically viable amounts of MABs where the target protein is the antigen. Immobilizing the antibody on a solid support creates an immunopurification matrix (4). It is also possible with recombinant DNA technology to produce a target protein in cell culture which has a specific affinity region, the tag, engineered into the product (5,6). The tag is used to purify the protein on

a relatively inexpensive and well-defined affinity matrix. Enzymic cleavage points may also be inserted between the protein and the tag sequences so that once purified from the culture, the tag can be removed. An example of this is a fusion containing the target protein with a hexahistidine sequence, which shows preferential binding to immobilized metal affinity matrices (7).

Two schools of thought exist regarding where an affinity chromatography step should lie in a purification scheme. One suggests that owing to the expense of producing affinity matrices, the material should be partially purified prior to the affinity step to protect the matrix from proteases, and so forth. Conversely, the affinity matrix can come first to purify rapidly what may be an unstable protein to near homogeneity in a single high-yielding step, while dramatically reducing the volume of material to be processed. With the increasing availability of affinity matrices manufactured using recombinant technologies to produce less costly ligands, the latter school of thought appears to be gaining popularity. However, the actual position of the affinity chromatography step within a purification scheme must be decided on a case-by-case basis.

1.2. Affinity Matrices

1. **Ligand:** The immobilized affinity ligand must be able to form a reversible complex with the target protein. The binding constant (the inverse of the dissociation constant) should be high enough to enable stable complex formation ideally at physiological conditions. However, the affinity must be sufficiently weak to facilitate the elution of the protein under relatively mild conditions, thereby avoiding denaturation of either the ligand or target protein. Binding constants of 10^5 – $10^{11} M$ are usually considered a good working range (*see* Note 1).

A detailed description of the immobilization of the ligand to generate an affinity support is beyond the scope of this chapter. However, reviews exist describing immobilization techniques (*see* ref. 8 and Note 2). In addition to immobilization in a form that retains biological function of the ligand, good chemical stability must be maintained in order to withstand the harsh elution and cleaning regimes used in affinity chromatography processes. Leaching of the ligand from the matrix must be minimized to avoid contamination of the purified protein.

2. **Matrix:** The matrix of choice should show the physical and chemical properties required for adsorption chromatography techniques (9,10). The matrix typically constitutes macroporous beads (50–400 μm) to produce a large surface area, which maximizes the capacity of the activated matrix

and prevents any size exclusion effects. It must be hydrophilic, but uncharged to prevent nonspecific ionic binding. It must demonstrate good mechanical rigidity and chemical stability under the conditions used for activation, derivatization, elution, and regeneration. Although the most commonly used matrix support is agarose, a range of other matrix supports, including silica and polyacrylamide, are available. Selection of the matrix should be made based on the physicochemical properties required.

3. **Spacers:** One of the common problems encountered when derivatizing a matrix for affinity chromatography is the steric hindrance observed when the ligand is too close to the solid phase. To minimize this affinity matrices are often activated with a spacer arm, commonly a six-carbon hydrophilic chain to distance the ligand from the solid phase. The length of the spacer arm should be optimized for each individual matrix. Although longer spacer arms may allow greater ligand availability, they can themselves lead to steric hindrance and confer unwanted hydrophobicity onto the support.
4. **Activation of the matrix:** The protein ligand is almost always covalently attached to the solid support. The matrix is activated with a reagent, such as cyanogen bromide, which reacts with the ϵ -amino groups of lysine residues in protein ligands, resulting in the protein being covalently attached. Other chemistries enable immobilization of proteins via carboxyl, hydroxyl, and thiol functionalities (*see ref. 8 and Note 2*)

1.3. Practical Aspects

Purification by affinity chromatography is a relatively straightforward technique because of the selective binding of the target protein (Fig. 1). The crude extract is passed through the column, the target material binds, and the matrix is then washed to remove the nonbound fraction. This may involve more than one wash step if nonspecific interactions or selective binding of impurities to the bound target protein are suspected. The conditions of the mobile phase are then modified usually by changes in pH or ionic strength, or by inclusion of either a general chaotropic agent or a competing soluble ligand to facilitate elution of the protein. The eluted protein is collected and the column regenerated under suitable conditions.

1. **Sample preparation:** Efficient binding is facilitated by ensuring that the sample and the column are equilibrated in a buffer that is optimal for binding; this requires a defined pH, ionic strength, and so forth. Since most biological affinities in mammalian systems occur at physiological pH, buffers such as phosphate-buffered saline (PBS) are common. The buffer should contain any elements, such as cofactors, required to maintain activity of the target protein. The sample can be conditioned for binding to the matrix by such methods as dialysis, desalting, ultrafiltration or simply by

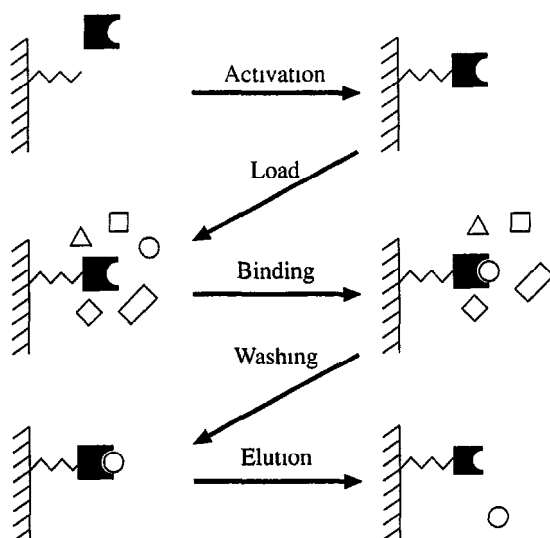


Fig. 1. Principle of affinity chromatography. The ligand is immobilized onto a solid support matrix. The crude extract is passed through the column. The target molecule for which the ligand possesses affinity is retained, whereas all other material is eluted. The bound target protein is eluted by alteration of the mobile-phase conditions.

pH adjustment. If crude material is to be loaded, then some form of pretreatment may be required, such as filtration or centrifugation. It may be necessary to include protease inhibitors, such as leupeptin or phenylmethylsulfonyl fluoride (PMSF), to prevent proteolytic degradation of the target (*see* Note 3 and Chapter 9).

2. Column: The column size and the degree of ligand substitution determine the capacity of the column for the target protein. The affinity constant of the ligand for the target protein under the operating conditions will also dictate column performance. This can be determined empirically by overloading the column or in a more sophisticated manner by producing a breakthrough curve to calculate the adsorption isotherm (Fig. 2). Capacities of 1–20 mg protein bound/mL matrix are common. In addition to standard chromatographic columns, other affinity separation methodologies exist, including activated filters (11) and stirred tanks or expanded beds (*see* ref. 12 and Note 4).

Under normal conditions short, wide columns are used to facilitate rapid separations. However, where the affinity is low and the material is only retarded, separation is enhanced by use of a thinner, longer column (13).

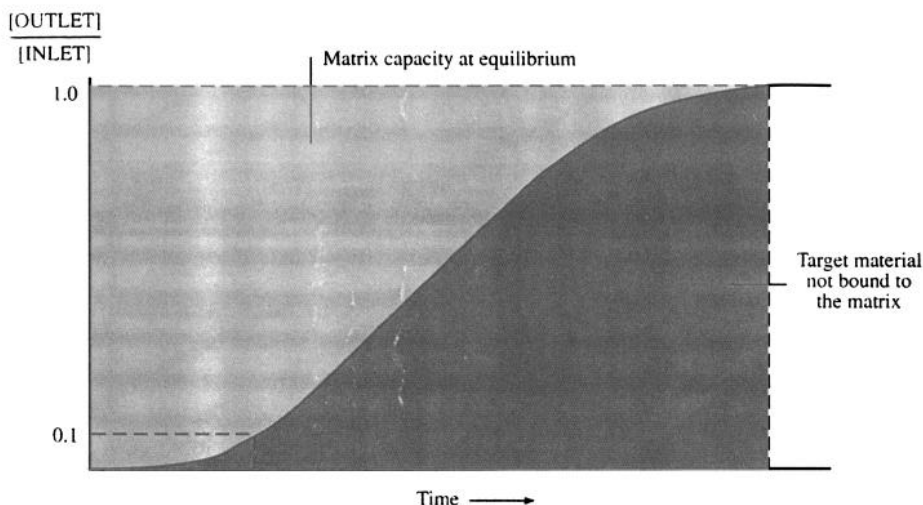


Fig. 2. Breakthrough curves can be used to establish the effective capacity of an affinity matrix. By specifically detecting or assaying the target protein appearing in the flowthrough from the column, the breakthrough of material can be plotted. Eventually, as the matrix reaches equilibrium (theoretical maximum capacity), the inlet concentration and outlet concentration are equivalent. The effective, or dynamic capacity of the matrix is reached sooner and is often taken when the ratio of outlet to inlet is 0.1.

Columns should be packed in accordance with the standard procedures described by the manufacturers. Briefly, the matrix should be packed in a glass or acrylic column with flow adaptors to ensure a minimum of dead space. For small-scale separations, cheaper disposable columns can be used under gravimetric flow.

The flow rate used for adsorption depends on the physical properties of the matrix support and the binding constant of the ligand for the target protein. Maximum adsorption usually occurs at a low flow rate, although flow rates tend to be a compromise between optimal binding and process time, which affects cost and target protein stability. In practice, linear flow rates of 30–50 cm/h are used for low-pressure preparative separations.

3. Loading: Loading volume is generally not critical as long as the total amount of target protein does not exceed the capacity of the column. The passage of proteins through the column is monitored most commonly using a UV detector. Following loading, the column is washed with equilibration buffer to remove any unbound material. Once baseline has been achieved on the detector, a more stringent wash can be made if necessary

to remove any weakly binding material that may have bound non-specifically. This wash is designed so as not to cause elution of the target protein. Often the wash buffers can contain low levels of detergents or moderate concentrations of salt.

4. Elution: The protein of interest is eluted by weakening the ligand protein interaction. This can be done either nonspecifically, for example, by using changes in pH, ionic strength, and so forth, or by the addition of a specific solute in the eluant to remove the target protein selectively (e.g., a competing ligand). In extreme circumstances chaotropic agents may be used, although these significantly increase the likelihood of denaturation. Owing to the selectivity of binding, it is common for affinity elutions to be performed in a stepwise mode (*see* Note 1). However, in certain circumstances, particularly where group-specific ligands are employed, a gradient elution may be used. This has been used successfully to separate antibodies of differing subclasses on protein A affinity matrices.

The conditions for elution are one of the primary reasons why affinity chromatography fails to yield active protein (*see* Note 5). The need to use harsh conditions to maximize recovery must be balanced against potential denaturation and inactivation of the target protein. A knowledge of the physiochemical properties of the target protein is invaluable when designing an elution buffer. In general, the pH of the buffer should not be near the pI of the protein. It is advisable to avoid buffering systems with chelation effects (e.g., citrate) for purification of metalloproteins.

The purified protein should be placed in an environment that promotes its stability as soon as possible after elution. Frequently in the case of pH elution, this involves titration to near neutrality. The material may be placed in a suitable buffer by using buffer-exchange techniques, such as desalting, ultrafiltration, or dialysis. Treatment of the eluted protein will be largely influenced by subsequent purification steps and the use for which the protein is designed (*see* Note 6).

5. Purity analysis: The performance of the affinity chromatography step is usually determined by comparison of the purity before and after purification by polyacrylamide gel electrophoresis (PAGE). Other techniques for assessing purity include gel filtration and reverse-phase high-performance liquid chromatography (HPLC). Where possible, specific assays should be used to detect recovery (% yield), such as Western blotting techniques or, in the case of enzymes, calculating the activity per mass of total protein using techniques, such as enzyme-linked immunosorbent assay (ELISA).

It is important to analyze the flowthrough to ensure maximum recovery. Several methods exist for analyzing the flowthrough material, such as rechromatography, to establish that the capacity of the column has not been

exceeded. This, however, will not reflect the inability to bind owing to denaturation of the ligand or inappropriate binding conditions.

6. Matrix regeneration and storage: It is often the case that the harshest conditions a solid-phase chromatography matrix is exposed to are those used for regeneration and sanitization. Where a labile protein is used as a ligand, the affinity matrix cannot withstand treatment with stringent agents, such as sodium hydroxide. Commonly used regeneration regimes involve the use of chaotropic agents, such as 6*M* guanidine hydrochloride or 3*M* sodium thiocyanate.

Affinity matrices should be stored under conditions that prevent bacterial and fungal growth. Matrices are typically stored in the presence of antimicrobial agents, such as 20% (v/v) ethanol, 0.02% (w/v) Thimerasol, or 0.01% (w/v) sodium azide at standard cold-room/refrigerator temperatures (< 8°C).

2. Materials

1. Protein A Sepharose 4B (Pharmacia, Uppsala, Sweden).
2. Buffer A: 100 mM Tris-HCl, pH 7.5, 3*M* NaCl.
3. Buffer B: 50 mM Tris-HCl, pH 7.5, 1.5*M* NaCl.
4. Buffer C: 0.1*M* citrate, pH 3.0.
5. Buffer D: 0.1*M* citrate, pH 2.0.

3. Methods

Immunoglobulin Gs (IgGs) bind to protein A from *S. aureus* via the immunoglobulins Fc region. The affinity of the antibody for protein A is both species- and subclass-dependent. A range of protein A affinity matrices can be bought commercially. The crude antibody extract can be prepared by dilution or by dialysis into the binding buffer. The matrix is equilibrated with the binding buffer before the extract is loaded. The unbound material is washed away with the buffer prior to elution with a low-pH buffer. An example purification strategy is given (see ref. 14 and Fig. 3):

1. Dilute 5 mL of goat polyclonal antiserum raised against human albumin with 5 mL of buffer A, and retitrate to pH 7.5.
2. Pack a 5 mL (1.0 × 6.4 cm) column with protein A Sepharose 4B, and equilibrate in buffer B. Apply the diluted serum to the column at a flow rate of 50 cm/h. Wash the column with 5 column volumes (25 mL) of buffer B.
3. Program the fraction collector to collect 1 mL fractions. Exchange the inlet and outlet tubing of the column to reverse the flowthrough the column.
4. Elute the antibody with 5 column volumes of buffer C. Wash the column with 5 column volumes of buffer D, followed by 2 column volumes of 6*M* guanidine hydrochloride.

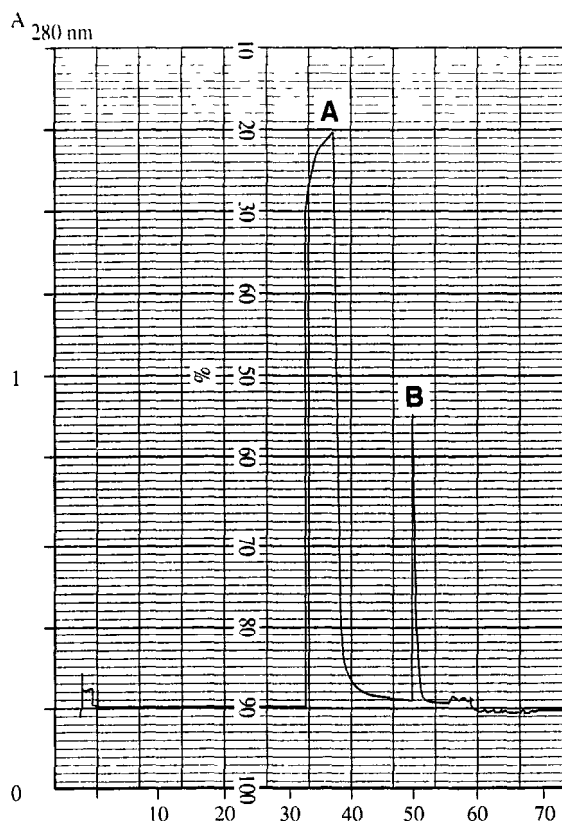


Fig. 3. Purification of goat immunoglobulin from serum by protein A affinity chromatography. The majority of the protein elutes in the flowthrough fraction (Peak A). The bound immunoglobulin is eluted by lowering the pH of the mobile phase (Peak B).

5. Pool fractions containing antibody, and adjust the pH to pH 7.5 with 1M untitrated Tris base. Analyze the antibody for purity by SDS-PAGE (Fig. 4).

4. Notes

1. It is a commonly held belief that the higher the capacity of a column, the better. Although a high capacity is clearly advantageous, it should be remembered that on adsorption, the protein of interest is being removed from a dilute solution and concentrating on the solid phase. On elution, the protein is often released in a relatively high concentration form in a dissociating buffer. This can lead to insolubility of the protein and dramatic losses. This concentration can be partly modulated by the elution flow rate

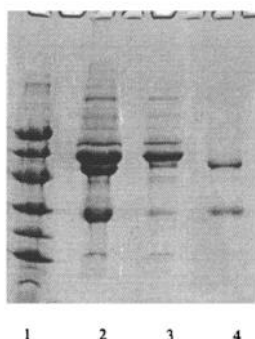


Fig. 4. Analysis of the purification of immunoglobulins from goat serum by sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE). The crude serum (track 2) was purified by protein A affinity chromatography. The unbound material (track 3) was eluted in the column flowthrough. The purified antibody (track 4) was eluted from the column by lowering the pH of the mobile phase. The SDS-PAGE is run under reducing conditions resulting in the antibody appearing as its constituent "heavy" chains (~50 kDa) and "light" chains (~25 kDa). The molecular weights are compared to standard marker proteins (track 1).

In general, low flow rates promote elution in a concentrated form, whereas higher flow rates produce more dilute solutions, which in certain cases may be advantageous. Where the binding constant is too high to facilitate elution of the protein in an active form, it may be necessary to attenuate the ligand affinity via chemical modification. This is, however, a relatively complicated process requiring extensive knowledge of the interaction between ligand and target molecule.

2. Knowledge of the nature of the interaction between ligand and target protein is invaluable when selecting the chemistry by which a ligand is to be immobilized. A common immobilization strategy is to immobilize proteins via lysine residues using cyanogen bromide-activated supports. If it is known that a lysine residue on the ligand is involved in the interaction that is being exploited, then it is clear that the matrix may show poor performance. A recent advance is "site-directed immobilization" (15). Many ligands are glycoproteins with defined regions of oligosaccharide commonly linked to asparagine residues. Use of hydrazine chemistry, which reacts specifically with the *cis* diols of the sugar moieties allows the ligand to be selectively immobilized. This has been used successfully in immunopurification matrices where antibodies are immobilized via their oligosaccharide-containing Fc regions, leaving their antigen binding regions

free to interact with their target protein. Ligands have also been engineered to include defined biotinylated residues, which can then be immobilized on streptavidin-activated matrices in an oriented manner.

3. An important factor in protein purification is the stability of the target protein. The target protein may be susceptible to inactivation or degradation owing to the physicochemical conditions employed during purification, proteolytic degradation by enzymes that may be inadvertently activated during purification, and the harsh conditions that may be used for elution of the bound material. All of these must be considered when planning a purification protocol.
4. Affinity chromatography has evolved to fit into the format of general protein purification and as such has been predominantly performed in columns. Since methodologies are required for preliminary purification of crude extracts, the use of the matrix in "batch mode" has begun to be developed. The advantage of batch mode is that the matrix can be applied to very crude cell cultures or tissue extracts with only minimal need for any form of clarification (removal of cell debris, etc.). At lab scale, matrices, such as silica, are popular owing to their density allowing rapid recovery of the matrix from the extract by centrifugation. At analytical scale, the methodology can be applied to separations in microcentrifuge tubes. Once the target protein has bound to the ligand, the matrix can be washed, and either eluted in batch mode or packed into a column format. On a larger scale, the method has been performed with purpose-designed equipment, such as "fluidized beds," e.g., Streamline® (Pharmacia).
5. If a highly concentrated eluate is required, then reverse elution can often be advantageous. When a column is loaded to under capacity, the protein occupies ligands at the top of the column preferentially. By eluting in the reverse mode, the material elutes in a tighter band. This is also advantageous if there is a heterogeneous pool of proteins with differing affinities, such as an extract of serum containing polyclonal antibodies with differing affinities for the immunogen that is acting as a ligand. The antibodies with the highest avidity for the epitope rest at the top of the column, so reverse elution aids recovery.
6. Purification is always a compromise between purity and recovery. The more steps that are to be used in the purification, the purer the end product, but the lower the recovery will be. Losses are incurred during the chromatography and in the manipulation of the protein between chromatography steps. The eluted protein fractions commonly undergo buffer exchange and concentration prior to subsequent steps. By selecting elution conditions that are consistent with subsequent chromatography steps, such as high salt for hydrophobic chromatography or defined pH changes for ion-

exchange chromatography, the target may be eluted in a form requiring minimal conditioning prior to further purification and, thereby maximizing the recovery.

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CHAPTER 17

Dye-Ligand Affinity Chromatography

D. Margaret Worrall

1. Introduction

Dye-ligand affinity is based on the ability of reactive dyes to bind proteins in a selective and reversible manner (1,2). The dyes are generally either monochloro-triazine or dichloro-triazine compounds (two example structures are shown in Fig. 1) and were originally developed in the textile industry. The reactive chloro group allows easy immobilization of the triazine dye to a support matrix, such as Sepharose or agarose, by nucleophilic displacement of the chlorine atom.

The initial discovery of the ability of these dyes to bind proteins came from the observation that Blue Dextran (a conjugate of Cibacron Blue FG-3A), used as a void volume marker on gel-filtration columns, could retard the elution of certain proteins (3). A number of studies have been carried out on the specificity of the dyes for particular proteins, mostly using the prototype Cibacron Blue dye. The dyes appear to be most effective at binding proteins and enzymes that utilize nucleotide cofactors, such as kinases and dehydrogenases. It has been proposed that the aromatic triazine dye structure resembles the nucleotide structure of NAD and that the dye interacts with the dinucleotide fold in these proteins (4). In many cases, bound proteins can be eluted from the columns by a substrate or nucleotide cofactor in a competitive fashion (5), and dyes have been shown to compete for substrate binding sites in free solution (6). However, other proteins with no nucleotide binding sites, such as human serum albumin, can also bind dye ligands tightly. It seems likely that these dyes can bind proteins by both electrostatic interactions and/or by more specific "pseudoaffinity" interactions with ligand bind-

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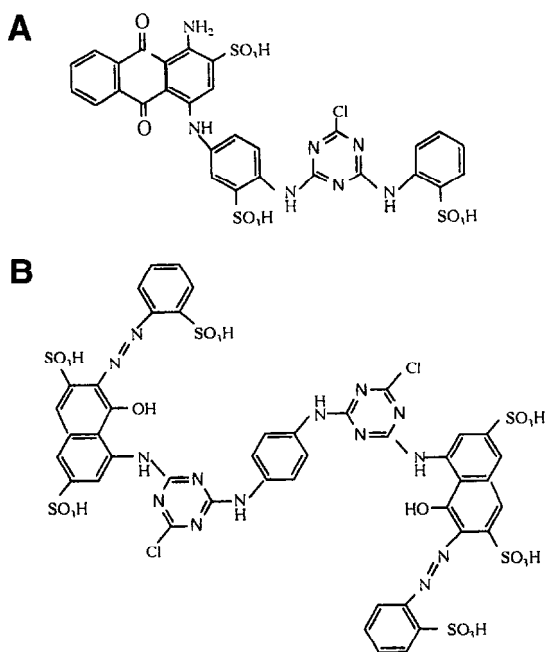


Fig. 1. **(A)** Structure of Cibacron Blue F3G-A (Procion Blue H-B, Reactive Blue 2) and **(B)** structure of Procion Red HE-3B (Reactive Red 120).

ing sites. In practical terms, the degree of purification achieved with dye-ligand chromatography is generally better than that obtained with less specific techniques, such as ion-exchange or gel-filtration chromatography. A further advantage is that the reactive dyes are relatively inert and unaffected by enzymes in crude cellular extracts.

Individual dyes show differences in binding profiles, and it is generally useful to carry out a screening procedure to determine the best dye ligand for a given purification protocol. It has been reported that NAD-utilizing enzymes bind Cibacron Blue, whereas NADP-utilizing enzymes preferentially bind to Procion Red HE-3B (7). In general, however, it is not possible to predict which ligand will give the best purification, and a detailed method for screening is given in Section 3.1.

Elution of the bound protein is generally carried out by increasing ionic strength or by using a competing ligand, such as the substrate or cofactor. The optimized conditions for purification of a mammalian bifunctional enzyme CoA synthase, using both of these methods, are described.

Table 1
Commercially Available Immobilized Dyes

Procion dye	Amicon	Sigma
Blue H-B (Cibacron Blue)	Matrex Gel Blue A	Reactive Blue 2
Blue MX-R		Reactive Blue 4
Red HE-3B	Matrex Gel Red A	Reactive Red 120
Yellow H-A	Matrex Gel Orange A	Reactive Yellow 3
Yellow MX-3R	Matrex Gel Orange B	
Green H-4G		Reactive Green 5
Green H-E4BD	Matrex Gel Green A	Reactive Green 19
Brown MX-5BR		Reactive Brown

2. Materials

1. Dye screening kits are available from Sigma (St. Louis, MO) and Amicon (Beverly, MA) containing a range of immobilized dyes as described in Table 1. Both of these suppliers use agarose as the support matrix.
2. Blue Sepharose CL-6B (Cibacron Blue F3G-A linked to Sepharose CL-6B) and Red Sepharose (Procion Red HE-3B-Sepharose CL-6B) are available from Pharmacia (Piscataway, NJ) (*see* Note 1).
3. Equilibration buffer: 20 mM Tris-HCl, pH 8.0, 0.5 mM DTT (*see* Notes 2 and 5)
4. Elution buffer: 20 mM Tris-HCl, pH 8.0, 1M KCl, 0.5 mM DTT
5. Blue Sepharose affinity elution buffer: 0.1 mM Coenzyme A, 0.1M KCl in equilibration buffer

All buffers contain analytical-grade reagents. DTT is added freshly before use.

3. Methods

3.1 Screening Dye Ligands

1. Prepacked columns are available to screen for binding of the protein of interest. Otherwise, pour individual columns of 1-mL packed bed volume of a range of adsorbants, such as those listed in Table 1. Wash the columns with 10 mL of equilibration buffer.
2. Adjust the pH of the protein sample to be applied to the columns to that of the equilibration buffer. The ionic strength of the protein sample should also be close to that of the equilibration buffer and preferably not exceeding a total of 0.05M. This can be achieved by simple dilution. The protein concentration should be about 10–20 mg/mL, and if necessary, the sample should be centrifuged or filtered to remove any particulate matter

3. Load 1 mL of the sample to the columns under gravity, and wash through with 5 mL of the equilibration buffer, collecting the unbound proteins in one fraction. Elute the bound protein with 5 mL elution buffer and collect in a fresh tube.
4. Assay both fractions from each column for total protein using a quantitative assay, such as the Bradford assay (8). Assay for the protein of interest using the most quantitative method available. Calculate the purification factor achieved in each case.
5. The best binding dye will be the one that effectively binds all the target protein, but allows much of the contaminants to pass through. Further development of this step will involve optimization of the elution conditions to remove contaminants by step or gradient elution. A dye that does not bind the protein of interest, but adsorbs some of the contaminating proteins can also be useful as a negative binding purification step (*see* Note 4).

3.2. Optimization of Purification

Further small-scale columns are recommended for optimization of purification conditions prior to scaling up.

1. For a negative binding step, vary the composition of the equilibration buffer with respect to both ionic strength and pH. Lowering of the ionic strength and lowering pH will generally increase the amount of total protein binding to the immobilized dyes. As in the dye screening step, collect the unbound material, and assay for total protein and protein of interest. Calculate the specific activity and the recovery of the protein of interest. The best buffer conditions will give the highest specific activity in the unbound fraction, provided a good yield of the protein is also achieved.
2. For a positive binding purification step, the equilibration buffer conditions should similarly be varied in order to maximize binding of the protein of interest and minimize binding of contaminant proteins.
3. The capacity of the dye for the protein of interest can be determined by frontal analysis. This is achieved by continuous loading of the sample solution onto the column until the protein of interest is detected in the eluate. This occurs when the target protein is displaced by proteins with higher affinities for the immobilized dye. Frontal analysis can be useful for examining the relative affinities of more than one protein in a mixture of proteins for a particular matrix. Optimum loading is 80% of the sample volume required for frontal detection.
4. Elution conditions can now be optimized. Wash the column with 5 vol of equilibration buffer to remove nonbinding proteins fully. Apply increasing salt concentrations to the column up to 1M NaCl (or KCl) in a stepwise

fashion with increases of 200 mM/step and with 2 column volumes/step. Alternatively, a salt gradient can be applied, and a 0–1M NaCl linear gradient could be tried in the first instance. If gradient programming is available as in FPLC, then nonlinear gradients can be used later to optimize purification and speed of elution.

5. The ability of specific ligands, such as a substrate, cofactor, or inhibitor, to elute the protein of interest can also be screened for, and this “affinity elution” can often provide a more powerful purification step.

As before, a small-scale column is loaded with sample and washed with equilibration buffer. Having already assessed the ionic strength conditions necessary for elution as described in the previous step, this information is used to optimize affinity elution conditions. First wash the column with buffer of an ionic strength just below that required to elute the protein in order to remove contaminants (try 20 mM < eluting buffer). Check that the eluate does not contain the protein of interest, and if so, lower the ionic strength of this wash solution. To affinity elute, lower the salt concentration of the buffer further (by 5–10 mM depending on the ionic strength of the specific ligand) and spike this solution with the specific ligand. In cases where the specific ligand will elute many proteins (e.g., ATP), better purification may be attained by applying a concentration gradient of the affinity ligand.

6. When scaling up of the purification step, keep the proportion of sample to column volume constant, although column size should be increased with larger diameters rather than larger column lengths. The linear flow rate should remain constant so that the sample protein will have the same contact time with the dye (see Notes 5 and 6).

3.3. Example Purification: Isolation of Coenzyme A Synthase

The bifunctional enzyme CoA synthase isolated and partially purified from pig liver was further purified to near homogeneity by dye-ligand chromatography on two types of immobilized dyes (9). The crude protein solution was loaded onto a Red Sepharose column equilibrated with 10 vol of equilibration buffer. The column was further washed with 2 column vol of equilibration buffer, and a linear gradient of 0–1M KCl in 20 mM Tris-HCl, pH 8.0, was applied to elute the protein. The fractions containing CoA synthase (Fig. 2A) were pooled and diluted with 2 vol of 0.5 mM DTT in order to reduce the ionic strength and allow binding to Blue Sepharose. This material was loaded onto a Blue Sepharose Cl-6B column equilibrated with 10 mM Tris-HCl, pH 8.0. The column was then washed with 0.12M KCl, in 10 mM Tris-HCl, pH 8.0, which does not

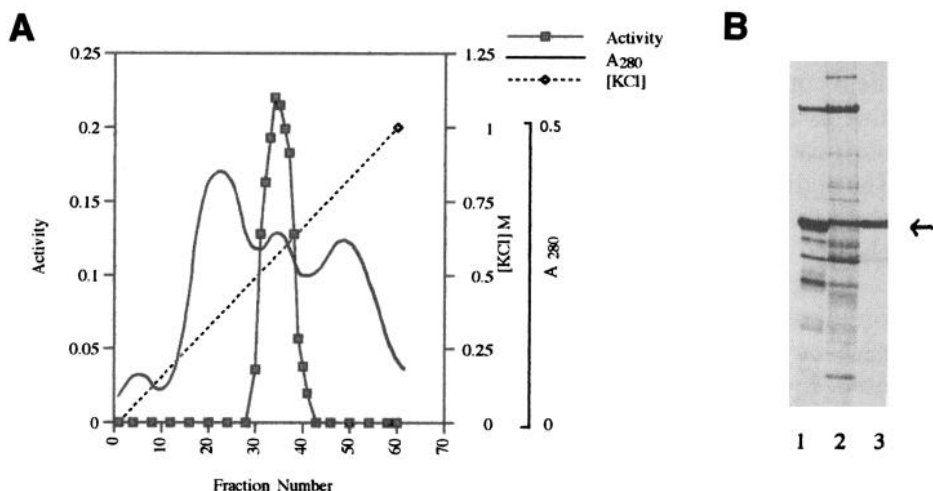


Fig. 2. Purification of CoA synthase on Procion Red Sepharose and affinity elution from Blue Sepharose. **(A)** Elution profile of CoA synthase from Red Sepharose CL-6B (Procion Red HE-3B) using a 0–1M KCl gradient. Fractions 30–40 were pooled, diluted with 2 vol 0.5 mM DTT, and applied to a Blue Sepharose CL-6B column. Affinity elution was carried out with 0.1 mM Coenzyme A. **(B)** SDS-PAGE of purification steps; lane 1, crude fraction (CoA synthase band of 60 kDa not visible); lane 2, pooled Red Sepharose fractions; lane 3, Blue Sepharose affinity-eluted protein.

elute the CoA synthase, but does elute most of the contaminating proteins. The enzyme was then eluted with a solution of 0.11M KCl, 10 mM Tris-HCl, pH 8.0, containing 0.1 mM Coenzyme A. As can be seen in Fig. 2B, the combination of these two steps can purify this protein from a crude extract where it is not visible on SDS-PAGE to near homogeneity.

4. Notes

1. Immobilization of dyes onto a support matrix can be readily carried out in the laboratory, and this may be desirable for cost-saving measures, particularly for large-scale column purification. The reactive dyes can be purchased relatively cheaply from Sigma and Polysciences. Detailed methods for coupling the dyes to the support matrix are given elsewhere (10). The major drawback from homemade columns is usually leakage of the dye from the matrix even after extensive washing. Novel purpose-made dye ligands have recently been synthesized that reduce leakage problems and have higher protein binding capacity (11).

2. The composition of the equilibration and elution buffers can be varied greatly. The examples given use Tris buffers at pH 8.0, which is relatively high, and so will reduce the amount of total protein binding. If binding of the protein of interest is low or weak in these conditions, it may be desirable to decrease the pH, in which case phosphate buffers may be more suitable.
3. The addition of divalent and trivalent metal ions (e.g., Mg^{2+} , Ca^{2+} , Zn^{2+} , Cu^{2+} , Mn^{2+} , Fe^{3+}) to the equilibration buffer can promote binding of certain proteins (12). Similarly, addition of EDTA to elution buffers can increase desorption of proteins.
4. Negative binding steps have proven to be very effective in the purification of proteins from plasma and from cell-culture extracts, which contain large amounts of fetal calf serum. This is owing to the high affinity of serum albumin, which is the major contaminating protein in such preparations, to Cibacron Blue and procion Red HE-3B dyes (13)
5. Regeneration of dye-ligand adsorbants can be carried out using either high-salt (2M NaCl) and/or high-pH (1M NaOH) solutions, and this is generally quite effective. However, if the column capacity is noticeably decreasing with continued use, then it may be improved by using chaotropic agents, such as 8M urea or 6M guanidinium hydrochloride, to denature and so remove residual protein.
6. HPLC-grade support matrices for dye-ligand chromatography have been used for purification of a number of proteins by immobilization of dyes onto silica-based matrices (14). Prepacked columns of Procion Blue MX-R, Procion Red HE-3B, and Cibacron Blue immobilized to silica (300 and 500 Å) are now commercially available from Serva Fine Chemicals.

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CHAPTER 18

Lectin Affinity Chromatography

Iris West and Owen Goldring

1. Introduction

Lectins are glycoproteins or proteins that have a selective affinity for a carbohydrate, or a group of carbohydrates. Many purified lectins are readily available, and these may be immobilized to a variety of chromatography supports.

Immobilized lectins are powerful tools which can be used to separate and isolate glycoconjugates (1-3), polysaccharides, soluble cell components, and cells (4) that contain specific carbohydrate structures. Glycopeptide mixtures can be fractionated by sequential chromatography on different lectins (5). Immobilized lectins can also be used to remove glycoprotein contaminants from partially purified proteins. In addition, immobilized lectins can be used to probe the composition and structure of surface carbohydrates (6). Subtle changes in glycoprotein structure owing to disease may be reflected in altered affinity for lectins (7). The predominant lectins used for affinity chromatography are listed in Table 1.

In principle, the mixture of glycoconjugates to be resolved is applied to the immobilized lectin column. The glycoconjugates are selectively adsorbed to the lectin and components without affinity for the lectin are then washed away. The adsorbed carbohydrate containing components are dissociated from the lectin by competitive elution with the hapten sugar, i.e., the best saccharide inhibitor. This may be a simple monosaccharide or an oligosaccharide (24,25).

There are many matrices suitable for coupling to lectins, and a number of reliable coupling methodologies have been developed. Many supports

Table 1
Common Lectins used in Affinity Chromatography

Lectin	Specificity	Useful eluants	Uses	Refs.
Concanavalin A	α -D-Mannopyranosyl with free hydroxyl groups at C3, C4, and C6	0.01–0.5M Methyl α -D-mannoside	Separation of glycoproteins	8
		D-Mannose	Purification of glycoprotein enzymes	9
		D-Glucose	Partial purification of IgM	10
			Separation of lipoproteins	11
<i>Lens culinaris</i>	α -D-Glucopyranosyl residues α -D-Mannopyranosyl	Methyl- α -D-glucoside	Purification of gonadotrophins	12
		0.1M Na Borate, pH 6.5	Purification of HeLa cells	13
		0.15M Methyl α -D-mannoside	Isolation of mouse H antigens	14
	Binds less strongly than Con A		Purification of detergent-solubilized glycoproteins	15
			<i>Schistosoma mansoni</i> surface membrane isolation	16
<i>Tritium vulgare</i>	N-acetyl-D-glucosamine	0.1M N-acetyl-glucosamine	Biochemical characterization of H4G4 antigen from HOON pre-B leukemic cell line	17
			Purification and analysis of RNA polymerase transcription factors	18
<i>Ricinus communis</i>	α -D-Galactopyranosyl residues	0.15M D-galactose	Fractionation of glycopeptide binding proteins	19
Jacalin	D-Galactopyranosyl residues	0.1M melibiose in PBS	Separating IgA1 and IgA2	20,21
			Purification of C1 inhibitor	22
<i>Bandeira simplicifolia</i>	α -D-Galactopyranosyl and N-acetyl-D-galactosamyl	PBS	Resolving mixtures of nucleotide sugars	23

Table 2
Commonly Available Activated Matrices

Commercial name	Supplier ^a
Affi 10, Affi 15	Bio-Rad
Affi Prep 10 and 15 (26)	Bio-Rad
CNBr agarose	Sigma
Epoxy-activated agarose	Sigma
CDI agarose	Sigma
Reacti 6X	Pierce
Polyacrylhydrazide agarose	Sigma

^aLocations Bio-Rad (Hercules, CA), Sigma (St Louis, MO); Pierce (Rockford, IL)

are now supplied preactivated. These are simple to use. Table 2 lists a few of the most popular activated matrices (see Note 1).

The lectin must be chosen for its affinity for a carbohydrate or sequence of sugars. Many of the lectins are not specific for a single sugar and may react with many different glycoproteins. Concanavalin A (Con A) has a very broad specificity, and will bind many serum glycoproteins, lysosomal hydrolases, and polysaccharides. The binding efficiency of Con A is reduced by the presence of detergents 0.1% sodium dodecyl sulfate (SDS), 0.1% sodium deoxycholate, and 0.1% cetyltrimethyl-ammonium bromide (CTAB) (27). *Lens culinaris* lectin has similar carbohydrate specificities to Con A, but binds less strongly. The lectin retains its binding efficiency in the presence of sodium deoxycholate. It is therefore useful for the purification of solubilized membrane glycoconjugates. Wheat germ lectin can interact with mucins that contain *N*-acetylglucosamine residues. It retains its binding efficiency in the presence of 1% sodium deoxycholate. Jacalin is a lectin isolated from the seeds of *Artocarpus integrifolia*, which has an affinity for D-galactose residues (22). It is reported to separate IgA₁ from IgA₂ subclass antibodies. It is therefore useful for removing contaminating IgA from preparations of purified IgG.

Two simple procedures (Sections 3.1. and 3.2.) for the immobilization of lectins are given, one for Con A onto carbonyldiimidazole (CDI)-activated agarose, and the other for Con A onto Affigel 15. Once the Con A has been immobilized to the matrices, the purification and elution procedures are common. A single protocol (Section 3.3.) is given detailing this procedure.

2. Materials

2.1. Immobilization of Con A on CDI-Activated Agarose

- 1 CDI-activated agarose (Sigma). The activated gel is stable in the acetone slurry for 1–2 yr at 4°C.
2. Affinity-purified Con A
3. Protective sugar. methyl- α -D-mannoside.
4. Coupling buffer: 0.1M Na₂CO₃, pH 9.5.
5. 0.1M Tris (hydroxymethyl) methylamine (Tris) pH 9.5.
6. Phosphate-buffered saline (PBS).
7. 0.1M Sodium acetate, pH 4.0.
8. 0.1M NaHCO₃, pH 8.0

2.2. Immobilization of Con A on Affigel 15

1. Affigel 15.
2. Affinity-purified Con A.
3. Protective sugar: methyl- α -D-mannoside.
4. Coupling buffer (PBS): 8.0 g NaCl, 0.2 g KH₂PO₄, 2.9 g Na₂HPO₄ · 12H₂O, 0.2 g KCl made up to 1 L with distilled water
5. Equilibration buffer: 5 mM Sodium acetate buffer containing 0.1M NaCl, 1 mM MnCl₂, 1 mM CaCl₂, 1 mM MgCl₂, and 0.02% sodium azide (*see* Note 4).

2.3. Purification of Glycoproteins on Immobilized Con A

1. Equilibration buffer: 5mM sodium acetate buffer, pH 5.5, containing 0.1M NaCl, 1 mM CaCl₂, 0.02% sodium azide.
2. 0.01M Methyl- α -mannoside in acetate buffer.
3. 0.3M Methyl- α -mannoside in acetate buffer.
4. 5 mM Sodium acetate buffer containing 1M NaCl.

3. Methods

3.1. Immobilization of Con A (see Note 3) on CDI-Activated Agarose

1. Place the gel in a sintered glass funnel. Apply gentle suction (a water pump is adequate).
2. Wash the gel free of acetone with 10 vol of ice-cold distilled water (*see* Note 2). Do not dry cake.
3. Dissolve Con A at 20 mg/mL in 0.1M Na₂CO₃ buffer, pH 9.5, containing the protective sugar 0.5M methyl- α -D-mannoside (*see* Note 3). Keep at 4°C.
4. Gently break up the gel cake, then add the cake directly to the coupling buffer containing the ligand, 1 vol of gel to 1 vol of buffer.

5. Incubate for 48 h at 4°C with gentle mixing.
6. Transfer the gel to a column (*see* Note 9), and then wash the gel with 0.1M Tris to block any remaining unreacted sites.
7. Wash the gel with 3 vol of 0.1M sodium acetate buffer, pH 4.0, and then with 3 vol of 0.1M NaHCO₃, pH 8.0.
8. Repeat step 7.
9. Wash the gel with PBS containing 0.02% sodium azide, and then store the gel at 4°C in buffer containing 0.02% sodium azide.

3.2. Immobilization of Con A (*see* Note 3) on Affigel 15 (*see* Notes 5, 7, and 8)

1. Place the gel in a sintered glass funnel. Apply gentle suction to remove the isopropyl alcohol. Do not let the gel dry out.
2. Wash the gel with 3 vol of cold distilled water (*see* Note 2).
3. Drain off excess water.
4. Dissolve Con A at 10 mg/mL in PBS containing 0.3M methyl- α -D-mannoside (*see* Note 3).
5. Transfer the gel to the coupling buffer, 1 vol gel:1 vol coupling buffer (*see* Note 6)
6. Mix gently for 2 h at 4°C.
7. Transfer the gel to a column (*see* Note 9) and wash with 5 vol of PBS.
8. Wash with equilibration buffer, and then store the column at 4°C in the presence of 0.2% sodium azide.

3.3. Purification of Glycoproteins on Immobilized Con A

1. Equilibrate the column with 5mM sodium acetate buffer containing 0.1M NaCl, 1 mM MnCl₂, 1 mM CaCl₂, 1 mM MgCl₂, 0.02% sodium azide (*see* Note 10), flow rate 0.1 mL/min.
2. Either dissolve solid glycoprotein in a little equilibration buffer, or if the glycoprotein is a solution, dilute 1:1 with equilibration buffer. Remove any particulate matter.
3. Apply to the column, and wash with equilibration buffer to remove nonbound components. Monitor effluent at 280 nm until the A_{280} is 0 (*see* Note 11).
4. Elute with 0.01M methyl- α -D-mannoside in acetate buffer (5 vol). Collect 1-mL fractions, and store on ice.
5. Elute with 0.3M methyl- α -D-mannoside in acetate buffer (*see* Notes 12 and 13). Collect 1-mL fractions, and store on ice.
6. Wash column with acetate buffer containing 1M NaCl, and then regenerate the column by extensive washing with equilibration buffer.
7. Pooled bound fractions can either be concentrated by ultrafiltration or dialyzed against a buffer to remove the hapten sugar

Table 3
Common Lectins and Their Protective Saccharides

Lectin	Hapten saccharide	References
Con A	Methyl- α -D-mannoside	28
<i>L. culinaris</i>	Methyl- α -D-mannoside	—
<i>T. vulgaris</i>	Chitin oligosacchrides	29
<i>R. communis</i>	Methyl- β -galactoside	18
Jacalin	D-Galactose	22

4. Notes

1. When using CNBr-activated matrices, charged groups may be introduced during the coupling step. This may lead to nonspecific adsorption owing to ion-exchange effects.
2. Adequate washing of the gel prior to coupling is required. Many proteins are sensitive to the acetone or alcohols used to preserve the activated gels.
3. Coupling methods are similar for different lectins; however, the lectin binding site must be protected by the presence of the hapten monosaccharide. Table 3 lists the common lectins and the saccharides used to protect the active site during immobilization.
4. Con A requires Mn^{2+} and Ca^{2+} to preserve the activity of the binding site.
5. Affigel 10 and 15 are *N*-hydroxysuccinimide esters of crosslinked agarose. Both are supplied as a slurry in isopropyl alcohol. The gels can be stored without loss of activity for up to a year at $-20^{\circ}C$.
6. Buffers containing amino groups should not be used for coupling. Suitable buffers include HEPES, PBS, and bicarbonate.
7. Affigel 10 couples neutral or basic proteins optimally. The coupling buffers used should be at a pH at, or below, the pI of the ligand. To couple acidic proteins, coupling buffers should contain 80 mM $CaCl_2$. Affigel 15 couples acidic proteins optimally. The pH of coupling buffers should be above or near the pI of the ligand. To couple basic proteins, include 0.3M NaCl in the coupling buffer.
8. Steps 1–4 should be completed in 20 min to ensure optimum coupling.
9. The size of column depends on the amount of sample to be purified. Preliminary experiments can easily be carried out using Pasteur pipets, with supports of washed glass wool. Alternatively, the barrels of disposable plastic syringes can be used. Long, thin columns will give the best resolution of several components. However, where the requirement is to remove only contaminants from the protein, a short, fat column will elute the proteins in a smaller volume.
10. Equilibration buffers used will depend on the stability of the glycoconjugate being purified, and where protease activity is suspected, the appropriate

protease inhibitors should be included. Some glycoconjugates are sensitive to acetate (18). Phosphate buffers will precipitate out any Ca^{2+} added as calcium phosphate. The optimal pH for the equilibration buffer should be established to ensure maximum adsorption to the affinity column. Binding may be enhanced in certain cases by the inclusion of NaCl in the buffer (30).

11. Sparingly reactive glycoconjugates may be retarded rather than bound by the column (31). Washing should continue for at least five bed volumes. The nonbound fractions can be retained and reapplied to the column on a subsequent occasion, when it is suspected that the load applied to the column is saturating.
12. Elution can be achieved either as a "step," i.e., a change to a buffer containing the hapten sugar, or using a gradient of the hapten sugar. Different inhibiting sugars can be used for elution, e.g., D-glucose, α -D-mannose, methyl- α -D-glucoside.
13. Occasionally, glycoconjugates may bind very tightly to the matrix and may not elute even with 0.5M hapten sugar. Sodium borate buffers offer an alternative: these may elute components with high affinity for the lectin without denaturing it. The eluant temperature can be raised above 4°C, dependent on the protein stability. Detergents, e.g., 0–5% SDS together with 6M urea will often strip components from the lectin, but may irreversibly denature the lectin. Occasionally, it may be expedient to use stringent denaturing conditions to desorb bound fractions with a high yield. Glycoproteins have been eluted from Con A Sepharose by heating in media containing 5% SDS and 8M urea (32). 1M NH_4OH has also been used as an eluant (13) with high yield, but irreversibly denaturing the lectin.

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CHAPTER 19

Immunoaffinity Chromatography

George W. Jack and David J. Beer

1. Introduction

Immunoaffinity chromatography (IAC) harnesses the specificity and avidity of the interaction between an antigen and its antibody to purify the antigen. The technique may employ either polyclonal or monoclonal antibodies (MAbs) (1), but MAbs are preferred for a variety of reasons. Polyclonal sera are never specific for a single antigen, but reflect the variety of immunological challenges sustained by the animal since birth. Furthermore, the best polyclonal sera are generated by immunization with a highly purified antigen, so if a purification method already exists for the protein, why raise antisera to develop a second purification method? Perhaps the main disadvantage of polyclonal sera lies in the wide range of avidity of the antibodies they contain, because although ensuring that such antibodies will capture their antigen, the dissociation of the antigen–antibody complex may prove impossible under conditions that retain the activity of both.

Conversely, MAbs may be generated even when pure antigen is not available and produced on an appropriate scale by growing as ascites or in cell culture. A unique and consistent product may be obtained by this means, whereas every animal immunized with a given antigen will produce a different set of antibodies, each with their own properties. Each MAb exhibits a unique association constant (K_a) for its antigen, so that from a panel of MAbs one can be chosen for use in IAC with sufficient avidity to ensure good binding of the antigen, while still allowing the MAb–antigen complex to be dissociated under relatively mild conditions.

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A rapid ELISA plate technique has been developed (2) to determine if a given MAb will withstand a range of potential elution conditions.

The technique of IAC comprises first the immobilization of an MAb on an insoluble support matrix followed by packing the sorbent as a column on which to perform adsorption/desorption chromatography. A wide range of matrices for immobilization is available commercially (3), but probably the most commonly employed are the Sepharoses produced by Pharmacia (Uppsala, Sweden) and the Affigel series from Bio-Rad (Hercules, CA). The Affigels are crosslinked agaroses with spacer arms for carbodiimide-induced coupling or derivatized with *N*-hydroxy-succinimide esters for spontaneous coupling of proteins through free amino groups via a 10- or 15-atom long spacer arm; *see* Fig. 1 Pharmacia markets a range of activated and derivatized Sepharoses, also agarose-based, the most commonly used of which is CNBr-activated Sepharose; *see* Fig. 2 for the coupling reaction. CNBr-activated agarose (4) may be synthesized in-house, but since the production of it and a range of other activated matrices involves handling highly toxic chemicals and may result in varying degrees of substitution between batches, it is probably best brought in as the prepared derivative.

The degree of substitution of the matrix influences the capacity of the sorbent for antigen (5), but in general, substitution levels over 10 mg MAb/mL of matrix result in inefficient utilization of the MAb. The choice of the MAb to be used in IAC must to some extent be determined by trial and error even after MAbs unstable in commonly used elution buffers have been eliminated. If the $K_{a,s}$ of a panel of MAbs are known, the number to be tested can be restricted to those with $K_{a,s}$ in the range 5×10^{-7} to $5 \times 10^{-8}M$, since this has been shown to be optimal for IAC (6). Pure MAb should be used to minimize the amount of nonfunctional immobilized protein. This will reduce the nonspecific binding capacity of the sorbent, since by virtue of their amphoteric nature, immobilized proteins may act as ion exchangers, whereas any exposed hydrophobic sequences may result in nonspecific hydrophobic interactions. The selection of appropriate buffers for the binding and washing of the MAb-antigen complex will minimize contamination of the eluted product.

The nature of the chemistry used to immobilize the MAb must also be determined by trial and error, since all chemistries will not necessarily result in an active sorbent, even though coupling of the MAb to the matrix has been achieved. A further constraint on the choice of coupling chemistry is the

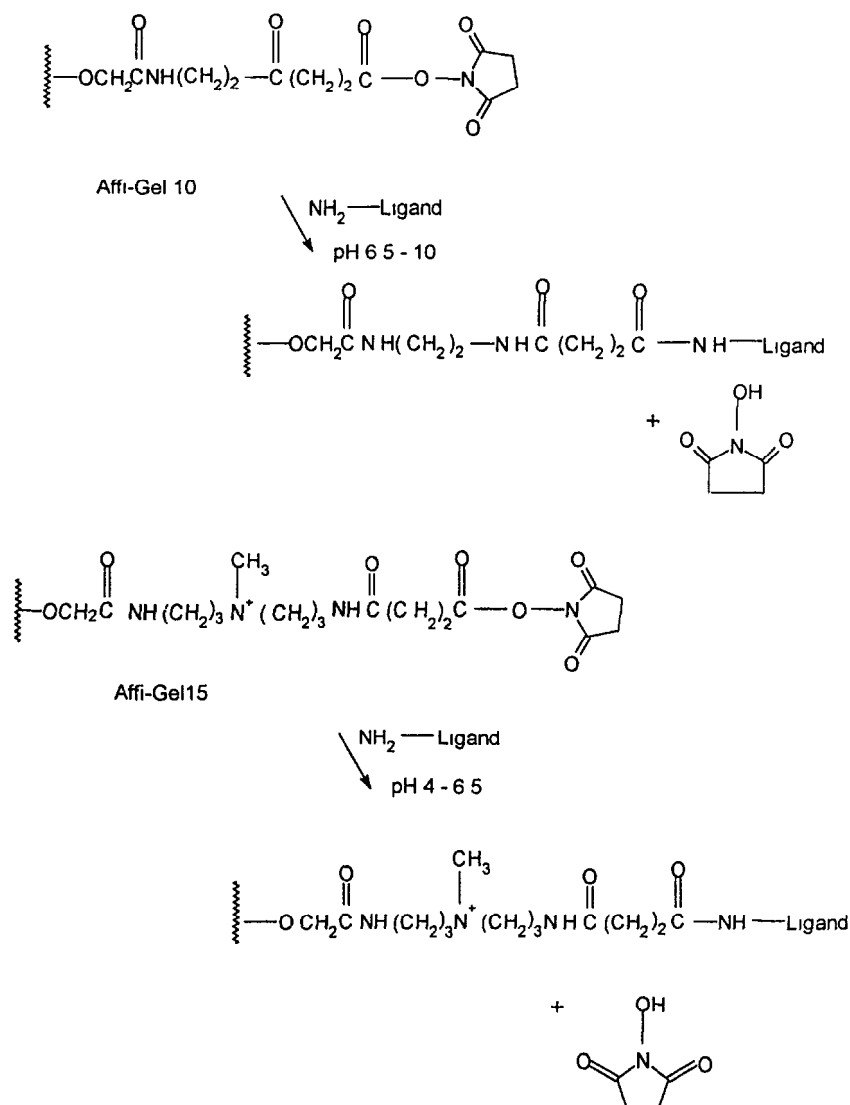


Fig. 1. The reactions of Affigels 10 and 15 with an amine containing ligand.

known leakage of ligand from sorbents generated by some chemistries. Although all IAC sorbents are known to leak ligand to some extent, this can be minimized by using those chemistries that have been demonstrated to leak least (7,8). The coupling of MAbs to Affigel 10 has been shown (9) to produce the most stable sorbents and, hence, is described in this example.

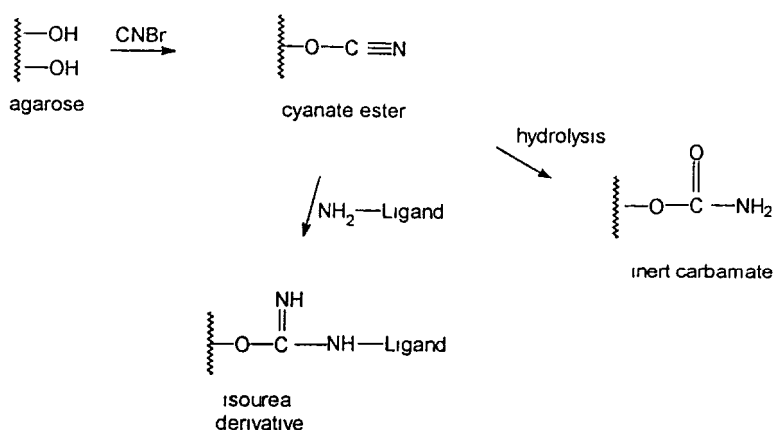


Fig. 2. The reaction of cyanogen bromide with an agarose matrix and the subsequent coupling of an amine containing ligand.

Among the advantages of IAC is its ability to produce virtually pure protein in a single step, thereby avoiding multistep procedures, which invariably result in low overall yields. Because of the high specificity and avidity of the adsorption stage, very dilute antigen solutions in the presence of large amounts of extraneous protein may be processed successfully, since the sorbent will only bind the antigen of the immobilized MAb. The capacity of sorbents for antigen is, however, low with capacities in excess of 1 mg antigen/mL of gel being something of a rarity. This factor, together with the high cost of producing IAC columns, has encouraged the use of relatively small columns operated in a cycling mode and automated by microprocessor or computer control of pumps, multiway solenoid valves, and fraction collectors (5,6). Commercially produced instruments to run this sort of chromatography are available from Amicon (Beverly, MA), Pharmacia, Sepracor (Marlborough, MA), Bio-Rad, and PerSeptive (Cambridge, MA). The use of short, fat columns at relatively high flow rates enables several cycles to be completed per day. The inclusion of protective devices such as pressure and level sensors in automated systems allows for unattended operation 24 h/d ensuring high productivity from the system despite a low yield/cycle. Furthermore, should the column become contaminated and lose activity, the replacement cost is not prohibitive.

For small-scale laboratory investigative use, several cycles of chromatography may be run per day using relatively unsophisticated equip-

ment, while the high resolving power of the technique ensures high product quality in a single step.

2. Materials

Use Analar- or even Aristar-grade reagents and distilled water for all solutions. All solutions must be sterilized either by filtration (0.2- μ m filter) or autoclaving (121°C for 30 min) where appropriate.

1. The apparatus required for running laboratory-scale IAC is commonly available and includes a glass column with adjustable flow adaptors, a peristaltic pump, a flowthrough UV-absorbance monitor and recorder, and a fraction collector. For synthesis of the IAC sorbent, a sintered glass funnel of porosity 2 to British Standard 1752, i.e., 40–90 μ m, with a Buchner flask attached to a vacuum pump is required.
2. Coupling buffer: 0.1M phosphate/citrate, pH 6.4. Dissolve Na_2HPO_4 (14.2 g) in water and adjust to pH 6.4 with solid citric acid then make to 1 L with water. Sterilize by autoclaving.
3. Blocking buffer: 0.1M ethanolamine pH 8.5. Dilute 61 mL ethanolamine in water, and adjust the pH to 8.5 by the careful addition of 6M HCl in a fume hood. Make to 1 L with water. Sterilize by filtration.
4. Binding buffer: borate/salt, pH 8.5. Dissolve boric acid (60 g), NaCl (140 g), and NaOH (10 g) in water and make to 1 L. Sterilize by autoclaving. For use, dilute 200 mL buffer concentrate with 800 mL sterile water.
5. Wash buffer (acid): acetate/salt, pH 4.0. Dissolve sodium acetate trihydrate (13.6 g) and NaCl (29.2 g) in water and adjust to pH 4.0 with glacial acetic acid, and then with water to 1 L. Sterilize by autoclaving.
6. Column cleaning buffer: 8M urea/acetate, pH 4.0. Dissolve Aristar-grade urea (480 g) and sodium acetate trihydrate (13.6 g) in water, adjust to pH 4.0 with glacial acetic acid, and then to 1 L with water. Sterilize by filtration.

3. Method

1. Take 10 mL of a 50% (v/v) suspension of Affigel 10 (*see* Note 1), place in a sintered glass funnel, and suck through the suspending isopropanol. Ensure that the gel is not allowed to go dry during washing.
2. Wash the gel rapidly with 200 mL of cold water/mL of gel in several aliquots to remove the isopropanol.
3. Suck the liquid through between successive additions of water. Rapidly wash the gel with 100 mL of coupling buffer, and then immediately transfer the gel to a flask containing approx 40 mg of MAb in coupling buffer.
4. Stir the gel gently to obtain an even suspension, and then set the flask to shake at 4°C for 4 h.

5. Take a sample of the supernatant above the gel beads, and assay it for protein. This cannot be done by a simple $A_{280\text{nm}}$ measurement owing to the absorbance of the leaving group from the matrix. If more than 80% of the MAb is no longer in solution, proceed; if not, allow the reaction to continue at 4°C.
6. When coupling is complete, add to the suspension 0.1 mL of blocking buffer/mL of suspension, and shake at room temperature for 30 min
7. Return the gel to the sintered funnel, and wash it with alternate 100-mL lots of binding buffer, and wash buffer (acid). Five cycles is adequate.
8. Pack the gel in a 1.6-cm diameter glass column with the flow adaptors in contact with the gel bed. At a flow rate of 40 mL/h, wash the column with 3 column volumes (15 mL) of binding buffer followed by an equal volume of wash buffer. Repeat the process three times and then wash the gel with 1 column (5 mL) of urea/acetate before equilibrating the gel with 3 column vol of binding buffer.
9. If the purification protocol is known, subject the column to five blank cycles with no sample application before the first purification cycle is run
10. Equilibrate the extract to be chromatographed with the binding buffer either by dialysis or by buffer exchange through Sephadex G-25. Filter-sterilize the extract (*see Note 2*).
11. Pump the extract through the IAC sorbent at the above flow rate, and then wash the column with binding buffer until the UV absorbance of the column effluent falls back to the baseline level. Use a sensitive range on the absorbance monitor, such as 0–0.1 absorbance units, for full-scale recorder deflection (*see Note 3*).
12. Wash the column with 2 column volumes of a solution of low ionic strength. This could be the binding buffer omitting the salt, or even water will suffice.
13. Stop the flow through the column, and alter the configuration of the plumbing to reverse the direction of flow through the column. This may be also achieved by the use of a number of three-way valves. Elute the column with 3 column volumes of a solution (*see Notes 4 and 5*), which will dissociate the MAb–antigen complex.
14. Wash the column with 1 column volume of urea/acetate followed by 2 column volumes of binding buffer. Again, reverse the direction of flow through the column, and equilibrate the sorbent with 3 column volumes of binding buffer before applying the next sample of extract. The entire chromatographic process should take between 2 and 3 h.

4. Notes

1. Affigels 10 and 15 are supplied as 50% (v/v) suspensions in isopropanol and may be stored at –20°C for extended periods without loss of binding capac-

ity if kept dry. Adherence to the manufacturer's recommendations should result in successful coupling of MAb to matrix. If the pI of the MAb is known, then a rational choice can be made between the 10 and 15 series gels, since at neutral pH, Affigel 10 couples proteins of neutral to alkaline pI, whereas Affigel 15 couples proteins of acidic pI with high efficiency. In general, however, MAbs couple more efficiently to Affigel 10.

2. All proteins are susceptible to proteolysis even when afforded the stabilizing influence of immobilization. An IAC column run with nonsterile buffers and extracts will rapidly lose its antigen binding capacity because of bacterial or fungal proteolysis of the immobilized antibody. The use of sterile buffers and extracts is essential for the long-term activity of an IAC column. The choice of sterilization method for buffers is also important, since some buffers will change pH (e.g., carbonate/bicarbonate buffers) or decompose (buffers containing urea) on autoclaving, and for those, filtration is the sterilization method of choice. If operated carefully under sterile conditions, columns have been used for over 50 cycles of operation spread over many months, but such longevity is enhanced if the column is operated in a "clean room" environment. The ease with which a column can become contaminated and inactivated is exacerbated by the inability to sanitize the column owing to the protein nature of the ligand. This reinforces the recommendation to perform multiple cycles on a small column since you have less MAb to lose following accidental contamination.
3. All IAC columns will bind protein nonspecifically, although appropriate selection of the buffer for the absorption stage can reduce this phenomenon quite markedly. In the purification of human thyrotropin (10), altering the binding buffer from Tris to borate, and then adding NaCl to the borate sequentially reduced the level of contamination of the thyrotropin by lutropin by approx 40-fold. Since protein binds nonspecifically to IAC columns by a mixture of ionic and hydrophobic interactions, washing the sorbent with solutions of both high and low ionic strength will minimize these opportunistic contaminants.
4. The nature of the eluting agent to be used in IAC must be determined empirically in each instance. A change to more acidic conditions is one of the most commonly used elution strategies, but to be successful requires some knowledge of the pH stability of the protein under purification; human thyrotropin, for example, starts losing activity below pH 3.5 (10). Immobilized MAbs have much greater pH stability than antibodies in free solution and can withstand acid pH down to pH 2.0, but many appear susceptible to inactivation above pH 10.0. They are also able to withstand the effects of chaotropes and protein denaturants. Protein denaturants, such as 8M urea or 6M guanadine hydrochloride at neutral pH have been used as eluents if

the antigen retains its activity on removal of the denaturant. High concentrations of chaotropic agents, such as thiocyanate and iodide ions and MgCl_2 at 2.5M, are commonly used as eluents. A commonly used ploy with chaotropes is to pump just <1 column volume of eluent onto the column and then stop pumping for 30 min before continuing the elution. The above are only a small selection of the compounds that have been employed as eluents in IAC (2). The nature of the eluent requires careful consideration in the case of potential therapeutic agents so as not to compromise their administration to humans.

5. Whenever practical, reverse-flow elution is to be preferred, since it can result in elution volumes down to one-third of those obtained by unidirectional flow through the sorbent. It also has the advantage of clearing debris away from the net or sinter of the normal inlet flow adaptor of the column.
6. Leakage of antibody occurs from all IAC columns. The extent to which such leakage results in product contamination can be minimized both by careful column washing and by choosing a coupling chemistry known to generate a stable linkage. A regime as outlined in Section 3 for washing a new gel will produce a sorbent from which antigen may be eluted with very low, yet still detectable, contamination. During thyrotropin purification, the first three cycles on a newly synthesized sorbent yielded thyrotropin contaminated with MAb at 24, 12, and then <6 ng/mL, showing that leakage from a new gel decreases with use (10). Using more sensitive assays for MAbs (11), Affigel 10-based sorbents have been shown to leak <0.3 ng MAb/mL of eluent (9). If minimum contamination levels are sought from the first cycle of operation, the precycling protocol described in Section 3. should be employed as well as after the column has been unused for some time.

Leakage results largely from solvolysis of the matrix–ligand bond. However, the antibody is bound to the gel by multipoint attachment, and more than one bond must hydrolyze before leakage occurs. Mechanical fragmentation and chemical degradation of the matrix do not appear to contribute significantly (9) to leakage. The presence of proteases in the extracts applied to IAC sorbents can induce leakage of antibody fragments (9) and can be minimized by treating the extract with protease inhibitors prior to the chromatography. The surest way to achieve minimal leakage is to test a range of immobilization chemistries and determine which results in least leakage in the case of the MAb of choice. In general, however, the order of stability of coupling chemistries is *N*-hydroxysuccinimide > CNBr > 1,1'-carbonyldiimidazole > tressyl > hydrazido, for crosslinked agarose as matrix (9).

7. MAbs are produced by cell lines derived from a cancerous cell, and as a consequence, oncogenic nucleic acid may be present in crude MAb prepa-

rations. Before use in IAC sorbents, MAbs should be purified both to eliminate oncogenic material and to diminish the amount of nonfunctional protein immobilized. This may be achieved by a variety of means, but the most common methods currently in use are chromatography on either ion exchangers or immobilized protein A or G (12). Both protein A and protein G are receptors for the Fc region of antibodies derived from staphylococcal and streptococcal strains respectively. Both may be purchased already immobilized from suppliers, such as Bio-Rad and Pharmacia, with instructions for use. Each sorbent does not bind all subclasses of mouse and rat antibodies, but one or the other should purify almost any MAb. The product, however, is likely to contain small amounts of protein A or G leaking off the matrix.

As an alternative, a two-step procedure of cation- and anion-exchange chromatography will purify virtually all MAbs to homogeneity. A 0–0.3M NaCl gradient in 20 mM acetate, pH 5.0, on S-Sepharose followed by a 0–0.3M NaCl gradient in 20 mM Tris-HCl, pH 7.6, on Q-Sepharose should provide pure MAb.

- 8 Of necessity, this is a brief overview of IAC. For a more comprehensive treatment of the subject, the reader is directed to recent books (3) and reviews (5,13–15) of what has become an extensive literature

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CHAPTER 20

Immobilized Metal Ion Affinity Chromatography

Tai-Tung Yip and T. William Hutchens

1. Introduction

Immobilized metal ion affinity chromatography (IMAC) (1,2) is also referred to as metal chelate chromatography, metal ion interaction chromatography, and ligand-exchange chromatography. We view this affinity separation technique as an intermediate between highly specific, high-affinity bioaffinity separation methods, and wider spectrum, low-specificity adsorption methods, such as ion exchange. The IMAC stationary phases are designed to chelate certain metal ions that have selectivity for specific groups (e.g., His residues) in peptides (e.g., 3–7) and on protein surfaces (8–13). The number of stationary phases that can be synthesized for efficient chelation of metal ions is unlimited, but the critical consideration is that there must be enough exposure of the metal ion to interact with the proteins, preferably in a biospecific manner. Several examples are presented in Fig. 1. The challenge to produce new immobilized chelating groups, including protein surface metal-binding domains (14,15) is being explored continuously (16). Table 1 presents a list of published procedures for the synthesis and use of stationary phases with immobilized chelating groups. This is by no means exhaustive, and is intended only to give an idea of the scope and versatility of IMAC.

The number and spectrum of different proteins (Fig. 2, p. 200) and peptides (Fig. 3, p. 201) characterized or purified by use of immobilized metal ions are increasing at an incredible rate. The three reviews listed (8,19,20) barely present the full scope of activities in this field.

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Edited by S. Doonan Humana Press Inc., Totowa, NJ

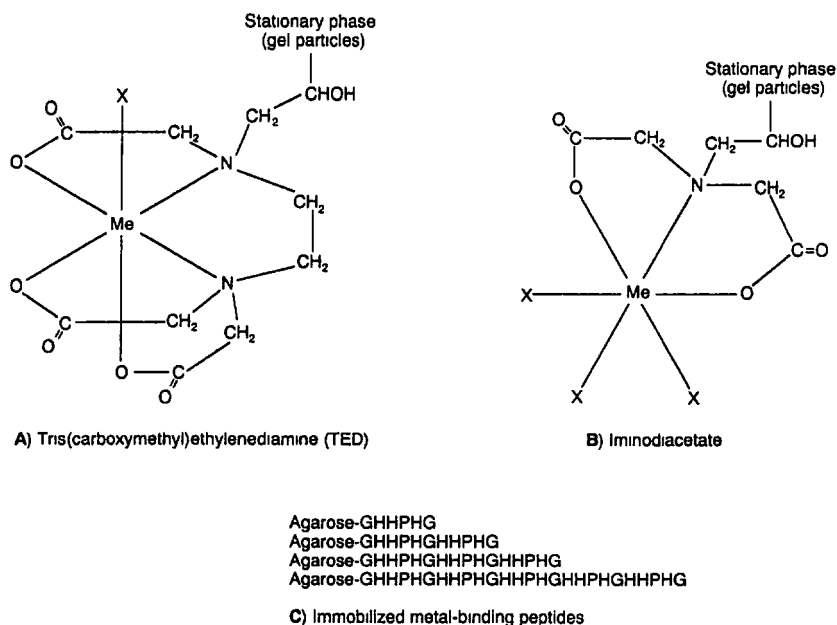


Fig. 1. Schematic illustration of several types of immobilized metal-chelating groups, including, iminodiacetate (IDA), tris(carboxymethyl) ethylenediamine (TED), and the metal-binding peptides (GHHPH)_nG (where $n = 1, 2, 3$, and 5) (14,15).

Beyond the use of immobilized metal ions for protein purification are several analytical applications, including mapping proteolytic digestion products (5), analyses of peptide amino acid composition (e.g., 5,6), evaluation of protein surface structure (e.g., 8–11), monitoring ligand-dependent alterations in protein surface structure (Fig. 4, p. 202) (12,16), and metal ion exchange or transfer (e.g., 14,21).

The versatility of IMAC is one of its greatest assets. However, this feature is also confusing to the uninitiated. The choice of stationary phases and the metal ion to be immobilized is actually not complicated. If there is no information on the behavior of the particular protein or peptide on IMAC in the literature, use a commercially available stationary phase (immobilized iminodiacetate), and pick the relatively stronger affinity transitional metal ion, Cu(II), to immobilize. If the interaction with the sample is found to be too strong, try other metal ions in the series, such as Ni(II) or Zn(II), or try an immobilized metal chelating group with a

Table 1
Immobilized Chelating Groups and Metal Ions
Used for Immobilized Metal Ion Affinity Chromatography

Chelating group	Suitable metal ions	Refs.	Commercial source ^a
IDA	Transitional	1,2	Pharmacia LKB Pierce Sigma Boehringer Mannheim TosoHaas Not available
2-Hydroxy-3[<i>N</i> -(2-pyridylmethyl)glycine]propyl	Transitional	3	Not available
α -Alkyl nitrilotriacetic acid	Transitional	4	Not available
Carboxymethylated aspartic acid	Ca(II)	13	Not available
TED	Transitional	2	Not available
(GHHPH) _{<i>n</i>} G ^b	Transitional	14,15	Not available

^aLocations Pharmacia LKB, Uppsala, Sweden, Pierce, Rockford, IL, Sigma, St Louis, MO, Boehringer Mannheim, Mannheim, Germany, TosoHaas, Philadelphia, PA

^bLetters represent standard one-letter amino acid codes (G = glycine, H = histidine, P = proline) The number of internal repeat units is given by *n* (*n* = 1, 2, 3, and 5)

lower affinity for proteins (2,22). An important contribution to the correct use of IMAC for protein purification is a simplified presentation of the various sample elution procedures. This is especially important to the first-time user. There are many ways to decrease the interaction between an immobilized metal ion and the adsorbed protein. Two of these methods are efficient and easily controlled; they will be presented in detail in this chapter. Interpretation of IMAC results for purposes other than separation (i.e., analysis of surface topography and metal ion transfer) has been discussed elsewhere and is beyond the scope of this chapter.

2. Materials

The following list of materials and reagents is only representative. Other stationary phases, metal ions, affinity reagents, and mobile phase modifiers are used routinely.

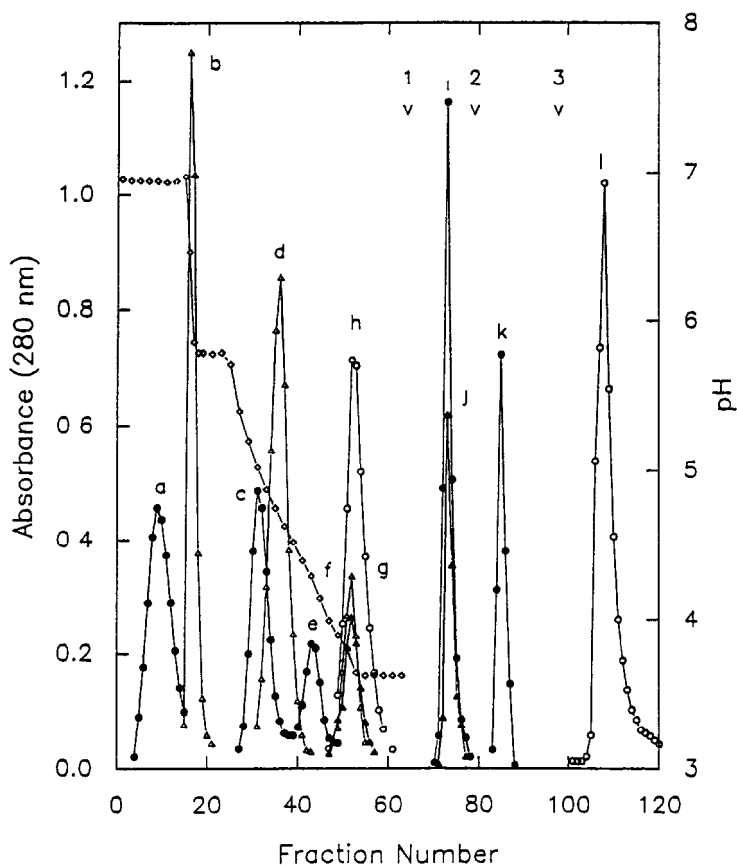


Fig. 2. Protein elution from immobilized (IDA) Cu(II) ions as a function of decreasing pH and increasing imidazole concentration. Proteins were eluted in the following order: chymotrypsinogen (a), chymotrypsin (b), cytochrome *c* (c), lysozyme (d), ribonuclease A (e), ovalbumin (f), soybean trypsin inhibitor (g), human lactoferrin (h), bovine serum albumin (i), porcine serum albumin (j), myoglobin (k), and transformed (DNA-binding) estrogen receptor (l). Open triangles represent pH values of collected fractions. Arrows 1–3 mark the introduction of 20, 50, and 100 mM imidazole, respectively, to elute high-affinity proteins resistant to elution by decreasing pH. Protein elution was evaluated by absorbance at 280 nm. In the case of the [³H]estradiol-receptor complex, receptor protein elution was determined by liquid scintillation counting. Except for the estrogen receptor (l), protein recovery exceeded 90%. Only 50–60% of the DNA-binding estrogen receptor protein applied at pH 7.0 was eluted with 100–200 mM imidazole. Reproduced with permission from ref. 17.

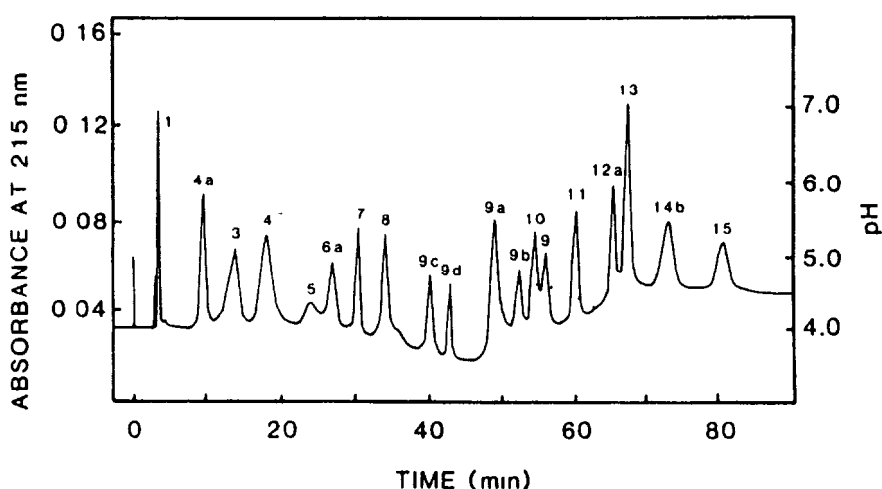


Fig. 3. Separation of bioactive peptide hormones by IMAC. The pH-dependent separation of a synthetic peptide hormone mixture (19 peptides) was accomplished using a TSK chelate-5PW column (8×75 mm, 10- μ m particle diameter) loaded with Cu(II). A 20- μ L sample (1–4 μ g of each peptide) was applied to the column equilibrated in 20 mM sodium phosphate containing 0.5M NaCl, pH 7.0. After 10 min of isocratic elution, pH-dependent elution was initiated with a 50-min gradient to pH 3.8 using 0.1M sodium phosphate containing 0.5M NaCl at a flow rate of 1 mL/min. Peptide detection during elution was by UV absorbance at 215 nm (0.32 AUFS). The pH profile of effluent is indicated by the dotted line. Sample elution peaks were identified as: 1, neurotensin; 4a, sulfated [leu⁵] enkephalin; 3, oxytocin; 4, [leu⁵] enkephalin; 5, mastoparan; 6a, tyr-bradykinin; 7, substance P; 8, somatostatin; 9c, [Asu¹⁻⁷] eel calcitonin; 9d, eel calcitonin (11–32); 9a, [Asu¹⁻⁷] human calcitonin; 9b, human calcitonin (17–32); 10, bombesin; 9, human calcitonin; 11, angiotensin II; 12a, [Trp (for)^{25,26}] human GIP (21–42); 13, LH-RH; 14b, human PTH (13–34); 15, angiotensin I. Reproduced with permission from ref. 18.

2.1. Stationary Phase for IMAC

2.1.1. Conventional Open Column Stationary Phases (Agarose)

One example is Chelating Sepharose Fast Flow (Pharmacia), which uses the iminodiacetate (IDA) chelating group. Another example is Tris(carboxymethyl)ethylenediamine (TED) (*see* Table 1 and Fig. 1). This stationary phase is used for proteins whose affinity for IDA-metal groups is too high (2,22).

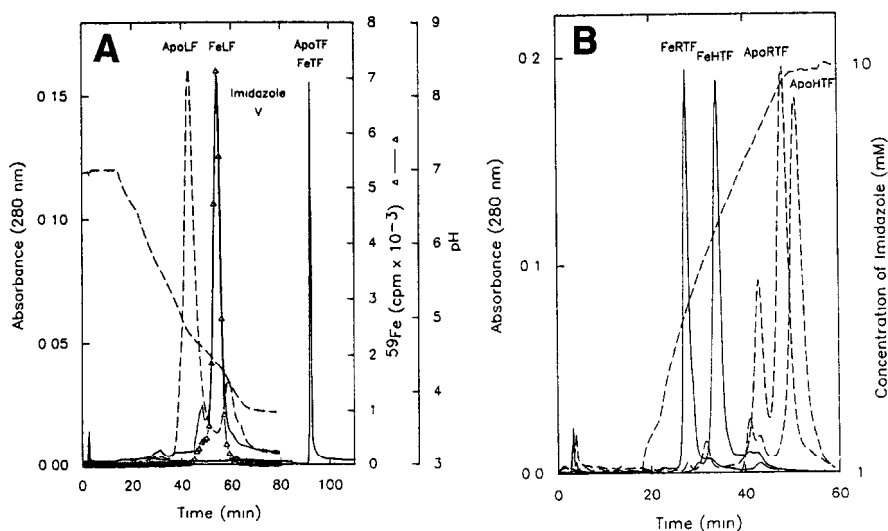


Fig. 4. (A) Separation of apolactoferrin (dashed line) and iron-saturated (solid line) human lactoferrins on high-performance IDA-Cu(II) columns with a phosphate-buffered gradient of decreasing pH in the presence of 3M urea. Both apotransferrin (ApoTF) and iron-saturated human serum transferrins (FeTF) were eluted only on introduction of 20 mM imidazole (imidazole-labeled arrow). UV absorbance was monitored at 280 nm. The elution of iron-saturated human lactoferrin was determined by measuring protein-bound ^{59}Fe radioactivity (open triangles). (B) Separation of iron-free (dashed line) and iron-saturated (solid line) transferrins on a high-performance IDA-Cu(II) affinity column using an imidazole elution gradient protocol. The apo and holo forms of human serum transferrin (hTF) and rabbit serum transferrin (rTF) are shown. The Fe and Apo prefixes to these abbreviations designate iron-saturated and metal-free transferrins, respectively. Reproduced with permission from ref 12

2.1.2. High-Performance Stationary Phases (Rigid Polymer)

One example is TSK Chelate 5PW (TosoHaas) (IDA chelating group)

2.1.3. Immobilized Synthetic Peptides as Biospecific Stationary Phases

Stationary phases of this type are designed based on the known sequence of protein surface metal-binding domains (14,15). The metal-binding sequence of amino acids is first identified (e.g., 14,23). The synthetic protein surface metal-binding domain is then prepared by solid-phase methods of peptide synthesis (14) and verified to have metal-binding

properties in solution (14,24). Finally, the peptides are immobilized (e.g., to agarose) using chemical coupling procedures consistent with the retention of metal-binding properties (15) (Table 1 and Fig. 1).

The procedures outlined in this chapter emphasize the specific use of agarose-immobilized IDA metal chelating groups. In general, however, these procedures are all acceptable for use with a wide variety of different immobilized metal chelate affinity adsorbents.

2.2. Metal Ion Solutions in Water

1. 50 mM CuSO_4 .
2. 50 mM ZnSO_4 .
3. 50 mM NiSO_4 .

2.3. Buffers

1. Buffer 1: 20 mM sodium phosphate (7.8 mM NaH_2PO_4 , 12.2 mM Na_2HPO_4), 0.5M sodium chloride, pH 7.0.
2. Buffer 2: 0.1M sodium acetate, 0.5M sodium chloride, pH 5.8. Use 0.1M acetic acid and adjust the pH with 2–5M sodium hydroxide.
3. Buffer 3: 0.1M sodium acetate, 0.5M sodium chloride, pH 3.8. Use 0.1M acetic acid and adjust the pH with 2–5M sodium hydroxide.
4. Buffer 4: 50 mM sodium dihydrogen phosphate, 0.5M sodium chloride. Add concentrated HCl until pH is 4.0.
5. Buffer 5: 20 mM imidazole (use the purest grade or pretreat with charcoal), 20 mM sodium dihydrogen phosphate (or HEPES), 0.5M sodium chloride. Adjust to pH 7.0 with HCl.
6. Buffer 6: 2 mM imidazole, 20 mM sodium dihydrogen phosphate (or HEPES), 0.5M sodium chloride. Adjust to pH 7.0 with HCl.
7. Buffer 7: buffer 1 containing 50 mM EDTA.
8. Buffer 8: 20 mM sodium phosphate (3.2 mM NaH_2PO_4 , 16.8 mM Na_2HPO_4), 0.5M sodium chloride, 3M urea, pH 7.5.
9. Buffer 9: 50 mM sodium dihydrogen phosphate, 0.5M sodium chloride, 3M urea. Add concentrated HCl until pH is 3.8.
10. Milli-Q (Millipore) water or glass-distilled, deionized water.

Urea (1–3M) and ethylene glycol (up to 50%) have been found useful as additives to the abovementioned buffers (see Notes 7 and 8).

2.4. Columns and Equipment

1. 1-cm Inner diameter columns, 5–10-cm long for analytical and micropreparative scale procedures.
2. 5–10-cm Inner diameter columns, 10–50 cm long for preparative scale procedures.
3. Peristaltic pump

4. Simple gradient forming device to hold a vol 10–20X column bed vol (if stepwise elution is unsuitable).
5. Flow-through UV detector (280 nm) and pH monitor.
6. Fraction collector.

3. Methods

3.1. Loading the Immobilized Metal Ion Affinity Gel with Metal Ions and Column Packing Procedures (Agarose-Based IDA Chelating Gel)

1. Suspend the IDA gel slurry well in the bottle supplied. Pour an adequate portion into a sintered glass funnel. Wash with 10 bed vol of water to remove the alcohol preservative.
2. Add 2–3 bed vol of 50 mM metal ion solution in water. Mix well.
3. Wash with 3 bed vol of water (use buffer 3 for IDA-Cu²⁺) to remove excess metal ions.
4. Equilibrate gel with 5 bed vol of buffer 1.
5. Suspend the gel and transfer to a suction flask. Degas the gel slurry.
6. Add the gel slurry to column with the column outlet closed. Allow the gel to settle for several minutes; then open the outlet to begin flow.
7. When the desired volume of gel has been packed, insert the column adaptor. Pump buffer through the column at twice the desired end flow rate for several minutes. Readjust the column adaptor until it just touches the settled gel bed. Re-equilibrate the column at a linear flow rate (volumetric flow rate/cross-sectional area of column) of approx 30 cm/h.

3.2. Elution of Adsorbed Proteins

3.2.1. pH Gradient Elution (Discontinuous Buffer System)

1. After sample application, elute with 5 bed vol of buffer 1.
2. Change buffer to buffer 2, and elute with 5 bed vol (*see* Note 1).
3. Elute with a linear gradient of buffer 2 to buffer 3. Total gradient vol should be equal to 10–20 bed volume.
4. Finally, elute with additional buffer 3 until column effluent pH is stable and all protein has eluted (recovery should exceed 90%).

3.2.2. pH Gradient Elution (Continuous Buffer System)

1. After sample application, elute with 2.5 bed vol of buffer 1.
2. Start a linear pH gradient of buffer 1 to buffer 4. Total gradient vol should be equal to approx 15 bed vol (*see* Note 2).
3. Elute with additional buffer 4 until the column effluent pH is stable.
4. If the total quantity of added protein is not completely recovered, elute with a small vol (<5 bed vol) of buffer 4 adjusted to pH 3.5.

3.2.3. Affinity Gradient Elution with Imidazole

1. Equilibrate the column first with 5 bed vol of buffer 5. Now, equilibrate the column with 5–10 bed vol of buffer 6 (*see* Note 3).
2. After sample application, elute with 2.5 bed vol of buffer 6.
3. Now elute with a linear gradient of buffer 6 to buffer 5. Total imidazole gradient vol should equal 15 bed volumes (*see* Notes 4 and 5)

3.3. Evaluation of Metal Ion Exchange or Transfer from the Stationary Phase to the Eluted Peptide/Protein

1. Use trace quantities of radioactive metal ions (e.g., ^{65}Zn) to label the stationary phase (i.e., immobilized) metal ion pool (*see* Section 3.1.). After elution of adsorbed proteins (*see* Section 3.2.), determine the total quantity of radioactive metal ions transferred to eluted proteins from the stationary phase (by use of a gamma counter).
2. To avoid the use of radioactive metal ions, the transfer of metal ions from the stationary phase to apo (metal-free) peptides present initially in the starting sample may be determined by either of two methods of soft ionization mass spectrometry. Both electrospray ionization mass spectrometry (25) and matrix-assisted UV laser desorption time-of-flight mass spectrometry (14,23–25) have been used to detect peptide-metal ion complexes (Fig. 5). Both techniques are rapid (<10 min), sensitive (picomoles), and are able to address metal-binding stoichiometry

3.4. Column Regeneration

1. Wash with 5 bed vol of buffer 7.
2. Wash with 10 bed vol of water. The column is now ready for reloading with metal ions.

3.5. High-Performance IMAC

For example, use a TSK chelate 5PW column (7.5 mm id $\times$ 750 mm) 10- μm particle size.

1. High-performance liquid chromatography (HPLC) pump system status.
 - a. Flow rate: 1 mL/min
 - b. Upper pressure limit: 250 psi.
2. Metal ion loading.
 - a. Wash the column with 5 bed vol of water.
 - b. Inject 1 mL of 0.2M metal sulfate in water.
 - c. Wash away excess metal ion with 3 bed vol of water; for Cu(II), wash with 3 bed vol of buffer 3.
3. pH Gradient elution: phosphate buffers pH 7.0 and pH 4.0.
 - a. 100% buffer 1, 5–10 min.

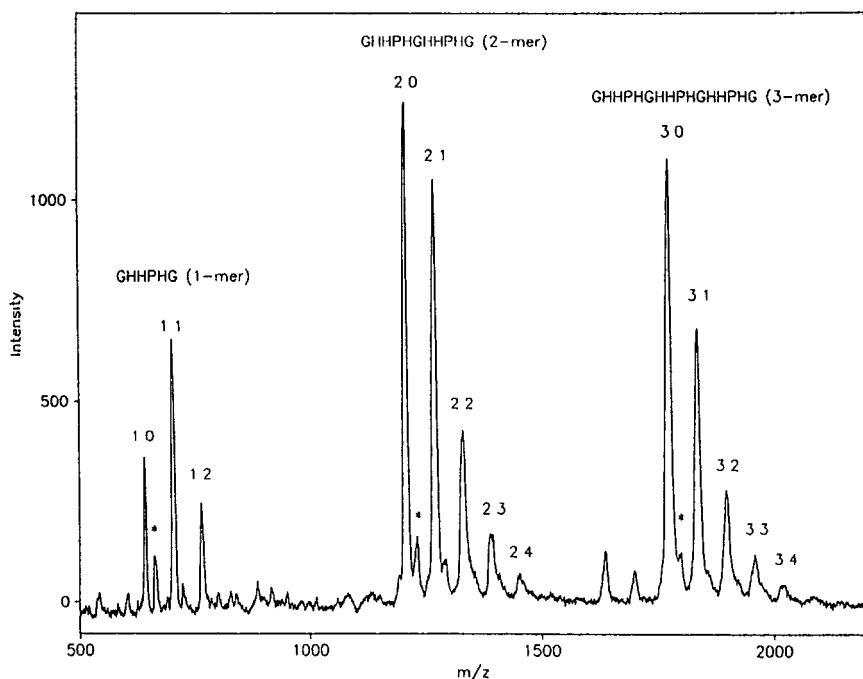


Fig. 5. Matrix-assisted UV laser desorption time-of-flight mass spectrometry (LDTOF) of a mixture of three synthetic metal-binding peptides (1-, 2-, and 3-mer) after elution from a column of immobilized GHHPHGHHHPHG (2-mer) loaded with Cu(II) ions (14). The synthetic peptide-metal ion affinity column used for metal ion transfer was prepared by coupling GHHPHGHHHPHG (2-mer) to Affi-10 (Bio-Rad, Hercules, CA). Cu(II) ions were loaded as described in Section 3.1. The column was equilibrated with 20 mM sodium phosphate buffer (pH 7.0) with 0.5M NaCl. An equimolar mixture of the three different synthetic peptides (free of bound metal ions) was passed through the column unretained. Flow-through peptide fractions were analyzed directly by LDTOF (23–25). The metal ion-free peptides GHHPHG (1-mer peak 1.0), GHHPHGHHHPHG (2-mer peak 2.0), and GHHPHGHHHPHGHHHPHG (3-mer peak 3.0) are observed along with peptides with 1, 2, 3, or 4 bound Cu(II) ions (e.g., 3.1, 3.2., 3.3., 3.4). The small peaks marked by an asterisk indicate the presence of a peptide-sodium adduct ion. A detailed description of these results is provided in ref. 14 (reproduced with permission).

- b. 0–80% buffer 4, duration 25 min.
- c. 80–100% buffer 4, duration 20 min.
- d. 100% buffer 4 until column effluent pH is constant or until all proteins have been eluted.

4. pH Gradient elution in 3M urea: phosphate buffers pH 7.5 to pH 3.8.
 - a. 100% buffer 8, 5 min.
 - b. 0–10% buffer 9, duration 10 min.
 - c. 10–80% buffer 9, duration 18 min.
 - d. 80–100% buffer 9, duration 25 min.
 - e. 100% buffer 9 until eluent pH is constant or all proteins have been eluted.
5. Imidazole gradient elution: 1 mM imidazole (5% buffer 5 in buffer 1) to 20 mM imidazole (100% buffer 5) (*see* Notes 5 and 6).
 - a. 5–10% buffer 5, immediately after sample injection, duration 10 min.
 - b. 10–100% buffer 5, duration 30 min.
 - c. 100% buffer 5 until all samples have been eluted.

4. Notes

1. The discontinuous buffer pH gradient is ideal for the pH 6.0–3.5 range. We have observed that acetate is also a stronger eluent than phosphate.
2. The phosphate buffer pH gradient is good for the pH 7.0–4.5 range. It has the advantage of UV transparency and is particularly suitable for peptide analysis (18).
3. HEPES can be used instead of phosphate for both discontinuous buffer pH gradients and in the imidazole gradient elution mode. HEPES is a weaker metal ion “stripping” buffer than phosphate. HEPES is also good for preserving the metal binding capacity of some carrier proteins such as transferrin (12).
4. For well-characterized proteins, a stepwise gradient of either pH or imidazole can be used to eliminate the need for a gradient-forming device.
5. For the affinity elution method with imidazole, the imidazole gradient actually formed must be monitored by, for example, absorbance at 230 nm or by chemical assay, if reproducible results are desired. Even when the column is presaturated with concentrated imidazole, and then equilibrated with buffers containing a substantial amount (2 mM) of imidazole, the immobilized metal ions can still bind additional imidazole when the affinity elution gradient is introduced. As a result, when simple (nonprogrammable) gradient-forming devices (typical for open-column chromatography) are used, a linear imidazole gradient is not produced; a small imidazole elution front (peak) at the beginning of the gradient will cause some proteins to elute “prematurely.” The multistep gradient described for use with the HPLC systems is designed to overcome this problem. However, we emphasize that this particular program is custom designed only for the high-performance immobilized metal ion column of given dimensions and capacity. The program must be adjusted for other column types.
6. The imidazole gradient of up to 20 mM is only an example. Quite often, much higher concentrations (up to 100 mM) are required to elute higher affinity proteins (9,10,17) (*see* Fig. 2)

- 7 To facilitate the elution of some proteins, mobile phase modifiers (additives), such as urea, ethylene glycol, detergents, and alcohols, can be included in the column equilibration and elution buffers (10,17,18,26).
- 8 To ensure reproducible column performance for several runs without a complete column regeneration in between, low concentrations of free metal ions can be included in the buffers to maintain a fully metal-charged stationary phase. This will not affect the resolution and elution position of the proteins (7,17,27). On the other hand, if free metal ions are not desired in the protein eluent, a separate metal ion scavenger column (e.g., a blank or metal-free IDA-gel or TED-gel column) may be connected in series.
- 9 Batch-type (i.e., nonchromatographic) equilibrium binding assays have been described (17,28). This is useful in screening different types of immobilized metal ion-protein interaction variables. These variables include the selection of appropriate immobilized metal ion type (17), the effects of temperature and mobile phase conditions (17,26), and free metal ions (7) on protein-immobilized metal ion interaction capacity and affinity.
10. Immobilized metal ions may be useful for the reversible site- or domain-specific immobilization of functional receptor proteins or enzymes. Data collected with the estrogen receptor protein suggest that receptor immobilization on IDA-Zn(II) (29) and IDA-Cu(II) (unpublished) does not impair receptor function (ligand-binding activity).

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CHAPTER 21

Chromatography on Hydroxyapatite

Shawn Doonan

1. Introduction

Hydroxyapatite (HT/HTP) has achieved only limited popularity as a chromatographic material for the purification of proteins. This is for a variety of reasons, including difficulties in predicting its chromatographic behavior, its relatively low capacity, and the fact that its handling properties are not ideal. For most applications, other materials with superior chromatographic properties are to be preferred. That being said, because its modes of protein adsorption and desorption are different from those of techniques such as ion-exchange chromatography (*see next paragraph*), it is sometimes possible to achieve fractionation using HT/HTP when other techniques have failed; the example described in Section 3. is a case in point. Aficionados of HT/HTP may consider this relegation of the material to "last chance" status as unjustified, but it is nonetheless the case that most practitioners of protein purification will try something else first.

HT/HTP is a crystalline form of calcium phosphate with the molecular formula $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$. The mechanism of interaction of proteins with the material has been investigated and reviewed by Gorbunoff (*1*). When the HT/HTP is equilibrated with phosphate buffer (the most common mode of use), it then appears that positively charged proteins interact nonspecifically with the general negative charge on the column produced by immobilized phosphate ions. The proteins can then be eluted by increasing the phosphate concentration, addition of a salt, such as sodium chloride, or by use of Ca^{2+} or Mg^{2+} , which complex with phosphate ions on the column and effectively neutralize it. In the case of negatively charged

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proteins, interaction is a balance between electrostatic repulsion by the negative charge on the column and specific complexation between protein carboxylic acid groups and column calcium sites; the latter effect will depend on the disposition of carboxylate side chains in the protein rather than simply on their number. Elution is effected by ions that complex more strongly with calcium than do carboxylic acids (e.g., phosphate or fluoride); Cl^- is not effective because it does not complex with calcium.

These considerations have led to the formulation of set of guidelines for the use of HT/HTP to fractionate an unknown mixture of proteins (1). The column should be equilibrated with dilute phosphate buffer (e.g., 1 mM, pH 6.8) if the objective is to retain basic proteins and some of the acidic ones from a mixture; alternatively equilibration can be in an unbuffered NaCl solution (1 mM) if the protein of interest is acidic and is to be retained on the column. Elution should then carried out by using an initial wash with 5 mM MgCl_2 (or a gradient of 1–5 mM) to remove basic proteins, a wash with 1M NaCl (or a gradient of 0.01–1M) to elute proteins with isoelectric points around neutrality, and finally a gradient of 0.1–0.3M phosphate to elute acidic proteins. Clearly for particular protein mixtures, it may be possible to use a more simple elution schedule depending on the composition, but the procedures outlined above provide a basis for initial examination of an unknown protein sample from which simpler protocols can be derived. Some typical uses of HT/HTP chromatography are given in refs. 1–5.

The particular application described in Section 3. concerns the separation of the cytosolic and mitochondrial isoenzymes of fumarase (5). These two proteins are products of the same gene and differ only in a small number (so far undefined) of amino acid residues at the N-terminus. They are not separable by conventional ion-exchange chromatography, but can be easily resolved by chromatography on HT/HTP.

2. Materials

1. HT/HTP from Bio-Rad (Watford, UK) (*see* Note 1).
2. Buffer A: 20 mM potassium phosphate, pH 7.0, containing 10 μM phenylmethanesulfonyl fluoride (PMSF) (*see* Note 2).
3. Buffer B: 350 mM potassium phosphate, pH 7.0, containing 10 μM PMSF.

3. Method

1. Suspend about 25 g of HT/HTP in 100 mL of buffer A. Stir gently, allow to settle, and then remove fines by aspiration. Repeat the last step (*see* Note 3).
2. Pack the HT/HTP into a column (2.0 $\times$ 10 cm), and wash with five bed volumes of buffer A at a flow rate of 30 mL/h (*see* Note 4).

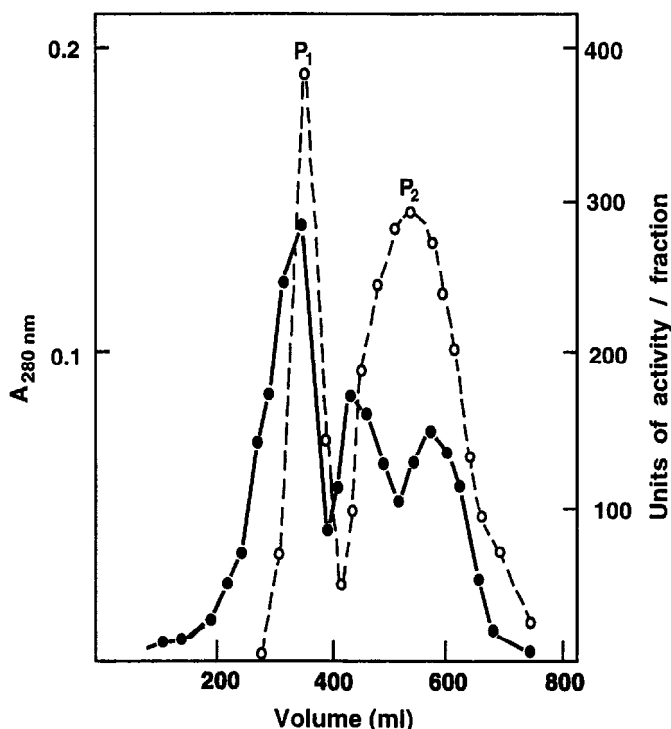


Fig. 1. Separation of cytosolic and mitochondrial fumarases from baker's yeast by HT/HTP chromatography. Full and dashed traces show absorbance at 280 nm and enzyme activity, respectively. (Reprinted from ref. 5 with kind permission from Elsevier Sci., Ltd.)

3. Apply the protein sample previously dialyzed in buffer A, and continue washing the column with the same buffer until no more protein is eluted as judged from A_{280} measurements.
4. Elute bound protein by application of a linear gradient formed from 400 mL each of buffers A and B.
5. Regenerate the column by washing with two column volumes of buffer B, followed by five column volumes of buffer A. Store in buffer A containing 0.02% sodium azide or in buffer A saturated with toluene.

The elution profile obtained is shown in Fig. 1. The source of the partially purified mixture of fumarases is described in Note 5.

4. Notes

1. HT/HTP suitable for chromatography can be prepared in the laboratory. The procedure involves mixing calcium chloride and sodium phosphate to

form the amorphous compound brushite, and then converting this to crystalline hydroxyapatite by boiling in sodium hydroxide or ammonia. A detailed procedure is given by Bernardi (6). Although this product may be superior to commercial HT/HTPs (7), it is likely that the occasional user will prefer to purchase the material.

There are several sources of supply, and the nature of the material varies from one to another. BDH (Poole, Dorset UK) markets a spheroidal HT/HTP that has similar properties to that used here, but may have greater mechanical strength. Other products consist of matrices with surface coatings of HT/HTP. For example, HA-Ultrogel from IBF (Luton, UK) consists of coated agarose. The binding capacities of these products vary from 2.5–10 mg BSA/mL of packed column. It is not possible to predict which of these commercial materials will perform best in any particular application; the important point is to stick to one product, and if following a procedure from the literature, to use the same product as described in the original work. For HPLC applications, Bio-Rad also markets a high-performance HT/HTP (Bio-Gel HPHT) packed in a stainless-steel column.

2. PMSF was included in this particular application because it was vital to the objective of the investigation to preclude the possibility of proteolysis (5). The necessity to add protease inhibitors and which inhibitors to add will depend on the problem at hand (*see* Chapter 9).

Stock solutions of PMSF (50–100 mM) should be made in dry propan-2-ol, methanol, or ethanol; in this form, they are stable for some months at 4°C. The half-life in aqueous solutions is short, and hence fresh PMSF should be added to protein solutions or to buffers daily.

3. Crystalline HT/HTPs are not very stable mechanically, and it is important to avoid fragmentation owing to rough handling and to be sure to remove fines before packing columns. Failure to do so will result in high back-pressures and low flow rates. Similarly, the flow rates of columns of HT/HTP decrease with repeated use, and when this occurs, it is necessary to repack them. Absorption of carbon dioxide and consequent formation of a plug at the top of the column is one reason for development of back-pressure; this can be avoided by boiling buffers before use to remove dissolved CO₂ or, more simply, by replacement of the top 1–2 cm of the column bed if plugging occurs.
4. Use of high flow rates, particularly with crystalline HT/HTP, will lead to compaction of the material and necessitate repacking of the column.
5. The fumarase isoenzymes were extracted from 1 kg of baker's yeast, and then partially purified by precipitation with 60% saturation ammonium sulfate, followed by ion-exchange chromatography on carboxymethyl cellulose CM-23 at pH 5.8. It was this fraction that was subjected to chroma-

tography on HT/HTP with the results shown in Fig. 1. Peak 1 eluted from the column at about 120 mM phosphate buffer and contained mainly the cytosolic fumarase (8). Peak 2 eluted at about 250 mM phosphate buffer and contained mainly the mitochondrial form. The separation was not complete, so fractions from the front of peak 1 and from the back of peak 2 were combined to give the respective pools for the two isoenzymes; the central fractions were rejected. Final purification of the separated isoenzymes was achieved by affinity chromatography using pyromellitic acid-Sepharose 4B (9). Apart from HT/HTP chromatography, the only other technique that has been found to be effective in separating fumarase isoenzymes is chromatofocusing, which was used for the isoenzymes from rat liver (10).

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CHAPTER 22

Affinity Precipitation Methods

Jane A. Irwin and Keith F. Tipton

1. Introduction

1.1. Overview

Affinity chromatography (*see* Chapter 16) is a powerful protein purification technique, that exploits the specific interaction between a biological ligand (e.g., a substrate, coenzyme, hormone, antibody, or nucleic acid) or its synthetic analog and its complementary binding site on a protein. One of the variations on this technique (*see* refs. 1–3 for reviews) was that of affinity precipitation. As in affinity chromatography, the protein binds to a specific ligand, but the latter is free in solution, rather than bound to an insoluble support. Ligand binding gives rise to the precipitation of the protein from solution, which is then followed by centrifugation. The pellet contains the protein of interest and the ligand, whereas the other components of the mixture remain in the supernatant, allowing easy separation.

There are two main approaches to affinity precipitation. The first of these is called the “bis-ligand” or “homobifunctional ligand” approach. The ligand is bifunctional, bearing two identical ligands connected by a spacer arm. If the spacer is long enough, each ligand can bind to a ligand binding site on a different protein molecule. Oligomeric proteins can bind two or more bis-ligands, with the consequent formation of crosslinked lattices. When the lattice becomes large enough, it will precipitate from solution.

The second approach to affinity precipitation differs from this in that the affinity ligand has two functions, one of which binds the target protein and a second to promote the precipitation of the aggregate. In some

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cases, precipitation is induced by the addition of a third component, e.g., a lectin or metal ion. This technique is referred to in the literature as affinity precipitation, but in contrast to the first approach, the precipitation of the complex does not occur as a direct consequence of the affinity interaction between the protein and its ligand, but as a result of adding a third component. Table 1 gives a list of examples of both "bis-ligand" and this second form of affinity precipitation.

This chapter is confined to a discussion of the "bis-ligand" form of affinity precipitation, that can be described as "true" affinity precipitation, occurring solely as a result of the direct interaction between the ligand and the protein, and not as a result of a third component of the system.

1.2. Bis-Coenzymes and Their Applications

Larsson and Mosbach were the first to synthesize a bifunctional affinity precipitation reagent (4). This affinity precipitation reagent was N_2, N_2' -adipodihydrazido-bis-(N^6 -carbonylmethyl) NAD^+ (bis- NAD^+). It consisted of two molecules of the NAD^+ derivative N^6 -carboxymethyl- NAD^+ , linked by an adipic acid dihydrazide spacer arm (Fig. 1). This compound was used to affinity precipitate purified lactate dehydrogenase from bovine heart in up to 90% yield. Bovine liver GDH and yeast alcohol dehydrogenase were also precipitated by this technique (5–7). The latter enzyme also required high ionic strengths for precipitation to occur. Further applications of bis- NAD^+ are mentioned in Table 1, and some limitations on its use are described in Note 1.

An alternative synthesis of a range of bis- NAD^+ affinity reagents has been developed (28). It involves the carbodiimide-mediated condensation of two molecules of N^6 -(2-aminoethyl) NAD^+ with different dicarboxylic acids (Fig. 2). The synthesis of bis-coenzyme derivatives has not been limited only to NAD^+ ; a synthesis of a bis-ATP derivative, N_2, N_2' -adipodihydrazido-bis-(N^6 -carbonylmethyl)ATP (10) is described here.

1.3. Structural and Kinetic Requirements for Affinity Precipitation with Bis-Coenzymes

The technique of bis-ligand affinity precipitation with bis- NAD^+ only works under certain conditions (29), which may be summarized as follows:

1. The enzyme of interest has to contain more than one coenzyme binding site.
2. The bis-ligand has to have a strong affinity for the enzyme

Table 1
Published Examples of Protein Purification by Affinity Precipitation

		Bis-ligands ^a		
Protein	Bis-ligand			Refs
LDH	Bis-NAD ⁺			4-8
GDH	Bis-NAD ⁺			5-7,9
YADH	Bis-NAD ⁺			5,7
Isocitrate dehydrogenase	Bis-NAD ⁺			7
PFK	Bis-ATP			10
LDH (rabbit)	Bis-Cibacron Blue F3G-A			11,12
Bovine serum albumin	Bis-Cibacron Blue F3G-A			11,12
LDH (rabbit)	Methoxylated <i>p</i> -sulfonated isomer of Procion Blue H-B			13,14
GDH (recombinant)	EGTA (Zn) ₂			15
Human hemoglobin, sperm whale myoglobin	Cu(II) ₂ EGTA, Cu(II) ₂ polyethylene glycol-(iminodiacetic acid) ₂			16
Hetero-bifunctional ligands ^b				
Protein	Ligand	Carrier	Precipitant	Refs
Trypsin	<i>p</i> -Aminobenzamidine	<i>N</i> -acryloyl-aminobenzoic acid	Low pH	17
Trypsin	Soybean trypsin inhibitor	Chitosan	High pH	18
Wheat germ agglutinin	<i>N</i> -acetyl-D-glucosamine	Chitosan	High pH	19
Protein A	IgG	Hydroxymethyl cellulose	Low pH	20
		Acetate succinate		
Recombinant protein A	IgG	Eudragit S100	Low pH	21
IgG	Protein A	Galactomannan	Potassium borate	22
LDH	Cibacron Blue	Dextran	Concanavalin A	23
Trypsin	Soybean trypsin inhibitor	Alginate	Ca ²⁺ ions	24
Endo-polygalacturonase	Alginate	Alginate	Ca ²⁺ , low pH	25
Protein A	IgG	<i>N</i> -isopropylacrylamide polymer	Temperature increase	26
Trypsin	<i>p</i> -Aminobenzamidine	<i>N</i> -isopropylacrylamide polymer	Temperature increase	27

^aThese do not require a second component to effect affinity precipitation

^bThese require a ligand carrier and a third component to promote the precipitation of the ligand-protein complex

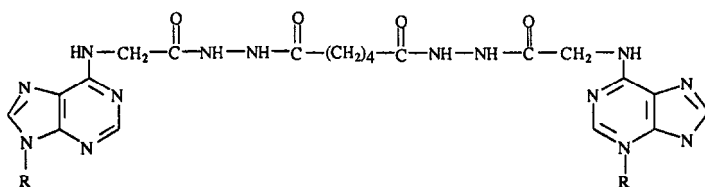


Fig. 1. Structure of bis-NAD⁺, first synthesized by Larsson and Mosbach (4), and used as an affinity ligand for several dehydrogenases. R represents nicotinamide mononucleotide.

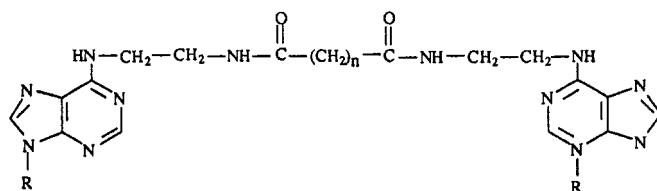


Fig. 2. General structure of bis-NAD⁺ derivatives, based on *N*⁶-(2-aminoethyl)-NAD⁺ as a starting material. These include *N,N'*-bis(*N*⁶-ethylene-NAD⁺) glutaramide ($n = 3$); *N,N'*-bis(*N*⁶-ethylene-NAD⁺) adipamide ($n = 4$), and *N,N'*-bis(*N*⁶-ethylene-NAD⁺) pimelamide ($n = 5$). R represents nicotinamide mononucleotide phosphoribose.

3. The spacer connecting the two ligands has to be long enough to bridge the distance between two ligand binding sites on two different enzyme molecules. In the case of *N*₂, *N*₂'-adipodihydrazido-bis-(*N*⁶-carbonylmethyl)-NAD⁺ (bis-NAD⁺), the spacer length was approx 1.7 nm, which permitted easy simultaneous access for a molecule of bis-NAD⁺ to two different molecules of a dehydrogenase.

A further limitation is the ratio of coenzyme derivative to enzyme subunit. If this is low, a lattice will not form since there are not enough crosslinks; if it is too high, each dehydrogenase subunit can be occupied by a separate molecule of bis-NAD⁺ and no crosslinks will form. Maximum crosslinking occurs at an optimum ratio of NAD⁺ eq/enzyme subunit (assuming two NAD⁺ Eq/bis-NAD⁺), which in the case of tetramers has been found to be approximately unity. This can vary; for example, the hexameric mammalian glutamate dehydrogenase (GDH) precipitated over a broad range of NAD⁺ equivalents to active site ratios (0.3:10), with up to 70% precipitation occurring at a ratio as low as 0.16, although

this probably involves the existence of higher GDH polymers (5). The ratio of approximately unity for tetramers is similar to the behavior observed in immunoprecipitation, in which two antigen molecules/antibody give rise to the optimum precipitation of immune complexes (30).

Adding bis-NAD⁺ alone to a crude extract containing many different dehydrogenases will not in itself cause specific affinity precipitation, since it is a "general ligand," binding to most NAD⁺-dependent dehydrogenases. In addition to being nonspecific, bis-NAD⁺ forms a weak binary complex with several dehydrogenases, e.g., lactate and horse liver alcohol dehydrogenases. The addition of a substrate analog (usually a competitive inhibitor, relative to the enzyme's second substrate) strengthens the binding interaction. Since NADH binds, on average, one order of magnitude more tightly to the active sites of many dehydrogenases than NAD⁺, it displaces bis-NAD⁺ and is commonly used to dissolve crosslinked aggregates.

The specificity of bis-NAD⁺ of affinity precipitation for any given dehydrogenase is conferred by a property described by O'Carra as the "locking-on" effect (31). This was originally developed to increase the strength of enzyme binding to an immobilized ligand by adding analogs of substrates specific to the enzyme under study. In the case of lactate dehydrogenase, for example, the strength of adsorption to an immobilized NAD⁺ derivative was increased by adding oxalate, a structural analog of lactate, to the irrigating buffer. The enzyme was eluted by simply leaving out the oxalate.

This "locking-on" property does not occur with all enzymes. It is confined to those with sequential mechanisms (*see* refs. 3 and 31 for more details). Most coenzyme-dependent enzymes, including the majority of dehydrogenases, have ordered sequential kinetic mechanisms, in which the coenzyme binds before the second substrate. If this is the case, the "locking-on" effect occurs. The addition of an unreactive, competitive inhibitor of the second substrate will displace the coenzyme binding equilibrium by ternary complex formation (sometimes described as an "abortive" complex) and thereby increase the strength of binding. In affinity precipitation, the bis-coenzyme is "locked" into place by the substrate analog, provided it is saturating. Some enhancement of bis-ligand binding can occur for an enzyme with a random sequential mechanism, provided that the equilibrium of the reaction under the conditions employed in affinity precipitation is such that the binding of the substrate analog then favors coenzyme binding.

The identity of the substrate analog will determine which dehydrogenase is precipitated from a crude extract. For example, adding bis-NAD⁺ and glutarate will lead to the precipitation of GDH from a crude extract; addition of pyrazole or oxalate would favor alcohol dehydrogenase or lactate dehydrogenase precipitation, respectively. This is the key to making the technique specific for one enzyme.

In the case of the only reported bis-ATP derivative, which was used to affinity precipitate bovine heart phosphofructokinase, precipitation was not accomplished with the aid of a second substrate analog, but by the use of the allosteric inhibitor citrate. ATP is both a substrate and an allosteric inhibitor of this enzyme, and this inhibition is potentiated by citrate (10)

1.4. Affinity Precipitation with Other Bifunctional Ligands

Examples of bis-ligand affinity precipitation in the literature are few, and the technique has not gained widespread popularity since it has only been found to work with a few proteins. In order to extend its use, other ligands have been synthesized, but these have been limited so far to triazine dyes and immobilized metal ions. Triazine dyes have been widely used as pseudo-affinity ligands (*see* Chapter 17 for a discussion of dye-ligand chromatography). A bis-derivative of Cibacron Blue F3GA was found to precipitate bovine serum albumin and lactate dehydrogenase (11), and a monofunctional synthetic analog of Procion Blue H-B was found to precipitate rabbit muscle lactate dehydrogenase selectively (13,14). The synthesis of this analog is reported in ref. 13, and the method for large-scale LDH purification is described in ref. 14. Only two examples of metal ion affinity precipitation have been reported to date, one of which consisted of the precipitation of recombinant galactose dehydrogenase containing a pentahistidine affinity tail by a (Zn)₂-EGTA chelate (15). In the second case, myoglobin and hemoglobin were precipitated by an EGTA-Cu(II) chelate or, alternatively, by a bis-ligand consisting of Cu(II) cations chelated by molecules of iminodiacetic acid immobilized on each end of a molecule of polyethylene glycol (17).

2. Materials

2.1. Synthesis of Coenzyme Derivatives

2.1.1. Synthesis of Bis-NAD⁺

This compound is commercially obtainable, but is quite expensive (\$90.00 for 5 mg, Sigma, St. Louis, MO, 1995). The starting material is

N^6 -carboxymethyl- NAD^+ . This is also obtainable from Sigma, but may be readily synthesized. The following reagents are required.

2.1.1.1. N^6 -CARBOXYMETHYL- NAD^+

1. NAD^+ (98%, free acid).
2. Iodoacetic acid.
3. 2M LiOH.
4. 96% (w/v) Ethanol.
5. Sodium dithionite.
6. 0.24M $NaHCO_3$.
7. Yeast alcohol dehydrogenase (crystalline, Sigma or Boehringer Mannheim, Mannheim, Germany).
8. A sintered funnel (fairly large, at least 10 cm in diameter).
9. AG 1 $\times$ 2 anion-exchange resin (200–400 mesh, Cl^- form, Bio-Rad, Hercules, CA).
10. $CaCl_2$.

2.1.1.2. Bis- NAD^+

1. Adipic acid dihydrazide dichloride (Sigma).
2. 2M NH_4OH .
3. NH_4HCO_3 .
4. DEAE-cellulose (Whatman DE52, equilibrated with 10 column volumes of 1M NH_4HCO_3 , and washed with water).

2.1.2. N_2, N_2' -Adipodihydrazido-Bis-(N^6 -Carboxymethyl-ATP)

As for N^6 -carboxymethyl- NAD^+ , except that NAD^+ is replaced by ATP and LiCl is used in place of $CaCl_2$. Neither this ATP derivative nor N^6 -carboxymethyl-ATP is commercially available.

2.1.3. Other Bis- NAD^+ Derivatives

1. NAD^+ (Sigma, Boehringer; 98% free acid)
2. Ethyleneimine (Serva, Heidelberg, Germany). **Caution: This is toxic and carcinogenic.**
3. 70% $HClO_4$.
4. 96% Ethanol.
5. LiCl.
6. Bio-Rex 70 cation exchange resin (100–200 mesh, Na^+ form, Bio-Rad).
7. 1M LiOH.
8. 1M HCl.

Bio-Rex 70 can be converted to the H^+ form by washing it exhaustively with 0.5–1.0M HCl. The absence of Na^+ ions can be tested by

flame photometry. The column is then washed with 1 mM HCl, pH 3.0, and equilibration is checked by pH and conductivity measurements.

For bis-NAD⁺ synthesis: Glutaric, adipic, and pimelic acid (all available from Sigma) have been found to serve as satisfactory, water-soluble spacers. Also required: 1-ethyl-3-(3-dimethylaminopropyl)-carbodiimide hydrochloride (EDC), 5M NaOH, 1M HCl, DE52 cellulose, equilibrated with 0.5M ammonium acetate, pH 7.0, to convert it to the acetate form; this is then equilibrated for chromatography by washing it with 10 column volumes of 0.05M ammonium acetate, pH 7.0.

In addition to this, a rotary evaporator and chromatographic columns (dimensions variable) are needed for all described syntheses. Three different TLC solvent systems have been used for monitoring the products of these syntheses:

System A: (NH₄)₂SO₄· 0.1M potassium phosphate, pH 6.8 1-propanol (60:100:2 [w/v/v]).

System B: isobutyric acid: 1M aqueous NH₃ (5:3 [v/v]), saturated with Na₂EDTA.

System C: isobutyric acid: water: 25% aqueous NH₃ (66.33:1 [v/v/v]).

Plates: aluminum-backed, silica gel 60, fluorescent indicator F₂₅₄, layer thickness 0.2 mm (Merck). A source of fluorescent light (wavelength 254 nm) can be used to visualize the spots, which appear purple on a green background.

HPLC system: reverse-phase, μ Bondapak C-18 column (Waters; 3.9 $\times$ 300 mm), equilibrated with 0.1M potassium dihydrogen phosphate, pH 6.0, containing 10% (v/v) methanol for separating monosubstituted NAD⁺ derivatives and 20% methanol for separation of bis-NAD⁺ derivatives. All solutions used in HPLC must be degassed using a vacuum pump and filtered with a 0.22- μ m filter to exclude particulate matter and avoid air bubbles. In addition, all samples must be centrifuged for 2 min in a minifuge to pellet particles before application to the column. The absorbance should be monitored at 254 nm.

2.2. Pilot Affinity Precipitation Studies

1. A source of the protein of interest, e.g., a crude tissue extract supernatant, or a commercially available preparation of the protein to ensure the reagent is effective.
2. The bis-ligand.
3. Stock solutions of a substrate analog, e.g., 560 mM oxalate for lactate dehydrogenase, 560 mM pyrazole for alcohol dehydrogenase, and 700 mM glutarate for GDH.

4. 0.4M Potassium phosphate buffer (62.4 g/L NaH_2PO_4 , titrated to pH 7.4 with 5M KOH), for some applications.
5. 1.5-mL Polypropylene minifuge tubes.
6. Assay reagents for enzymes (*see, e.g., ref. 32*).

2.3. Enzyme Purification

Apart from the reagents and apparatus mentioned in Section 2.2., reagents and apparatus for electrophoresis, e.g., PAGE/SDS-PAGE are needed to check the purity of the affinity-precipitated protein (33). In addition, reagents for a protein assay (e.g., by the Lowry or Bradford method) are required to determine the protein concentration.

For the separation of LDH isoenzymes (*see* Section 3.3.3.), materials for starch gel electrophoresis are appropriate. These include hydrolyzed potato starch (Sigma), electrophoresis-grade Trizma base to buffer the gel, and grade III NAD^+ (Sigma). The activity stain for LDH contains 1.21 g Tris, 7.72 mL 70% Na DL-lactate (8 g), 50 mg nitroblue tetrazolium, 50 mg grade III NAD^+ , and 4 mg phenazine methosulfate. (**Caution: Nitroblue tetrazolium and phenazine methosulfate are toxic and the latter is light sensitive**). Make the volume up to 200 mL with water, and adjust the pH to 7.1 with 6M HCl.

3. Methods

3.1. Synthesis of Coenzyme Derivatives

3.1.1. Bis- NAD^+

The synthesis of N^6 -carboxymethyl- NAD^+ is described in ref. 34, but some modifications are given here. The synthesis of bis- NAD^+ is a modified and more detailed version of that described in ref. 4.

1. Dissolve 9 g of fresh iodoacetic acid in approx 1 mL of water, neutralize the solution with 2M LiOH, and add 3 g of NAD^+ . Adjust the pH to 6.5 with 2M HCl (the total volume should be about 30 mL), and leave the solution in darkness for 7 d at room temperature (approx 20°C) or, alternatively, at 37°C for 2 d. The pH should be checked daily (or every 4–6 h at the higher temperature), and readjusted to 6.5 with 2M HCl as required. The progress of the reaction should also be followed by TLC and/or HPLC.
2. When the reaction is complete, adjust the pH to 3.0 with 6M HCl, and add 2 vol of 96% (v/v) ethanol. This gives a milky, pink-tinged suspension, which precipitates on the addition of a further 10 vol of cold 96% ethanol. (The water/ethanol ratio is important; using a wet vessel can give rise to the formation of a brown, sticky substance that is water-soluble. This also applies to the corresponding ATP derivative.) Filter the crude $N(1)$ -carboxymethyl- NAD^+

- on a sintered funnel, wash with ethanol and diethyl ether, and dry under vacuum (average yield 2.9 g). Store the product at -20°C under vacuum.
3. Dissolve the crude *N*(1)-carboxymethyl- NAD^{+} in 0.24M NaHCO_3 (90 mL), which gives a pale orange solution, and adjust the pH to 8.5 with 1M NaOH . Deoxygenate the solution by bubbling N_2 gas through it for 2 min, add 1.5 g of sodium dithionite, and leave the solution in the dark until maximum reduction is achieved. This depends on the dithionite—monitor the reaction by taking samples, diluting them 1:50 or 1:100, and measuring the increase in A_{340} .
 4. Terminate the reaction by stirring vigorously for 10 min to oxygenate the solution, and then bubble N_2 gas through for 2 min. Adjust the pH to 11.5 with 1M NaOH , and leave in a water bath at 75°C to allow the Dimroth rearrangement from the *N*(1) to the *N*⁶-substituted derivative to occur. (See ref. 35 for a reaction mechanism.) Monitor the rearrangement by HPLC or TLC. Cool the reaction mixture to room temperature, and add 6 mL of 2M Tris and 1.5 mL of redistilled acetaldehyde. Adjust the pH to 7.5 with 1M HCl , and add 8–24 mg of yeast alcohol dehydrogenase (2500–7500 U). Monitor the reaction at 340 nm as in step 3. When a minimum A_{340} is reached, add 1 vol of 96% ethanol, and pour the milky flocculant precipitate into 10 vol of vigorously stirred 96% ethanol. Leave for 30 min (or overnight, if desired), and collect the precipitate by filtration. The crude *N*⁶-carboxymethyl- NAD^{+} can be stored at -20°C under vacuum for up to a month (yield approx 2.6 g).
 5. The product from step 4 can be purified by dissolving the crude powder (2.6 g) in 30 mL water, adjusting the pH to 8.0 with 1M LiOH , and applying this solution to a column of Dowex AG 1X2 (200–400 mesh, Cl^{-} form, 4×10 cm). Wash the column beforehand with 1 L of 3M HCl , followed by exhaustive washing with at least 20 L of water until neutral pH is reached. After applying the coenzyme solution, wash the column with 0.5 L of water, followed by 1 L of 5 mM CaCl_2 , until the pH of the effluent is 2.8. Apply a linear gradient (2×1 L) from 5 mM CaCl_2 , pH 2.7, to 50 mM CaCl_2 , pH 2.0. Collect 10–20 mL fractions, and monitor the absorbance at 260 nm. The composition of the fractions can be monitored by TLC R_f values: system A, NAD^{+} , 0.41, *N*(1)-carboxymethyl- NAD^{+} , 0.31, *N*⁶-carboxymethyl NAD^{+} , 0.22; system B, NAD^{+} , 0.44, *N*(1)-carboxymethyl- NAD^{+} , 0.27, *N*⁶-carboxymethyl NAD^{+} , 0.22. The values can vary slightly with changes in the solvent composition over time (see Note 2). Pool the fractions containing the *N*⁶ derivative, adjust the pH to 7.0 with 2M $\text{Ca}(\text{OH})_2$, and concentrate to 5–10 mL by rotary evaporation at 40°C . Precipitate with 96% ethanol as in step (4), and dry under vacuum. This gives a pale yellow compound (yield 0.82 g, 25%). $\epsilon = 19,300/\text{M}/\text{cm}$ at 266 nm

6. To synthesize bis-NAD⁺, dissolve 0.82 g of purified *N*⁶-carboxymethyl-NAD⁺ and 105 mg of adipic acid dihydrazide dihydrochloride in 20 mL water to give a brownish solution. Adjust the pH to 4.6 with 1M HCl, and then add 0.5M EDC at 0°C in 15 100-μL aliquots over a period of 35 min. Monitor the pH, and readjust it to 4.6 before each addition. Add water (2 L) to dilute the solution 100-fold, adjust the pH to 8.0 with 2M NH₄HCO₃, apply the solution to a DE52 column (2.5 × 30 cm) equilibrated with 1M NH₄HCO₃, pH 8.0, and then wash with water. Wash the column with water until the *A*₂₆₀ is < 0.1, and apply a linear gradient (2 × 1 L) from 0–0.25M NH₄HCO₃, pH 8.0. Monitor the *R*_f values of the fractions by TLC (*R*_f of bis-NAD⁺: system A, 0.09; system B, 0.05; see Note 2). The yield from pure *N*⁶-carboxymethyl-NAD⁺ is approx 14%. ε = 42,800/M/cm at 266 nm.

3.1.2. Bis-ATP (36,37)

1. Dissolve 3.4 g of fresh iodoacetic acid in 1 mL of water, and adjust the pH to 7.0 with 2M LiOH. Dissolve 1.16 g of ATP in the solution, and leave it at pH 6.5 in the dark for 4 d at 30°C, adjusting the pH to 6.5 every day. Monitor the reaction by HPLC and/or TLC, or follow the decrease in absorbance at 640 nm. When the reaction is complete, as judged by TLC, HPLC, or absorbance change, add 1 vol of cold 96% ethanol to give a milky suspension, and add this to 6–8 vol of stirring absolute ethanol at 0°C. Filter, and wash as for the corresponding NAD⁺ derivative. (Typical yield 82–87%, about 1.4 g.)
2. Dissolve 1.4 g of product in 30 mL of water to give a reddish solution. Adjust the pH to 8.5 with 1M LiOH and incubate at 90°C for 100 min, with pH adjustment every 20 min. Monitor the reaction over this time by TLC or HPLC. The solution can be cooled on ice and stored overnight at 4°C, if required. HPLC retention times: ATP, 4.5 min, *N*(1)-carboxymethyl-ATP 3.1 min, *N*⁶-carboxymethyl-ATP, 4.1 min.
3. Adjust the pH of the solution at 20°C to 2.75 with 6M HCl, and apply it to an AG 1 × 2 column (200–400 mesh, Cl[−] form, 4 × 9 cm), equilibrated with water. Wash the column with 1 L of 0.3M LiCl, pH 2.75, until the *A*₂₆₀ is < 1.0, and then apply a 2 × 1 L linear gradient, 0.3M LiCl, pH 2.75–0.5M LiCl, pH 2.0. Monitor the *R*_f values of the fractions by TLC as previously. (*R*_f values: system A, ATP, 0.54, *N*(1)-carboxymethyl-ATP, 0.73, *N*⁶-carboxymethyl-ATP, 0.54. system B, ATP, 0.35, *N*(1)-carboxymethyl-ATP, 0.27, *N*⁶-carboxymethyl-ATP, 0.21; see Note 2.) Pool the fractions containing the *N*⁶ derivative, and adjust the pH with 2M LiOH. Yield: 32–38% from ATP. ε = 17,300/M/cm at 266 nm. This procedure should be carried out as quickly as possible, preferably within 1 d, since ATP is unstable under acidic conditions.

4. Bis-ATP: Dissolve 105 mg of adipic acid dihydrazide dihydrochloride in a solution containing 0.42 g of *N*⁶-carboxymethyl-ATP in 100 mL. Adjust the pH to 4.0 with 1M HCl, and add 3 1-mL aliquots of 1M EDC over a period of 45 min, while stirring continuously, and monitoring the pH. Dilute to 2 L with water to stop the reaction, and adjust the pH to 8.0 with 2M NH₄OH. This solution can be stored at -20°C for about a week.
5. The compound can be purified by applying the dilute solution to a DE-52 column (2.5 × 25 cm), equilibrated with 2–4 L 1M NH₄HCO₃, and then washed with 8 L water. Elute the bis-ATP with a 0–0.4M NH₄HCO₃ gradient, pH 8.0 (2 × 1 L). Collect 10–20 mL fractions and pool and freeze-dry the fractions containing bis-ATP, as judged by HPLC or by affinity precipitation of phosphofructokinase (see Section 3.3.4). Ensure that the compound is completely dry before storage at -20°C, as it is very hygroscopic. Yield: 0.06 g from 0.42 g *N*⁶-carboxymethyl-ATP (yield approx 2% from ATP). $\epsilon = 39,600/M/cm$ at 266 nm (see Note 3).

3.1.3. Other Bis-NAD⁺ Derivatives

1. The synthesis of *N*⁶-(2-aminoethyl)-NAD⁺ is carried out essentially as described in ref. 38. However, *N*(1)-(2-aminoethyl)NAD⁺ can be rearranged to the *N*⁶-substituted derivative without removal of the unreacted NAD⁺ by ion-exchange chromatography (39), and the following procedure is a modification of this. Dissolve crude *N*(1)-(2-aminoethyl)-NAD⁺ (contains approx 70% *N*(1)-(2-aminoethyl)NAD⁺; 3.05 g, 4.3 mmol) in 850 mL of water. Adjust the pH to 6.5 with 1M LiOH. Place the solution in a water bath at 50°C, and allow the rearrangement to proceed for 6–7 h. Monitor the reaction by HPLC/TLC/change in absorbance maximum over this time. The wavelength of maximum absorbance should change from 260 nm to at least 264 nm, since the λ_{max} for *N*⁶-(2-aminoethyl)NAD⁺ is 266 nm. Terminate the reaction by cooling the solution to room temperature, and adjust the pH to 5.5 with 1M HCl. Concentrate the solution by rotary evaporation at 35°C to approx 20 mL. The compound is stable in concentrated solution for at least 1 wk at 4°C.
2. Apply the concentrated reaction mixture to a Bio-Rex 70 cation-exchange column (H⁺ form, 2.7 × 73 cm), pre-equilibrated with 1 mM HCl, pH 3.0. Wash the column with this solution, and collect fractions of 10–20 mL, in a total volume of approx 4 L. Monitor the A_{260} as an indicator of coenzyme concentration. The last peak to elute contains the *N*⁶-(2-aminoethyl)NAD⁺. Some overlap may occur with the previous peak, containing the tricyclic NAD⁺ derivative 1, *N*⁶-ethanoadenine-NAD⁺. Pool and rotary evaporate the fractions containing the desired compound to approx 5 mL. Lyophilization gives a fluffy pale yellow compound, stable for at least 12 mo at -20°C in an air-tight container. $\epsilon = 21,600/M/cm$ at 266 nm. HPLC retention times

- NAD⁺, 6.3 min; *N*(1)-(2-aminoethyl)NAD⁺, 4.4 min; *N*⁶-(2-aminoethyl)NAD⁺, 5.7 min; 1, *N*⁶-ethanoadenine-NAD⁺, 4.7 min. TLC *R_f* values: system C, NAD⁺ 0.28, *N*(1)-(2-aminoethyl)NAD⁺, 0.10, *N*⁶-(2-aminoethyl)-NAD⁺, 0.16, 1, *N*⁶-ethanoadenine-NAD⁺, 0.08 (28)
3. Dissolve the spacer (glutaric, adipic, or pimelic acid) in water to give a 50-mM solution. Adjust the pH to 7.0 with 5M NaOH, and add 700 μL of this solution to 50 mg (67 μmol) of *N*⁶-(2-aminoethyl)NAD⁺ to give a 2:1 molar ratio of coenzyme to spacer arm. Adjust the pH from approx 3.0–5.4 with 5M NaOH. Add solid EDC (34 mg, 335 μmol) to the solution in six to seven portions over 20 min, while monitoring the reaction by TLC/HPLC. The pH tends to rise, readjust it to 5.5–6.0 when it reaches a value of 6.9–7.0. Stop the reaction after 40 min by freezing the reaction mixture at –20°C. TLC indicates that at least four side products are formed in addition to bis-NAD⁺
 4. To purify the bis-NAD⁺, dilute the sample twofold with 50 mM ammonium acetate, pH 7.0, and apply it to a Whatman DE-52 anion-exchange column (2.3 × 16 cm), equilibrated as described in Section 2.1.3. Wash the column with this buffer. The last major peak contains the bis-NAD⁺ derivative. This can be checked by TLC, HPLC, or by the ability of the fractions to cause affinity precipitation of GDH. Pool and rotary evaporate the fractions containing bis-NAD⁺ to 3–5 mL, and freeze-dry these (stable for at least 4 mo at 4°C or –20°C). The compound can also be stored for approx 1 mo in solution at 4°C, although bacterial contamination can result. TLC *R_f* values (system C) for bis-NAD⁺ derivatives: *N,N'*-bis(*N*⁶-ethylene-NAD⁺)glutaramide, 0.09, *N,N'*-bis(*N*⁶-ethylene-NAD⁺) adipamide, 0.12, *N,N'*-bis(*N*⁶-ethylene-NAD⁺)pimelamide, 0.15. $\epsilon = 44,200 \pm 1400/\text{M}/\text{cm}$. HPLC retention times (in 0.1M potassium phosphate buffer, pH 6.0, containing 20% methanol) are 7.8, 11.2, and 13 min for the compounds, respectively.

3.2. Pilot Precipitation Study

1. Take a source of the enzyme of interest (e.g., a crude supernatant), and assay it for enzyme activity (see ref. 32 for a wide range of assays) to determine the specific activity.
2. Calculate the approximate concentration of enzyme based on the specific activity.
3. Add 100–200 μL of the enzyme solution to a series of minifuge tubes. If the study is to be carried out with purified enzyme, dilute a stock solution of 400 mM potassium phosphate buffer, pH 7.4, to give a final concentration of 20 mM. Set up all samples in duplicate, including two control tubes to which water has been added in place of the coenzyme derivative. Set up duplicate controls in addition to these, which contain no bis-ligand or substrate analog. This enables the percent inhibition owing to the presence of the substrate analog to be calculated.

4. Calculate the concentration of a stock solution of bis-coenzyme spectrophotometrically, using the extinction coefficients given in Section 3.1
5. Add bis-NAD⁺ (or bis-ATP) to give a range of coenzyme Eq/enzyme subunit from approx 0.1–20, calculating the concentration from the extinction coefficient
6. Add the appropriate substrate analog, and leave at 4°C for at least 30 min (or overnight, if required).
7. Centrifuge in a minifuge for 5 min, and assay the supernatant. Calculate the precipitation as a percentage of the activity remaining in the control sample. The tube giving the minimal residual activity shows the maximum affinity precipitation. This indicates the appropriate concentration of bis-coenzyme to be added to obtain maximum affinity precipitation.

3.3. Enzyme Purification Protocols

The following are a selection of protocols for purifying enzymes with bis-ligands.

3.3.1. Purification of Yeast Alcohol Dehydrogenase (YADH; EC 1.1.1.1)

1. The yeast lysis is carried out according to a modification of the procedure in ref. 40. Crumble 40 g of fresh baker's yeast into 21 mL of toluene, preheated to 45°C in a water bath. (**Caution: Keep the vessel covered, or incubate in a fume cupboard.**) Allow the mixture to liquefy over a period of 90 min with occasional stirring. (see Note 4)
2. Leave the mixture at room temperature for 3 h, and add 42 mL of 1 mM Na₂EDTA as a protease inhibitor. Stir for 2 h at 4°C, and leave the mixture at that temperature overnight.
3. Centrifuge the lysate at 47,800g for 10 min, remove the top fatty layer by aspiration, and retain the supernatant. Add solid (NH₄)₂SO₄ slowly, until 60% saturation is reached (7.22 g to 20 mL), and centrifuge for 10 min at 47,800g. The ammonium precipitation step removes some proteins that would otherwise precipitate on addition of bis-NAD⁺ and gives a purification of approx three- to fourfold. Dissolve the pellet in approx 3 mL of 20 mM potassium phosphate buffer, pH 7.4, containing 1 mM Na₂EDTA and dialyze against two changes of 2 L of this buffer to remove the salt (alternatively, gel filter on a column of Sephadex G-25, at least 1 × 50-cm long).
4. Carry out a pilot precipitation of this supernatant. This can be performed before or after removing the (NH₄)₂SO₄. However, if the (NH₄)₂SO₄ has been removed, it must be added back to a concentration of at least 0.5M, since it is required to promote precipitation. Dilute the supernatant 1:4 with buffer, and add pyrazole to a final concentration of 28 mM. A suitable range of concentrations of NAD⁺ equivalents is 0–20 μM. Allow precipitation to occur for at least 3 h (preferably overnight).

5. Assay the supernatants as described in ref. 32, diluting the enzyme solution if necessary, and determine which concentration of bis-NAD⁺ gives maximum precipitation. Add the appropriate concentration of bis-NAD⁺ to whatever vol of supernatant is required, along with (NH₄)₂SO₄ and pyrazole. Allow precipitation to occur as described above, and centrifuge the precipitate for 10 min at 12,000g to pellet the crosslinked enzyme aggregate.
6. Resolubilize the pellet by adding 0.6 mM NADH in 20 mM potassium phosphate buffer, pH 7.4 (the volume can be varied, depending on what final enzyme concentration is required). Allow resolubilization to proceed for at least 3 h, preferably 12 h, at 4°C on a rocking tray or with occasional stirring (see Note 4). The enzyme should be essentially homogeneous, as shown by SDS-PAGE on a 12.5% gel.

3.3.2. Purification of GDH (EC 1.4.1.3) from Beef and Rat Liver (9)

1. If beef liver is used, transport this from the abattoir on ice.
2. Homogenize the tissue, and carry out ammonium sulfate precipitation and DEAE-cellulose chromatography as described in ref. 41
3. Pool the fractions from the DE-52 column that contain GDH activity (32), and concentrate them to a volume of approx 10 mL by ultrafiltration through an Amicon XM-50 membrane. Dialyze the solution overnight at 4°C against 200 mL of sodium or potassium phosphate buffer, pH 7.4, with at least one change of buffer.
4. Carry out a pilot precipitation as described in Section 3.2. Incubate 100-μL samples of the dialyzed solution in the presence of 12.7 μL of 0.7M glutarate, pH 7.0, and bis-NAD⁺ for 15 min. Assume a subunit *M_r* for GDH of 56,700. The optimum ratio of NAD⁺ Eq/enzyme subunit may vary with the preparation used; for example, with a preparation of beef liver enzyme that had a specific activity of 2.7 U/mg, about half the activity was precipitated at a ratio of 2 NAD⁺ Eq/GDH subunit. However, a preparation with the lower specific activity of 0.8 U/mg required approx 8 NAD⁺ Eq/GDH subunit to precipitate half the enzyme activity. The precipitation yield also depends on the protein concentration.
5. Take the dialyzed solution, and add glutarate to a final concentration of 79 mM. Add the appropriate amount of bis-NAD⁺ to achieve maximum precipitation. Keep the mixture on ice overnight, and centrifuge at 10,000g for 15 min. Redissolve the pellet by stirring for 6 h in 1 mL of 20 mM sodium or potassium phosphate buffer, pH 7.4, containing 0.6 mM NADH. Dialyze the redissolved pellet against 200 mL of the same buffer as in step 3. A purification summary is given in Table 2. This method appears to avoid some of the proteolysis encountered in more conventional methods of purifying this enzyme. Although affinity precipitation was obtained with

Table 2
Purification of Beef and Rat Liver GDH^a

Step	Volume, mL		Total protein, mg		Total activity, U		Specific activity, U/mg		Purification factor		Yield, %	
	Beef	Rat	Beef	Rat	Beef	Rat	Beef	Rat	Beef	Rat	Beef	Rat
Homogenate	370	90	8580	2720	1920	760	0.2	0.3	—	—	100	100
Resuspended (NH ₄) ₂ SO ₄ precipitate	75	35	2010	980	1640	505	0.3	0.5	4	2	86	67
DEAE-cellulose chromatography	8.4	9.8	139	130	408	195	2.9	1.5	13	5	21	26
Affinity precipitation	1.2	1.0	9.4	5.1	376	190	40	37	180	140	20	25

^aReproduced from Graham et al. (9). The first three steps were carried out according to the method of Mc Carthy et al. (41). The purification of the enzyme from beef liver used 46 g and that from rat used 10 g liver.

preparations that had not been subjected to chromatography on DEAE-cellulose, the yield was much lower and the precipitated material was not completely pure.

3.3.3. Separation of Lactate Dehydrogenase (LDH; EC 1.1.1.27) Isoenzymes

This technique is based on the principle that both M and H isoenzyme subunits of LDH form abortive complexes with NAD^+ and oxalate, whereas the H form gives rise to abortive ternary complexes with NAD^+ and oxamate. In a mixture of H and M subunits, ternary complexes and, thus, crosslinks will form only with the H subunits, and tetramers with predominantly M subunits will tend not to be precipitated.

1. Select a source of LDH isoenzymes (e.g., a crude supernatant or Sigma Type X bovine muscle LDH, which contains varying proportions of all five isoenzymes). Determine the specific activity, and hence, estimate the concentration of the enzyme (32)
2. Carry out a pilot precipitation as described in Section 3.2. The precipitation solution should contain 20 mM potassium phosphate buffer, pH 7.4, and potassium oxamate to a final concentration of 20 mM. Optimum affinity precipitation should be obtained with a ratio of approx 1 NAD^+ Eq/LDH subunit. Allow precipitation to occur overnight at 4°C. Centrifuge for 10 min and retain the supernatant for further analysis.
3. Resolubilize the precipitated enzyme by adding 50 μL of 0.6 mM NADH in the above buffer and incubate overnight. Centrifuge the samples for 10 min at 12,000g to pellet any remaining crosslinked enzyme
4. The isoenzymic composition of the pellet and supernatant can be analyzed by starch gel electrophoresis. This is carried out by the method described in ref. 42. Make an 11% starch gel. Apply 10- μL samples of the enzyme solution to the gel on small squares of filter paper (about 0.5×0.5 cm), and carefully insert these, using a forceps, in wells cut across the middle of the gel. Place a piece of polyethylene wrap over the gel, and put a cooling plate on top. Run the gel for 4 h at 550 V and 60 mA in a cold room at 4°C. After electrophoresis, stain the gel with an activity stain for LDH (see Section 2.3.) until the bands appear. Wash the gel exhaustively with water, otherwise overstaining can easily occur, and treat the gel gently, since starch gels are fragile. Figure 3 shows a starch gel on which the separation pattern can be seen.

3.3.4. Purification of Beef Heart Phosphofructokinase (PFK; EC 2.7.1.11)

1. Obtain a fresh portion of beef heart from a freshly slaughtered animal, and keep it on ice. Remove the ventricular muscles and adipose tissue, dice the

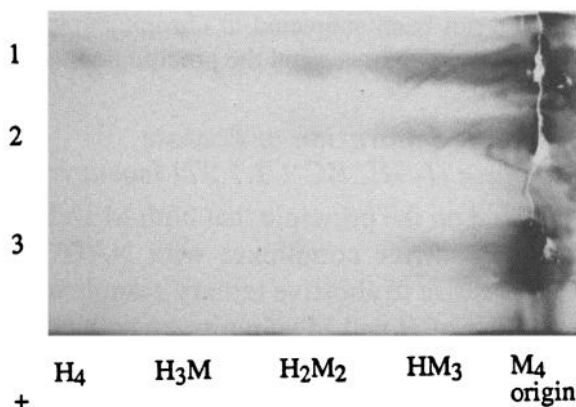


Fig. 3. Starch gel electrophoresis of LDH isoenzymes. The gel was stained with an activity stain for LDH, as described in the text. Lane 1 consists of Sigma Type X bovine muscle LDH (1.4 U) containing all isoenzymes of LDH with M and H subunits. Lane 2 contains the supernatant after affinity precipitation in the presence of bis-NAD⁺ (1 NAD⁺ Eq/LDH subunit) and 20 mM oxamate. Lane 3 contains the resolubilized pellet from the same sample as lane 2. The pellet was resolubilized in 20 mM potassium phosphate, pH 7.4, containing 0.6 mM NADH. Affinity precipitation and resolubilization were carried out in a final vol of 100 μ L.

remaining tissue, and wash it in distilled water. Homogenize it in a pre-cooled blender for 10 s, take 500 g of mince, and add 10 mM Tris-HCl buffer containing 2mM EDTA, pH 8.2, to a total vol of 1.25 L. Homogenize this for 90 s at high speed, and then centrifuge the homogenate at 2,600g for 10 min. Discard the supernatant.

2. Resuspend the pellet in 10 mM Tris-HCl containing 0.5 mM ATP, 50 mM MgSO₄, and 5 mM 2-mercaptoethanol, pH 8.0, to a total vol of 1 L. Pre-warm the buffer to 37°C. Stir the suspension at 37°C for 30 min. This extraction procedure gives rise to a decrease in pH of 0.6–0.8 pH units. Remove solid material by centrifugation for 15 min at 8000g, followed by a further centrifugation at 12,000g. The supernatant contains about 90% of the PFK activity.
3. Set up a pilot precipitation (*see* Section 3.2.) to determine the concentration of bis-ATP giving optimum precipitation. The following volumes are appropriate: 100 μ L enzyme solution, 30 μ L of 30 mM citrate, pH 7.0, 60 μ L 0.16M MgSO₄. Add bis-ATP to give a range of ATP Eq/PFK subunit ratios (0–8 is suitable, at an enzyme concentration of 0.38 mg/mL PFK). Make the volume up to 500 μ L with extraction buffer. A ratio of 4–6 is usually found

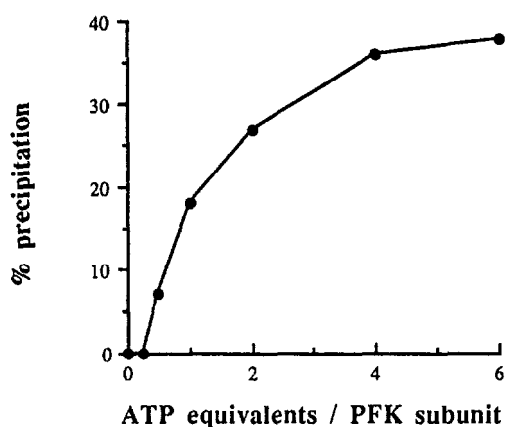


Fig 4. Affinity precipitation of PFK. A pilot precipitation was set up as described in the text. The precipitation mixture contained 0.1 mL enzyme supernatant, 20 μ L 0.5M citrate, pH 7.0, bis-ATP (0.25–6 ATP Eq/ATP subunit), and 20 μ L 0.5M MgSO_4 . The tubes were left overnight at 0°C, and centrifuged for 10 min to pellet the enzyme.

Table 3
Purification of PFK from Bovine Heart (200 g Tissue) (36)^a

Step	Volume, mL	Total protein, mg	Total activity, U	Specific activity, U/mg	Purification, factor	Yield, %
Crude extract						
Resuspended						
Precipitate	490	3675	970	0.26	1	100
Supernatant						
After extraction	548	376	883	3.2	12.3	91
Affinity precipitation	3	7.1	214	30.1	115.7	22

^aThe method used is that given in Section 3.3.4

to be optimum for precipitation (see Fig. 4). A similar pilot experiment varying the enzyme concentration can also be set up at a fixed concentration of bis-coenzyme. A sample purification table is given (Table 3).

- 4 The volume for precipitation can be scaled up as required. The enzyme can be recovered by dialysis, which causes resolubilization of the pellet.

4. Notes

- 1 Bis-NAD⁺ has been used as a ligand with several NAD⁺-linked dehydrogenases, but only for three enzymes (YADH, LDH, GDH) has it successfully

been applied for purification purposes. A partial purification of isocitrate dehydrogenase giving rise to a yield of 18% has been described (7). Horse liver alcohol dehydrogenase has also been observed to form aggregates with bis-NAD⁺ and pyrazole (5,8), but it does not affinity precipitate, despite the fact that it may form larger polymers. Other enzymes that do not successfully affinity precipitate include malate, alanine, and glyceraldehyde-3-phosphate dehydrogenases (28). Malate dehydrogenase is a dimer, and the other two enzymes, although tetrameric, may fail to form crosslinked ternary complexes, possibly owing to steric or kinetic factors.

2. The R_f values given for TLC of coenzyme derivatives are not exact. They can vary by up to ± 0.05 . This may be owing to slight changes in the solvent composition over time, possibly because of evaporation.
3. To date, bis-ATP has only been used in the purification of PFK. The yields obtained for this compound are low, and it is unstable, possibly owing to hydrolysis of the spacer arm, catalyzed intramolecularly by a phosphate group. It should be used within 4 d of synthesis. Because of this, the reagent must be regarded as being of limited practical use, and further studies on the synthesis of a more stable bis-ATP derivative are necessary.
4. YADH has essential thiol groups, and if these are oxidized, the enzyme loses activity. They should be kept reduced by the addition of 1 mM dithiothreitol or 1 mM 2-mercaptoethanol. The breaking of the yeast cells may also be carried out using glass beads.

Acknowledgments

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CHAPTER 23

Isoelectric Focusing

Reiner Westermeier

1. Introduction

Isoelectric focusing (IEF) is performed in a pH gradient in an electric field. The charged proteins migrate toward the anode or the cathode—according to the sign of their net charge—until they reach the position in the pH gradient where their net charges are zero. This pH value is the isoelectric point (pI) of the substance, an exactly defined physicochemical constant. Since the molecule is no longer charged, it stays there; the electric field does not have any influence on it. Should the protein diffuse away, it will gain a net charge and the applied electric field will cause it to migrate back to its pI. This concentrating effect leads to the name “focusing” and makes the method very useful for purification purposes.

The principle of isoelectric focusing is employed in varying technical approaches. The most complete information about these methods is found in the works of Righetti (1,2) and Andrews (3).

The procedure described here is performed in a horizontal flat bed of a granulated dextran gel. Large sample volumes can be mixed with the original gel slurry from which the gel bed is prepared. Labile samples are applied at a defined zone of the gradient. The separation is run at a constant power of 8 W for 14–16 h at controlled temperature. pH measurements and prints on filter paper can be made directly on the gel surface. The sample fractions are collected by sectioning the gel, and then eluting the proteins from these sections. The technique, introduced by Radola in 1969 (4), has been selected for several reasons:

1. It can be performed with standard horizontal electrophoresis equipment, which is also used for analytical methods.

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2. The method has a high loading capacity (up to gram quantities)
3. A large number of different applications and references are available for this technique; some examples are listed in refs. 5–15
4. The technique is much less sensitive to precipitation of proteins at their isoelectric points compared to preparative isoelectric focusing methods in a free liquid because the precipitate is trapped within the gel bed
5. The recovery of the protein fractions from a granulated dextran gel is much easier and gives a higher yield compared to compact gel media like agarose and polyacrylamide

The original method has been refined and optimized by Winter et al. (16).

2. Materials

2.1. Equipment

In addition to standard laboratory equipment, the following are needed:

1. Horizontal electrophoresis chamber (Multiphor II) and Preparative IEF kit (Pharmacia Biotech, Uppsala Sweden). The Preparative IEF Kit contains a tray with a 5-mm silicone rim, a fractionating grid frame with 20 fractionation blades, a sample applicator, sample elution columns, IEF electrode strips, and print papers.
2. Thermostatic circulator at 10°C.
3. Constant power supply (>1 kV).

2.2. Consumables, Chemicals, and Solutions

1. Desalting columns for sample preparation: PD-10 columns prepacked with Sephadex G-25 (Pharmacia Biotech, product code 17-0851-01).
2. IEF electrode strips (clean filter paper strips approx 5×2 mm).
3. Print paper (clean filter paper approx 110×250 cm).
4. Granulated gel media with a very low level of charged contaminants: Ultrodex® (Pharmacia Biotech, product code 80-1130-01) (*see* Note 1).
5. Carrier ampholytes of wide pH range (e.g., pH 3.5–9.5), and narrow pH ranges (2–3 pH units) depending on the pIs of the proteins to be purified: Ampholine® and Pharmalytes® (Pharmacia Biotech).
6. Detergent solution (0.1% [v/v] Triton X-100): 1 mL Triton X-100 made up to 1 L with distilled water.
7. Anode solution (1M phosphoric acid) 2.8 mL concentrated H_3PO_4 made up to 50 mL with distilled water.
8. Cathode solution (1M sodium hydroxide): 2 g NaOH made up to 50 mL with distilled water.
9. Fixing solution (10% [w/v] trichloroacetic acid): 100 g trichloroacetic acid made up to 1 L with distilled water.

Table 1
Volumes for the Carrier Ampholyte Solution

Ampholytes	pH	pH range, mL				
		2.5–5.0	3.5–5.5	5.0–7.5	6.0–8.5	7.0–10.0
Pharmalyte	2.5–5.0	5.5				
Ampholine	3.5–5.0		2.75			
Ampholine	4.0–6.0		2.75			
Ampholine	5.0–8.0			5.5		
Ampholine	6.0–8.0				2.75	
Ampholine	7.0–9.0				2.75	2.75
Ampholine	9.0–11.0					2.75
Distilled water (+ sample)		104.5	104.5	104.5	104.5	104.5
Total volume		110	110	110	110	110

10. Destaining solution (10% [v/v] methanol/20% [v/v] acetic acid): Mix 100 mL methanol with 200 mL acetic acid and make up to 1 L with distilled water.
11. Staining solution (0.2% [w/v] Coomassie blue): Dissolve 0.6 g Coomassie blue R 250 in 300 mL destaining solution.

3. Methods

3.1. Gel Preparation

When large volumes of stable sample solution have to be separated, the sample is included in the initial gel slurry. Labile samples are applied in a certain zone of the gel bed with a sample applicator.

3.1.1. Preparing the Carrier Ampholyte Solution

Prepare a total of 110 mL of a 2% (w/v) carrier ampholyte solution (100 mL for the gel bed and 10 mL for the electrode strips) (*see* Note 2). The ampholytes are diluted in distilled water or with sample solution if large volumes of sample are to be separated (*see* Section 3.1.). The pH range of the carrier ampholytes is selected so that the pI(s) of the protein(s) to be purified fall(s) approximately in the middle of the range. This pH value can be identified by a previous analytical experiment (17) or from information in the literature. A large number of proteins are listed with their isoelectric points in two tables that have been published by Righetti et al. (18,19). Some carrier ampholyte solution recipes are listed in Table 1.

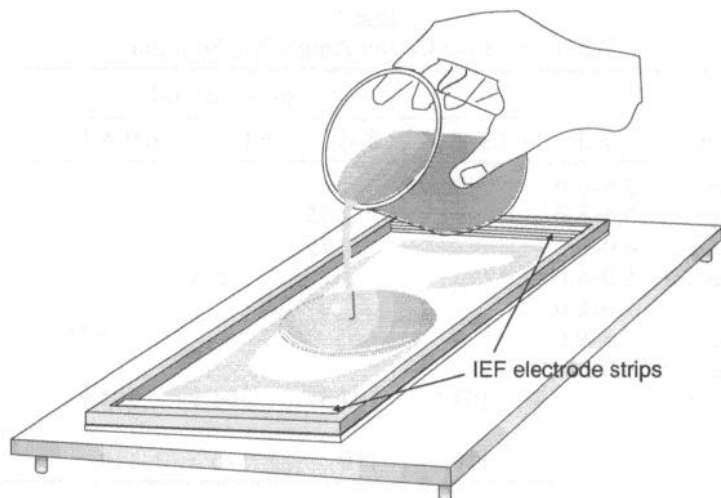


Fig. 1. Pouring the gel slurry into the tray.

3.1.2. Preparing the Gel Bed (see Note 3)

1. Clean the glass plate of the tray thoroughly before use in order to avoid uneven spreading of the gel layer.
2. Cut six IEF electrode strips to 10.5 cm, and soak them in 10 mL of the carrier ampholyte solution. Place three layers of strips at each end of the tray, weigh the tray, and place it on a horizontal table.
3. Weigh out 4 g of Ultrodex, and sprinkle it in small portions over 100 mL of carrier ampholyte solution in a 200-mL beaker. Let the gel sink down into the liquid before adding the next portion.
4. Weigh the beaker and its contents. Homogenize the suspension by gently stirring with a spatula (do not use a magnetic stirrer). When a basic gradient range is selected, degas the slurry with a water jet pump or comparable equipment, in order to remove the atmospheric carbon dioxide.
5. Pour the suspension into the tray (Fig. 1). Gently tap against the ends of the tray, so that the suspension spreads evenly.
6. Weigh the beaker and the remainder of the slurry.

3.1.3. Evaporating the Gel Bed to the Correct Water Content

1. Mount a small fan 70 cm above the tray, and evaporate excess water with a light stream of air. The speed and distance of the fan must be adjusted so that no ripples are formed on the surface.
2. Control the weight of the tray and its content from time to time. After 1.5–2 h, it should have reached the final weight, which was calculated from the

weight difference of the beaker, the weight of the tray with the strips, and the evaporation limit (*see* Notes 4 and 5).

3.1.4. Arranging the Tray on the Cooling Plate

1. Apply 2 mL of a 0.1% (v/v) Triton X-100 solution on the cooling plate of the Multiphor electrophoresis chamber, which has already been cooled to 10°C.
2. Transfer the tray onto the cooling plate avoiding air bubbles.
3. Place an electrode strip soaked in anode solution at the anodal side and another strip soaked in cathode solution at the cathodal side, each on top of the strips already in the tray. Cathodal and anodal sides are marked on the cooling plate.
4. Cut off the protruding parts so that the electrode strips fit exactly into the tray. No gaps must exist between the ends of the strips and the silicone rubber of the tray.

3.2. Sample Application

3.2.1. Preparing the Sample

1. The sample should not contain any particles or grease. For isoelectric focusing, salt and buffer concentrations over 50 mM should be avoided. Ideally, it should not be higher than 10 mM. The easiest and quickest way to desalt a sample is to perform gel filtration with a PD-10 column or another column packed with Sephadex G-25.
2. The maximum loading capacity is 5–10 mg of protein mixture/mL gel bed (2–4 mg/mL of one single protein) when a narrow pH range is employed, e.g., pH 4.0–6.0. The wider the pH range, the lower the loading capacity.

3.2.2. Loading the Sample

The sample can be already mixed with the initial slurry as described in Section 3.1. Labile samples are applied with the sample applicator as a narrow zone in the gel bed as follows (*see* Note 6):

1. The sample solution should have a volume of 3 mL. Otherwise dilute it with the remainder of the carrier ampholyte solution, which was used for soaking the electrode strips.
2. Press the sample applicator through the gel bed at the desired position (Fig. 2). Scrape off the gel in the applicator, and mix it with the sample in a small beaker.
3. Pour the sample into the applicator, remove the applicator, and allow the the bed to equilibrate for a few minutes.

3.3. Isoelectric Focusing

The resulting separation pattern is dependent on the temperature; conventionally 10°C is used (*see* Notes 7 and 8).

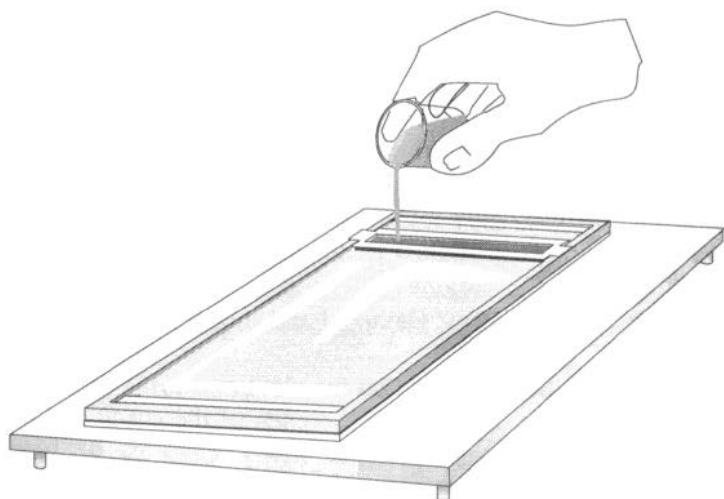


Fig. 2. Pouring the sample into the applicator.

1. Place the platinum electrodes on the respective electrode strips.
2. Perform electrofocusing at a constant current of 8 W for 14–16 h (overnight). The maximal current is 25 mA, and the maximal voltage 1.5 kV.

3.4. Detection of Protein Zones

1. Mark the anode (+) on a sheet of print paper.
2. Carefully roll the paper onto the gel surface avoiding air bubbles. Leave it for 1 min.
3. Carefully remove the print. Apply the electric field again to the gel during the staining of the print in order to avoid diffusion of the zones.
4. Treat the print as follows:
 - a. Three times washing out of the carrier ampholytes in fixing solution for 15 min.
 - b. Staining for 10 min in staining solution (*see* Note 9).
 - c. Destaining for 1 h in destaining solution (*see* Notes 10–12).
 - d. Dry the print with hot air.

If colored protein fractions are separated, no staining is necessary. Figure 3 shows the stained paper print of a hemoglobin separation from a pH gradient 5.0–7.5.

3.5. Measurement of the pH Gradient

1. Press the fractionating grid onto the gel, which is still on the cooling plate at 10°C (Fig. 4).

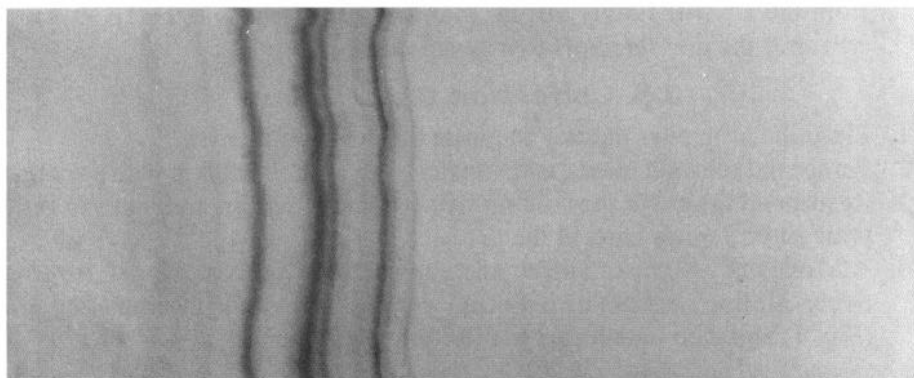


Fig. 3. Paper print of focused hemoglobin bands from a preparative bed of granulated gel with a pH gradient of 6.9–8.5. Reproduced with permission from Pharmacia Biotech AB.

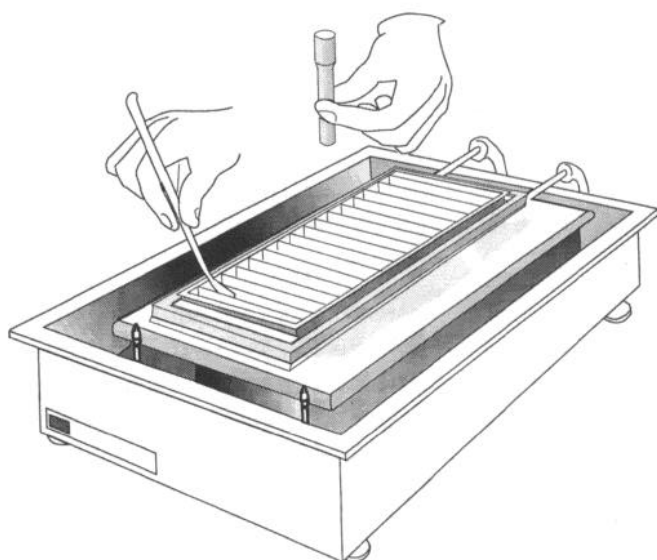


Fig. 4. Scraping the protein sections from the gel using the fractionation grid.

2. Measure the pH gradient by inserting a surface electrode into every second compartment of the 20 fractions of the grid. Wait until the reading has stabilized. For exact results, the electrode has to be calibrated at 10°C. Always start at the cathodal side of the gradient because carbon dioxide

from the air will slowly diffuse into the gel surface and the HCO_3^- ions will shift the measured pH to a lower value.

3.6. Collection of the Zones

1. Place the print near the tray to locate the zones of interest.
2. Scrape out relevant bands, and transfer to an elution column with a spatula.
3. Resuspend the gel in the column with a suitable buffer, and leave to settle until all buffer has entered the gel.
4. Add one gel volume of buffer, and elute the protein from the gel. Alternatively, all fractions can be collected with the help of the fractionating grid (Fig. 4) and then transferred to the elution columns.

3.7. Separation of the Protein from the Carrier Ampholytes

In most cases, it is not necessary to remove the carrier ampholytes from the protein fractions since:

1. They do not interfere with activity measurements and binding studies
2. They are not toxic.
3. The fraction can be directly injected for antibody production

If necessary, several methods can be employed for separating the protein from the carrier ampholytes, such as gel filtration (Chapter 11), ultrafiltration (Chapter 12), dialysis (Chapter 11), ammonium sulphate precipitation (Chapter 10), and electrophoresis (Chapters 33 and 34).

4. Notes

1. Ultrodex is a specially prepared dextran gel with very weak electroosmotic effects. Untreated Sephadex G-75 or G-100 superfine gels may contain varying amounts of charged, water-soluble contaminants that interfere with the formation of the pH gradient. These gels have to be washed with 10 vol of distilled water and filtered under gentle vacuum
2. Do not use < 2% (w/v) carrier ampholytes. Otherwise distortions and uneven bands appear in the gel during focusing. Using a higher concentration of carrier ampholytes generally does not improve the separation or the loading capacity
3. The water content of the final gel bed is very important. If it is too low, the gel cracks. If it is too high, the focused proteins sediment to the lower part of the gel bed. The optimal consistency of the gel bed is controlled by weighing the initial slurry and repeated weighing during an evaporation until a certain percentage of the original weight is reached. This percentage is calculated using the defined evaporation limit, which is indicated on each Ultrodex bottle. For example, a typical evaporation limit might be

34%. Suppose that the weights of the beaker plus slurry before and after filling the tray were 162.6 and 60.3 g, respectively, i.e., 102.3 g of slurry were added to the tray. The evaporation limit is 34% of this, i.e., 34.8 g, and the target weight of slurry is 67.5 g. If the initial weight of the tray and strips was 145.1 g, then the desired final weight after evaporation would be $145.1 + 67.5 = 212.6$ g.

4. Alternative to the weighing procedure, the water content of the gel bed can be adjusted in the following way: Cover the mouth of the bottle of Ultrodex with a fine gauze net, and fix it with an elastic rubber ring. Sprinkle the dry material onto the surface until the correct consistency has been achieved. This is controlled by tilting the tray to an angle of 45° ; when the gel surface does not move, the water content is correct.
5. If the gel cracks during the run in spite of all the precautions described, check the cooling efficiency of the system; the water flow should be 4–12 L/min at $4\text{--}10^\circ\text{C}$.
6. In the case of very labile proteins, sample application should be performed after a prefocusing step of 30 min at a constant 8 W. In this way, the sample can be loaded at a defined optimal pH value that can be predetermined from a previous analytical experiment. Often it is very helpful to check the literature to find out the optimal place of sample application.
7. When oxygen-sensitive enzymes or very basic proteins are being separated, flush the chamber with a light stream of nitrogen during the focusing experiment.
8. Some enzymes have to be separated at a lower temperature to maintain their biological activity, e.g., at 4°C . pH measurement is then almost impossible, because the electrode would respond extremely slowly at this low temperature. The pH values must always be measured at the focusing temperature because the pK_a values of the carrier ampholytes and of the proteins are highly dependent on temperature.
9. Alternatively or additionally to general protein staining of the paper print, specific staining for glycoproteins or zymogram techniques can be employed. The print can be cut into several strips and stained using different methods.
10. If skew bands are produced, then either the salt content of the sample could be too high or there could be too much electrolyte in the IEF electrode strips. Desalt the sample and/or blot the IEF electrode strips on a filter paper before placing them into the tray.
11. Irregular bands can also appear owing to overloading in a zone, reduce sample concentration, or choose a narrower pH range.
12. In some cases, band distortions can be cured by adding nonionic detergents (e.g., 0.5% [v/v] Triton X-100), 10% [v/v] glycerol, 7% [v/v] monoethyleneglycol, or 3–7M urea to the gel and the sample.

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CHAPTER 24

Chromatofocusing

Timothy J. Mantle and Patricia Noone

1. Introduction

Chromatofocusing (1) is essentially ion-exchange chromatography (conventionally using an anion exchanger) where the elution conditions are obtained by dropping the pH so that the proteins elute in order of their isoelectric points (pI). Although in principle this can be achieved by equilibrating an ion exchanger at a relatively alkaline pH and then titrating the resin with a buffer at a lower, more acidic pH, in practice it is more usual to utilize amphoteric buffers titrated to the lower pH to generate a more linear pH gradient. Since the pH gradient is generated gradually as the eluting buffer moves down the column, it is possible to add large volumes to the column or to add a second batch during the run and, in either case, single identical components with a particular pI will focus and elute as a single peak.

In practice this method is extremely straightforward since no gradient makers are required and the separation is normally superior to ion-exchange chromatography.

The protocol described here allows the purification of five major forms of pig liver glutathione *S*-transferase (GST) by chromatofocussing on polybuffer exchanger PBE 94 (*see* Note 1) following the preparation of a GST affinity pool using *S*-hexylglutathione-Sepharose. It is typical of the method used for proteins with pI values in the range 6.5–8.5 (*see* Note 2 for variations in the protocol for proteins with pI values outside this range).

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2. Materials

2.1. Buffers

Chromatofocusing buffers are available commercially as polybuffers (Pharmacia LKB, Uppsala, Sweden). This protocol uses Polybuffer 96 (*see* Note 1) which has a characteristic low absorbance at 280 nm, but can be detected at 254 nm. For this reason eluent from a chromatofocusing column must be measured at 280 nm. All other reagents are made up in distilled deionized water at 20°C, unless otherwise indicated.

1. Buffer A: 10% (v/v) Polybuffer 96, pH 6.0 (adjusted with glacial acetic acid).
2. Buffer B: 0.025M ethanolamine, pH 9.4 (adjusted with glacial acetic acid).
3. Buffer C: 0.1M sodium phosphate, pH 6.5.
4. Buffer D: 0.01M sodium phosphate, pH 7.2
5. Buffer E: 0.5% (w/v) SDS, 0.006% (w/v) bromophenol blue, 50% (v/v) glycerol, 63 mM Tris-HCl, pH 6.8, 0.5% (v/v) 2-mercaptoethanol added immediately prior to use.

2.2. Resins

Pack a 25 × 1.5-cm diameter column (total bed volume 44.3 mL, total capacity 3.5 meq/100 mL of resin) with PBE 94 (*see* Note 2).

2.3. Electrophoresis

For SDS-PAGE, Analar-grade chemicals are used with the buffer system of Laemmli (2). A 12% resolving gel was used in the present case.

3. Method

1. Pre-equilibrate the gel to be used (PBE 94) with start buffer (buffer B) and pack into the column.
2. Wash the column with buffer B until the pH of the eluent is the same as that of buffer B (approx 10 column volumes).
3. Precede the application of the sample by the application of 5 mL of the eluent buffer; this ensures the protein sample is never exposed to an extreme of pH.
4. Apply the sample at a flow rate of 1 mL/min (*see* Note 3).
5. Elute by the application of the eluent buffer to the column. The volume of the elution buffer depends on the strength of the eluent solution. For pH 6.0–9.0, an elution buffer volume of 10.5 column volumes is recommended (*see* Note 4).
6. Assay the fractions for the protein of interest (in this case GST; *see* Note 5).
7. Monitor the protein concentration by measuring the absorbance at 280 nm. It is assumed that an absorbance reading of 1 is equivalent to a protein concentration of 1 mg/mL (4). A typical elution profile is shown in Fig. 1.

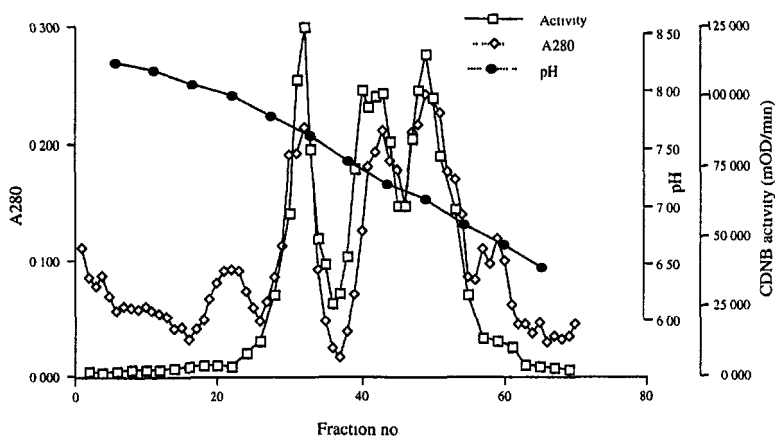


Fig 1. Chromatofocusing of hepatic porcine GSTs, pH 6.0–9.0.

8. Prepare the samples for electrophoresis by adding buffer D to give a final protein concentration of 0.1 mg/mL (*see* Note 6).
9. Add buffer E (5 μ L) to 20 μ L of each sample, mix, and boil on a heating block for 5 min to ensure complete inactivation of proteases and total protein denaturation by SDS.
10. Load the samples (10 μ L) on the gel using a Hamilton syringe. Run the gel at 30 mA until the dye front has run off the gel. An example is shown in Fig. 2.
11. Remove polybuffer, if required, by extensive dialysis against buffer D or by gel filtration on Sephadex G-25 (*see* Note 7 and Chapter 11).

4. Notes

1. For most work all that is required is the PBE 94 and the two amphoteric buffers, Polybuffer 96 and Polybuffer 74, which cover the pH ranges 9.0–6.0 and 7.0–4.0, respectively. If the protein of interest has a more basic pI, then it will be necessary to invest additionally in PBE 118 and Pharmalyte pH 8.0–10.5. All of these materials can be obtained from Pharmacia. Sophisticated columns are available commercially together with a range of peristaltic pumps and on-line monitors for A_{280} and pH. However we have always used successfully a long straight glass column with a sinter that allows a good flow rate and gravity.
2. For most separations a bed volume of approx 20 mL is sufficient for samples containing up to 200 mg of protein; however this is only a guide and may require adjusting. If the pI of the target protein is known (from isoelectric focusing) then Table 1 should be consulted for a suitable start buffer. If there is no information on the pI of the target protein, then it is probably sensible to start with PBE 94 and use the pH range 7.0–4.0

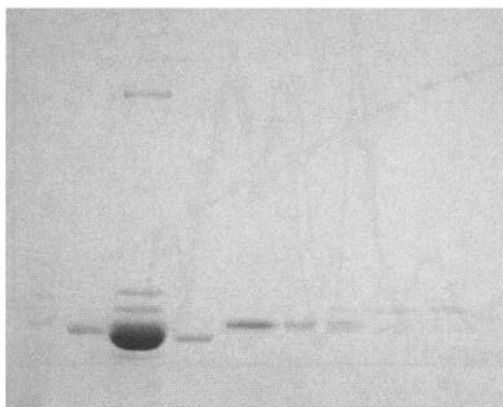


Fig. 2. Chromatofocusing of hepatic porcine GSTs, pH 6.0–9.0 analyzed by SDS-PAGE. Cytosolic GSTs were applied to a chromatofocusing column as described in Section 3. The peaks from the column were applied to a 12% SDS-PAGE gel and run as described. Lane 1, molecular weight markers; Lane 2, purified pig GST pi; Lane 3, mixture of cytosolic GSTs as purified on *S*-hexylglutathione-Sepharose; Lane 4, flow-through from the chromatofocusing column; Lanes 5–9, protein and activity peaks from the chromatofocusing column.

Table 1
Buffers Used in Chromatofocusing^a

pH range	Start buffer	Eluent	Dilution factor
10.5–8.0 ^b	pH 11.0, 0.25 <i>M</i> tri- ethylamine/HCl	pH 8.0, Pharmalyte pH 8.0–10.5/HCl	1:45
9.0–6.0	pH 9.4, 0.025 <i>M</i> ethanolamine/HCl	pH 6.0, Polybuffer 96/ acetic acid	1:10
7.0–4.0	pH 7.4, 0.025 <i>M</i> imidazole/HCl	pH 4.0, Polybuffer 74/ HCl	1:8

^aThe gradient volume should be 10 times the column volume in all cases.

^bThe exchanger used in this case is PBE 118; in the other cases it is PBE 94 (see Note 1).

(Polybuffer 74). If the target protein does not bind then try using the pH range 9.0–7.0 (Polybuffer 96). By using these two sets of conditions (pH range 9.0–4.0), the pI of at least 85% of all known proteins (3) will have been covered. If the target protein is still not binding to the resin, then it is a very basic protein and PBE 118 should be used. When the choice of

exchanger and start buffer has been made, equilibration with start buffer should be commenced. This can take place in the column or more rapidly in a sintered glass funnel. In either case equilibration usually requires 10–15 bed volumes of start buffer. Equilibration should always be checked so that the pH and conductivity of the eluent is identical with that of the start buffer.

3. The sample volume should normally be within 0.5 bed volumes, although the sample volume is unimportant as long as all of the sample is applied to the column before the component of interest has eluted. The sample should not contain large amounts of salt and the ionic strength should be $<0.05M$. If the buffer concentration of the sample is low then the pH of the sample is unimportant, however it is normal to load the sample in either start buffer or eluent buffer, whichever maintains the activity of the component of interest. The loading and elution sequence is:
 - a. 5 mL of eluent buffer;
 - b. Sample (in eluent or start buffer);
 - c. Eluent buffer
4. Elution is achieved simply by running 10 column volumes of eluent buffer through the column. The flow rate can be variable and may be regulated by a peristaltic pump or by gravity. Protein should be monitored at 280 nm and the pH of each fraction measured immediately. There is a dead volume of 1.5–2 column volumes so in practice it is usual to use 12 column volumes.
5. The assay conditions for the activity of GST have been described previously (4). Reactions are monitored by following a change in absorbance when substrate is conjugated to glutathione.

Assays are carried out in a final volume of 2 mL in a quartz cuvet at 30°C. The absorbance change at 340 nm is recorded with a Philips (Eindhoven, Holland) PU8625 UV/VIS spectrophotometer attached to a Philips PM 8261 Xt recorder. Glutathione (GSH; 40 mM stock solution) is prepared freshly in buffer C. 1-Chloro-2,4-dinitrobenzene (CDNB; 40 mM), the electrophilic substrate, is dissolved in ethanol. Buffer C (1.8 mL) incubated at 30°C is placed in the cuvet, glutathione (50 μ L) and CDNB (50 μ L) are then added. A blank rate is observed before eluent fraction (100 μ L) is added.

Alternatively the assays are carried out in a microwell plate (Nunc, Kamstrup, Denmark) with a final volume of 200 μ L. The absorbance change is monitored at 340 nm (30°C) using a Molecular Devices Thermo-max plate reader attached to a Macintosh SE 1/40. The results are analyzed using the Softmax[®] software. Buffer C (140 μ L) is applied to each well followed by protein (10 μ L) except for the blanks where buffer C (10 μ L) is added. CDNB (0.5 mL) and GSH (0.5 mL) are added to buffer C (9 mL). The reaction mix (50 μ L) is added to each well and the absorbance change

is observed for 2 min at 6-s intervals. This second method is particularly useful if only small quantities of protein are obtained.

6. If the protein concentration is <0.1 mg/mL, the sample can be concentrated using trichloroacetic acid (TCA). The desired amount of protein solution is mixed with an equal volume of ice-cold TCA (25% [w/v]). The samples are incubated on ice for 20 min and spun in a microcentrifuge at 12,000g for 3 min. The supernatant is discarded and the pellet is washed twice in ice-cold acetone (1 mL) and allowed to dry. The pellet is resuspended in buffer D (20 μ L).
7. There are a number of ways that removal of polybuffer from the sample can be achieved.
 - a. Precipitation with ammonium sulphate. Add sufficient solid ammonium sulphate to obtain 85% saturation (608 g/L). After 1 h precipitate the suspension using centrifugation, wash twice with saturated ammonium sulphate, and then store as an ammonium sulphate suspension or dialyse/gel filter to remove the ammonium sulphate. All of these steps should be conducted at 4°C.
 - b. Polybuffer can be removed from most proteins by gel filtration providing that the protein of interest has an $M_r > 15,000$.
 - c. If the protein of interest binds to an affinity matrix then this is the most ideal way of obtaining rapid and complete separation.
8. The following cautions should be noted:
 - a. In the absence of any data on the pI of the protein of interest, it is well worth a number of preliminary experiments to optimize the binding and eluting conditions.
 - b. Care should be taken with regard to the buffer counterion which should have a pK_a at least 2.0 pH units below the lowest point of the chosen gradient. For this reason chloride is the counterion of choice.
 - c. Atmospheric carbon dioxide may cause a plateau in the region pH 5.5–6.5 and if this is a problem all buffers should be degassed.

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CHAPTER 25

Size-Exclusion Chromatography

Paul Cutler

1. Introduction

Size-exclusion chromatography (also known as gel-filtration chromatography) is a technique for separating proteins and other biological macromolecules on the basis of molecular size. Size-exclusion chromatography is a commonly used technique owing to the diversity of the molecular weights of proteins in biological tissues and extracts. It also has the important advantage of being compatible with physiological conditions (1,2).

The solid-phase matrix consists of porous beads (100–250 μm) that are packed into a column with a mobile liquid phase flowing through the column (Fig. 1). The mobile phase has access to both the volume inside the pores and the volume external to the beads. The high porosity typically leads to a total liquid volume of >95% of the packed column.

Separation can be visualized as reversible partitioning into the two liquid volumes. Large molecules remain in the volume external to the beads, since they are unable to enter the pores. The resulting shorter flow path means that they pass through the column relatively rapidly, emerging early. Proteins that are excluded from the pores completely elute in what is designated the void volume, V_o . This is often determined experimentally by the use of a high-mol-wt component, such as Blue Dextran or calf thymus DNA. Small molecules that can access the liquid within the pores of the beads are retained longer and, therefore, pass more slowly through the column. The elution volume for material included in the pores is designated the total volume, V_t . This represents the total liquid volume of the column and is often determined by small molecules, such as vitamin B12.

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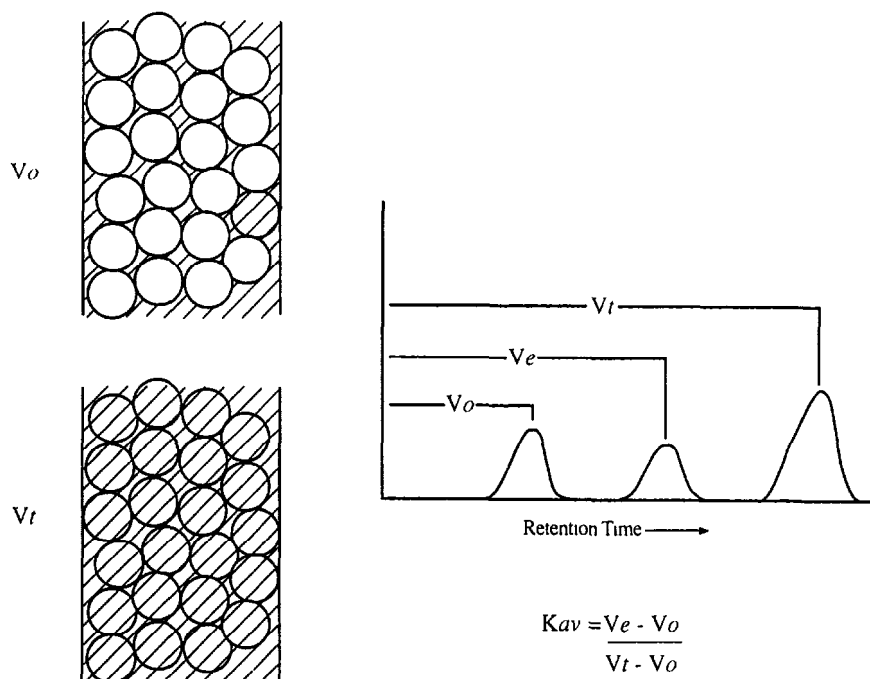


Fig. 1. The basic principle of size exclusion. Solutes are separated according to their molecular size. Large molecules are eluted in the void volume (V_o), and small molecules are eluted in the total volume (V_t). Solutes within the separation range of the matrix are fractionally excluded with a characteristic elution volume (V_e).

The elution volume for a given protein will lie between V_o and V_t and is designated the elution volume, V_e . Intermediate-sized proteins will be fractionally excluded with a characteristic value for V_e . A partition coefficient can be determined for each protein as K_{av} . In size exclusion, the macromolecules are not physically retained, unlike adsorption techniques. Therefore, the protein will elute in a defined volume between V_o and V_t (see Note 1). Size exclusion tends to be used at the end of a purification scheme when impurities are low in number, and the target protein has been purified and concentrated by earlier chromatography steps. An exception to this is membrane proteins, where gel filtration may be used first since concentration techniques are not readily used and the material will be progressively diluted during the purification scheme (3).

A range of different preparative and analytical matrices are available commercially. High-performance columns are used analytically for

studying protein purity, protein folding, protein-protein interactions, and so forth. (4). Preparative separations performed on low-pressure matrices are used to resolve proteins from proteins of different molecular weights, proteins from other biological macromolecules, and for the separation of aggregated proteins from monomers (*see* ref. 5 and Note 2).

Several parameters are important in size-exclusion chromatography. The pore diameter controlling the separation is selected for the relative size of proteins to be separated. Many types of matrices are available. Some are used for desalting techniques where proteins are separated from buffer salts (*see* ref. 6 and Note 3). Some matrices offer a wide range of mol-wt separations, and others are high resolution matrices with a narrow range of operation (Table 1).

2. Materials

2.1. Equipment

The preparative separation of proteins by size exclusion is suited to commercially available standard low-pressure chromatography systems. Systems require a column packed with a matrix offering a suitable fractionation range, a method for mobile-phase delivery, a detector to monitor the eluting proteins, a chart recorder for viewing the detector response, and a fraction collector for recovery of eluted proteins (Fig. 2). The system should be plumbed with capillary tubing with a minimum hold-up volume.

Early systems were less sophisticated with a gravimetric feed of the mobile phase from a suspended reservoir, whereas the most modern systems now have computers to control operating parameters and to collect and store data. The principle of separation, however, remains the same, and high resolution is attainable with relatively simple equipment.

1. Pumps: An important factor in size exclusion is a reproducible and accurate flow rate. The most commonly used pumps are peristaltic pumps, which are relatively effective at low flow rates, inexpensive, and sanitizable. Peristaltic pumps do, however, create a pulsed flow, and often a bubble trap is incorporated both to prevent air from entering the system and to dampen the pulsing effect. More expensive, yet more accurate pumps are syringe pumps, such as those seen on the FPLC[®] system (Pharmacia, Uppsala, Sweden).
2. Column: Size exclusion, unlike some commonly used adsorption methods of protein separation, is a true chromatography method based on continuous partitioning. Hence, resolution is dependent on column length. Col-

Table 1
Commonly Used Preparative Size-Exclusion Matrices

Supplier/matrix	Material type, pH stability ^a	Separation range ^b , kDa
Sephadex		
G25	Dextran, 2.0–10.0 ^d	1–5
G50		1.5–30
G75		3–80 ^c
G100		4–100 ^c
G150		5–150 ^c
G200		5–600 ^c
Sepharose		
6B	Agarose, 3.0–13.0	10–4000
4B		60–20,000
2B		70–40,000
Superdex		
30	Agarose/dextran, 3.0–12.0	0–10
75		3–70
200		10–600
Sephacryl		
S100HR	Dextran/bisacrylamide, 3.0–11.0	1–100
S200HR		5–250
S300HR		10–1500
Biogel		
P-2 gel	Polyacrylamide, 2.0–10.0	0.1–1.8
P-4 gel		0.8–4
P-10 gel		1.5–20
P-60 gel		3–60
P-100 gel		5–100

^aIn aqueous buffers

^bFor globular protein

^cDifferent grades are available, which affects performance

^dpH stability may vary depending on grade and exclusion limit

- umns tend toward being long and thin, typically 70–100 cm long (*see* Note 4). The column must be able to withstand the moderate pressures generated during operation and be resistant to the mobile phase. The use of columns with flow adapters is recommended to allow the packing volume to be varied and provide a finished support with the required minimum of dead space.
3. Detectors: Protein elution is most often monitored by absorbance in the UV range, either at 280 nm, which is suitable for proteins with aromatic amino acids, or at 206 nm, which detects the peptide bond. Detection at lower wavelengths may be complicated by the absorbance characteristics

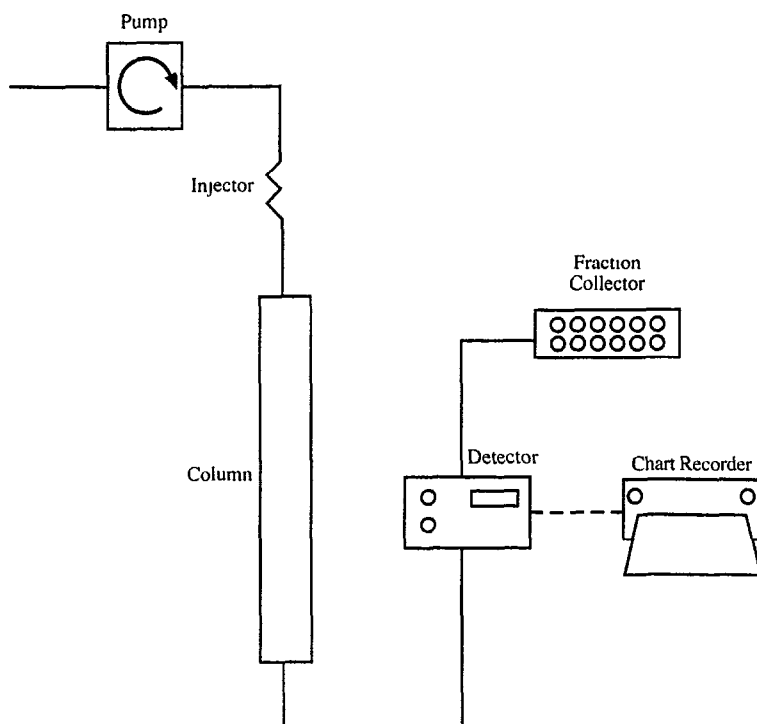


Fig. 2. Schematic diagram of the equipment for preparative size-exclusion chromatography.

of certain mobile phases. The advent of diode array detectors has enabled continuous detection at multiple wavelengths, enabling characterization of the elutes via analysis of spectral data.

Fluorescence detection either by direct detection of fluorescent tryptophan and tyrosine residues or after chemical derivitization (e.g., by fluorescein) has been used, as have refractive index, radiochemical and electrochemical methods, and molecular size (by laser light scattering). In addition to these nonspecific on-line monitoring systems, it is quite common, particularly when purifying enzymes to make use of specific assays for individual target molecules.

4. **Fraction collectors:** A key factor in preparative protein purification is the ability to collect accurate fractions. No matter how efficiently the column may have separated the proteins, the accurate collection of fractions is critical. For the detector to reflect as near as possible in real time the fraction collector, the volume between the detector and the fraction collector should be minimal.

2.2. Buffers

Size-exclusion matrices tend to be compatible with most aqueous buffer systems even in the presence of surfactants, reducing agents, or denaturing agents. Size-exclusion matrices are extremely stable with effective pH ranges of approx 2.0–12.0 (Table 1). An important exception to this are the silica-based matrices, which offer good mechanical rigidity, but low chemical stability at alkaline pHs. Some silica matrices have been coated with dextran, and so forth, to increase the chemical stability and increase hydrophilicity.

The choice of mobile phase is most often dependent on protein stability necessitating considerations of appropriate pH, solvent composition, as well as the presence or absence of cofactors, protease inhibitors, and so on, which may be essential to maintain the structural and functional integrity of the target molecule. All buffers used in size exclusion should ideally be filtered through a 0.2- μ m filter, and degassed by low vacuum or sparging with an inert gas, such as helium (*see* Note 5).

Many matrices retain a residual charge owing, for example, to sulfate groups in agarose or carboxyl residues in dextran. The ionic strength of the buffer should be kept at 0.15–2.0M to avoid electrostatic or van der Waals interactions, that can lead to nonideal size exclusion (7,8). Crosslinking agents, such as those used in polyacrylamide, may reduce the hydrophilicity of the matrix leading to the retention of some small proteins, particularly those rich in aromatic amino acid residues. These interactions have been exploited effectively to enhance purification in some cases, but are generally best avoided (*see* Note 6).

2.3. Selection of Matrix

The beads used for size exclusion have a closely controlled pore size, with a high chemical and physical stability. They are hydrophilic and inert to minimize chemical interactions between the solutes (proteins) and the matrix itself.

The performance and resolution of the technique have been enhanced by the development of newer matrices with improved properties. Historically gels were based on starch, although these were superseded by the crosslinked dextran gels (e.g., Sephadex). In addition, polystyrene-based matrices were developed for the use of size exclusion in non-aqueous solutions. Polyacrylamide gels (e.g., Biogel P series) are particularly suited to separation at the lower mol wt range owing to their

microreticular structure. More recently, composite matrices have been developed, such as the Superdex[®] gel (Pharmacia), where dextran chains have been chemically bonded to a highly crosslinked agarose for high-speed size exclusion (Table 1).

3. Methods

3.1. Flow Rates

In chromatography, flow rates should be standardized for columns of different dimensions by quoting linear flow rate (cm/h). This is defined as the volumetric flow rate (cm³/h)/unit cross-sectional area (cm²) of a given column. Since the principle of size exclusion is based on partitioning, success of the technique is particularly susceptible to variations in flow rates. Conventional low-pressure size-exclusion matrices tend to operate at linear flow rates of 5–15 cm/h. Too high a flow rate leads to incomplete partitioning and band spreading. Conversely, very low flow rates may lead to diffusion and band spreading.

3.2. Preparation and Packing

3.2.1. Preparation

Gel matrices are supplied as either preswollen gels or as dry powder. If the gel is supplied as a dry powder, it should be swollen in excess mobile phase as directed by the manufacturers. Swollen gels must be transferred to the appropriate mobile phase. This can be achieved by washing in a scintered glass funnel under low vacuum. During preparation, the gel should not be allowed to dry. The equilibrated gel should be decanted into a Buchner flask, allowed to settle, and then fines removed from the top by decanting. The equilibrated gel (approx 75% slurry) is then degassed under low vacuum.

3.2.2. Packing

Good column packing is an essential prerequisite for efficient resolution in size-exclusion chromatography. The column should be held vertically in a retort stand avoiding adverse drafts, direct sunlight, or changes in temperature. The gel should be equilibrated and packed at the final operating temperature. With the bottom frit or flow adaptor in place, degassed buffer (5–10% of the bed volume) is poured down the column side to remove any air from the system. In one manipulation, the degassed gel slurry is poured in to the column using a glass rod to direct the gel down the side of the column, avoiding air entrapment. If available, a packing

reservoir should be fitted to columns to facilitate easier packing. The column can be packed under gravity, although a more efficient method is to use a pump to push buffer through the packing matrix. The flow rate during packing should be approx 50% higher than the operating flow rate (e.g., 15 cm/h for a 10 cm/h final flow rate). The bed height can be monitored by careful inspection of the column as it is packing. Once packed, a clear layer of buffer will appear above the bed, and the level of the gel will remain constant. The flow should then be stopped and excess buffer removed from the top of the column, leaving approx 2 cm buffer above the gel. The outlet from the column should be closed, and the top flow adaptor carefully placed on top of the gel avoiding trapped air or disturbing the gel bed.

3.3. Equilibration

The packed column should be equilibrated by passing the final buffer through the column at the packing flow rate for at least one column volume. The pump should always be connected so as to pump the eluent onto the column under positive pressure. Drawing buffer through the column under negative pressure may lead to bubbles forming as a result of the suction. The effluent of the column should be sampled and tested for pH and conductivity in order to establish equilibration in the desired buffer.

3.4. Sample Application

Several methods exist for sample application. It is critical to deliver the sample to the top of the column as a narrow sample zone. This can be achieved by manually loading via a syringe directly onto the column, although this requires skill and practice. The material may be applied through a peristaltic pump, although this will inevitably lead to band spreading owing to sample dilution. The sample should never be loaded through a pump with a large hold-up volume, such as a syringe pump or upstream of a bubble trap. Arguably the best method of applying the sample is via a sample loop in conjunction with a switching valve, allowing the sample to be diverted manually or electronically through the loop and directly onto the top of the column.

3.5. Evaluation of Column Packing

The partitioning process occurs as the bulk flow of liquid moves down the column. As the sample is loaded, it forms a sample band on the column. In considering the efficiency of the column, partitioning can be

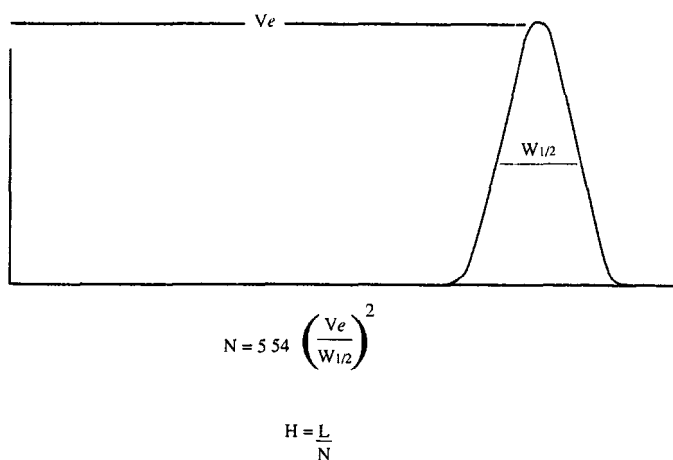


Fig. 3. Calculation of the theoretical plate height as a measure of column performance. The number of theoretical plates (N) is related to the peak width at half height ($W_{1/2}$) and the elution volume (V_e). The height of the theoretical plate (H) is related to the column length (L).

perceived as occurring in discrete zones along the axis of the column's length. Each zone is referred to as a theoretical plate, and the length of the zone termed the theoretical plate height (H). The value of H is a function of the physical properties of the column, the exclusion limit, and the operating conditions, such as flow rate, and so forth. Column efficiency is defined by the number of theoretical plates (N), which can be measured experimentally using a suitable sample, e.g., 1% (v/v) acetone (Fig. 3). Resolution, and hence, the number of theoretical plates are enhanced by increasing column length (*see* Note 7).

3.6. Standards and Calibration

Calibration is obtained by use of standard proteins and plotting retention time (V_e) against log molecular weight. Successful calibration requires accurate flow rates. The resultant plot gives a sigmoidal curve approaching linearity in the effective separation range of the gel.

It should be noted that it is not only the molecular weight that is important in size exclusion, but the hydrodynamic volume or the Stokes radius of the molecule. Globular proteins appear to have a lower molecular size than proteins with a similar molecular weight that are in an α helical form. These, in turn, appear smaller than proteins in random coil form. It

Table 2
Molecular-Weight Standards
for Size-Exclusion Chromatography

Vitamin B ₁₂	1350
Ribonuclease A	13,700
Myoglobin	17,000
Chymotrypsinogen A	25,000
Ovalbumin	43,000
Bovine serum albumin	67,000
Bovine γ globulin	158,000
Blue dextran 2000	~2,000,000

is critical to use the appropriate standards for size exclusion where proteins have a similar shape. This has been useful in some cases for studying protein folding and unfolding. Commonly used mol-wt standards are given in Table 2.

3.7. Separation of Proteins

The optimum load of size-exclusion columns is restricted to <5% (typically 2%) of the column volume in order to maximize resolution. Gel-filtration columns are often loaded at relatively high concentrations of protein, such as 2–20 mg/mL. The concentration is limited by solubility of the protein and the potential for increased viscosity, which begins to have a detrimental effect on resolution. This becomes evident around 50 mg/mL. It is important to remove any insoluble matter prior to loading by either centrifugation or filtration. Owing to the limitation on loading, it is often wise to consider the ability of the method for scale-up when optimizing the operating parameters (ref. 9 and Fig. 4).

3.8. Column Cleaning and Storage

Size-exclusion matrices can be cleaned in situ or as loose gel in a sintered glass funnel. Suppliers usually offer specific guidelines for cleaning gels. Common general cleaning agents include nonionic detergents (e.g., 1% [v/v] Triton X-100) for lipids, and 0.2–0.5M NaOH for proteins and pyrogens (not recommended for silica-based matrices). In extreme circumstances, contaminating protein can be removed by use of enzymic digestion (pepsin for proteins and nucleases for RNA and DNA). The gel should be stored in a buffer with antimicrobial activity, such as 20% (v/v) ethanol or 0.02–0.05% (w/v) sodium azide. NaOH is a good storage